SingleRAN

NSA Mobility Management Feature Parameter Description

Issue	Draft B
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Contents

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes
  - Revised the description of optimized non-overlapping between gap and NR resources. For details, see **Measurement Gap Configuration on the gNodeB**.
  - Revised the description of high-load-specific handover in **7.2.2 Impacts**.
  - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 05 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the "optimized non-overlapping between gap and NR resources" function. For details, see **Measurement Gap Configuration on the gNodeB**.	Modified parameter on the NR side: Added the **GAP_AVOID_NR_RES_OPT_SW** option to the **NRDUCellMeas**.***MeasAlgoSwitch*** parameter.	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the service-based inter-frequency handover enhancement function. For details, see **6.1.1.4.2 Reporting Configuration**.	Added the **NRCellInterFHoMeaGrp**.***InterFreqServHoA2ThldRsrp*** parameter on the NR side.	NR FDD Low-frequency NR TDD High-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Excluded the BTS5929R from the base station models that are compatible with NSA networking based on EPC. For details, see **7.3.3 Hardware**.	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added the high-load-specific handover function in NSA networking. For details, see: • **7.1.15 High-Load-specific Handover** • **7.2.2 Impacts** • **7.3.2.3 Function Impacts** • **7.4.1.1 Data Preparation** • **7.4.1.2 Using MML Commands**	Added the **gNBScenarioDiffGrp** MO on the NR side. Added parameters on the NR side: • **NRCellQciBearer**.***ScenarioDiffGrpId*** • **NRCellMobilityConfig**.***HoMeasAlgoSwitch*** • **NRCellMobilityConfig**.***HighLoadDiffHoUeNumThld*** • **NRCellMobilityConfig**.***HighLoadDiffHoDlPrbThld*** • **NRCellMobilityConfig**.***HighLoadDiffHoUlPrbThld***	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the gap-assisted scheduling compatibility function. For details, see **Measurement Gap Configuration on the gNodeB**.	Modified parameter on the NR side: Added the **GAP_SCH_COMP_SW** option to the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter.	Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs". For details, see **6 PSCell Change in NSA Networking**.	Modified parameter on the LTE side: Added the **NSA_REEST_FLOW_CONFLICT_OPT_SW** option to the **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** parameter.	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added blacklist control for non-gap-assisted coverage-based inter-frequency PSCell change. For details, see: • **7.1.14 Non-Gap-Assisted Coverage-based Inter-Frequency PSCell Change** • **7.4.1.1 Data Preparation** • **7.4.1.2 Using MML Commands**	Modified parameter on the NR side: Added the **INTERF_HO_NO_GAP_SW_OFF** option to the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter.	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for customizing the threshold for event A4 and threshold 2 for event A5 related to RFSP-specific frequency-priority-based measurements for UEs. For details, see **Measurement Event** in **6.1.1.4.2 Reporting Configuration**.	Added the **gNBRfspConfig**.***FreqPriIFHoA4RsrpThldOfs*** parameter on the NR side.	NR FDD Low-frequency NR TDD High-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the RSRP- and SINR-based joint handover function. For details, see **6.1.1.6 Target Cell or Frequency Evaluation**.	Added parameters on the NR side: • **NRCellInterFHoMeaGrp**.***RsrpSinrBasedNecSinrThld*** • **NRCellInterFHoMeaGrp**.***RsrpSinrBasedUnecSinrThld*** Modified parameters on the NR side: • Added the **RSRP_SINR_HO_SW** option to the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter. • Added the **RSRP_SINR_HO_SW** option to the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter.	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
  - Revised the structure of chapters "PCell Change in NSA Networking" and "PSCell Change in NSA Networking" with sections "Network Analysis", "Requirements", and "Operation and Maintenance" added. For details, see **5 PCell Change in NSA Networking** and **6 PSCell Change in NSA Networking**.
  - Revised the descriptions of the principle of service-based inter-frequency handover. For details, see **5.1.2 PCell Change Policies**.

- Revised the descriptions of the principles for the serving cell's RSRP offset related to LTE-coverage-based LTE inter-frequency handover for NSA DC. For details, see **5.1.2 PCell Change Policies**.
- Revised the license requirements for NSA mobility management. For details, see **5.3.1 Licenses**, **6.3.1 Licenses**, and **7.3.1 Licenses**.
- Adjusted the positions of some prerequisite functions, mutually exclusive functions, and function impacts in the document. For details, see **5.3.2 Functions**, **6.3.2 Functions**, and **7.3.2 Functions**.
- Revised the descriptions of the principle of the "NR+LTE-coverage-based LTE inter-frequency handover for NSA DC" function with the NR-coverage-based PCC anchoring function moved to **5.3.2.1 Prerequisite Functions**.
- Added **5.4.1.1 Data Preparation**, **5.4.1.2 Using MML Commands**, **6.4.1.1 Data Preparation**, and **6.4.1.2 Using MML Commands**.
- Added **5.4.2 Activation Verification** for PCell change in NSA networking.
- Revised the descriptions of the principle of PSCell change in NSA networking. For details, see **6 PSCell Change in NSA Networking**.
- Moved the descriptions of activation verification for PSCell change policies from **6.1.2 PSCell Change Policies** to **6.4.2 Activation Verification**.
- Added descriptions related to the "full configuration during inter-gNodeB handovers" function. For details, see **6.1.1.7 Handover Execution**, **6.2.2 Impacts**, **6.4.1.1 Data Preparation**, and **6.4.1.2 Using MML Commands**.
- Deleted some contents from **7.4.1.1 Data Preparation**, **7.4.1.2 Using MML Commands**, and **7.4.2 Activation Verification**. For details about the deleted contents, see *EPC-based NSA Basics*.
- Revised the description of step 5 in the MeNB-initiated SgNB addition procedure. For details, see **8.1 MeNB-Initiated SgNB Addition**.
- Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-131122	NSA Networking based on EPC (LTE FDD)	5 PCell Change in NSA Networking 7 Functions Related to NSA Mobility Management
LTE TDD	MRFD-131132	NSA Networking based on EPC (LTE TDD)	
NR	MRFD-131162	NSA Networking based on EPC (NR)	6 PSCell Change in NSA Networking 7 Functions Related to NSA Mobility Management
NR	FBFD-010014	Mobility Management	

2.4 Differences

Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
PCell change in NSA networking	None	6 PSCell Change in NSA Networking
Functions related to NSA mobility management	None	7 Functions Related to NSA Mobility Management

Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
PSCell change in NSA networking	None	6 PSCell Change in NSA Networking
Functions related to NSA mobility management	None	7 Functions Related to NSA Mobility Management

Differences between NSA and SA

Function Name	Difference	Chapter/Section
PSCell change in NSA networking	NSA networking involves PSCell change and PCell change, while SA networking involves only serving cell change. This document describes PSCell change and PCell change in NSA networking. For details about serving cell change in SA networking, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. ● The principles and mechanisms of basic functions related to serving cell change in SA networking are similar to those of PSCell change in NSA networking. The differences are as follows: – The maximum number of non-serving NR frequencies that can be delivered in NSA networking is different from that in SA networking. For details, see **6.1.1.4.1 Measurement Object**. – For inter-frequency measurement, the method of QCI-specific measurement parameter configuration is different between SA UEs and NSA UEs. For details, see **6.1.1.4.2 Reporting Configuration**. – SSB time-domain position delivery configuration is supported only in high-frequency scenarios in NSA networking and is not supported in SA	**6 PSCell Change in NSA Networking**

Function Name	Difference	Chapter/Section
	networking. For details, see **SSB Time-Domain Position Delivery**. – For other basic principles and mechanisms that are applicable only to SA networking, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. ● 5G supports multiple handover functions to guarantee UE mobility in various scenarios. The applicable function scenarios of serving cell change in SA networking are similar to those of PSCell change in NSA networking. The differences are as follows: – Coverage-based intra-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same in SA and NSA networking, except for the differences in activation verification and performance monitoring. – Coverage-based inter-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same, with the following differences: In SA networking, measurement-based inter-frequency handover, measurement-based redirection, and	

Function Name	Difference	Chapter/Section
	blind redirection are supported, but blind inter-frequency handover is not supported. In NSA networking, inter-frequency PSCell change can be implemented in measurement-based mode or blind mode, but redirection is not supported. Activation verification and performance monitoring differ between SA and NSA networking for measurement-based inter-frequency handover that is supported by SA and NSA networking. – Frequency-priority-based inter-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same in SA and NSA networking, except for the differences in activation verification and performance monitoring. – Operator-specific-priority-based inter-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same in SA and NSA networking, except for the differences in activation verification and performance monitoring.	

Function Name	Difference	Chapter/Section
	– Service-based inter-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same in SA and NSA networking, except for the differences in activation verification and performance monitoring. – Uplink-interference-based inter-frequency handover: This function is supported in both SA and NSA networking. The implementation is the same in SA and NSA networking, except for the differences in function switches, activation verification, and performance monitoring.	

Differences between high frequency bands and low frequency bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

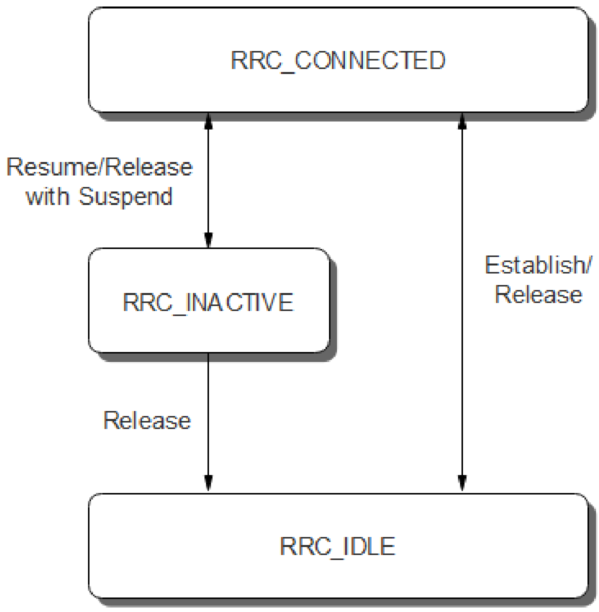
Function Name	Difference	Chapter/Section
PSCell change in NSA networking	Supported in both high and low frequency bands, with the following differences: • In low frequency bands, the SMTC duration is configurable. In high frequency bands, it is automatically calculated by the system. • SSB time-domain position delivery configuration is supported only in high frequency bands. For details, see **SSB Time-Domain Position Delivery**. • Coverage-based inter-frequency PSCell change: Measurement-based inter-frequency PSCell change is supported only between low frequencies as well as between high and low frequencies, not between high frequencies. Blind inter-frequency PSCell change is supported only from high frequencies to low frequencies, not between high frequencies, between low frequencies, or from low frequencies to high frequencies. • Uplink-interference-based inter-frequency PSCell change: not supported in high frequency bands.	**6 PSCell Change in NSA Networking**

3 Overview

Mobility management is a basic function used to ensure service continuity for UEs on the move.

In an NR network, a UE can be in one of the following radio resource control (RRC) states at a time: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states. **Figure 3-1** provides the details about the RRC states.

Figure 3-1 RRC states



- RRC_CONNECTED: An RRC connection is established between the UE and the gNodeB.
- RRC_IDLE: No RRC connection is established between the UE and the gNodeB.
- RRC_INACTIVE: In this state, the UE suspends data processing. However, the gNodeB still maintains the UE's context information, and as a result, the core network considers that the UE remains in the CM-CONNECTED state (as if it were still in the RRC_CONNECTED state). When data transmission is required, the UE can quickly transit to the RRC_CONNECTED state.

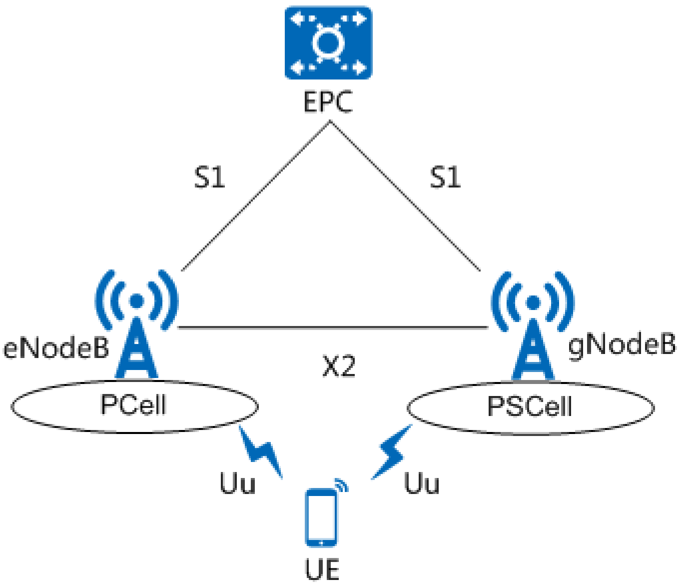
For details about the RRC states, see section 4.2.1 "UE states and state transitions including inter RAT" in 3GPP TS 38.331 V15.5.1.

5G supports standalone (SA) and non-standalone (NSA) networking options. Therefore, mobility management is classified into SA mobility management and NSA mobility management. This document describes NSA mobility management. For details about SA mobility management, see *SA Mobility Management in Connected Mode*, *SA Mobility Management in Idle Mode*, and *SA Mobility Management in Inactive Mode* in 5G RAN feature documentation.

NSA Mobility Management

In NSA networking, UEs supporting E-UTRA-NR dual connectivity (EN-DC) can connect to both an LTE base station (eNodeB) and a 5G New Radio (NR) base station (gNodeB), using the radio resources of both base stations for transmission. Specifically, a UE can connect to both a primary cell (PCell) served by an eNodeB and a primary SCG cell (PSCell) served by a gNodeB, as shown in **Figure 3-2**. For details about NSA networking, see *EPC-based NSA Basics*.

Figure 3-2 NSA EN-DC networking (Option 3x as an example)



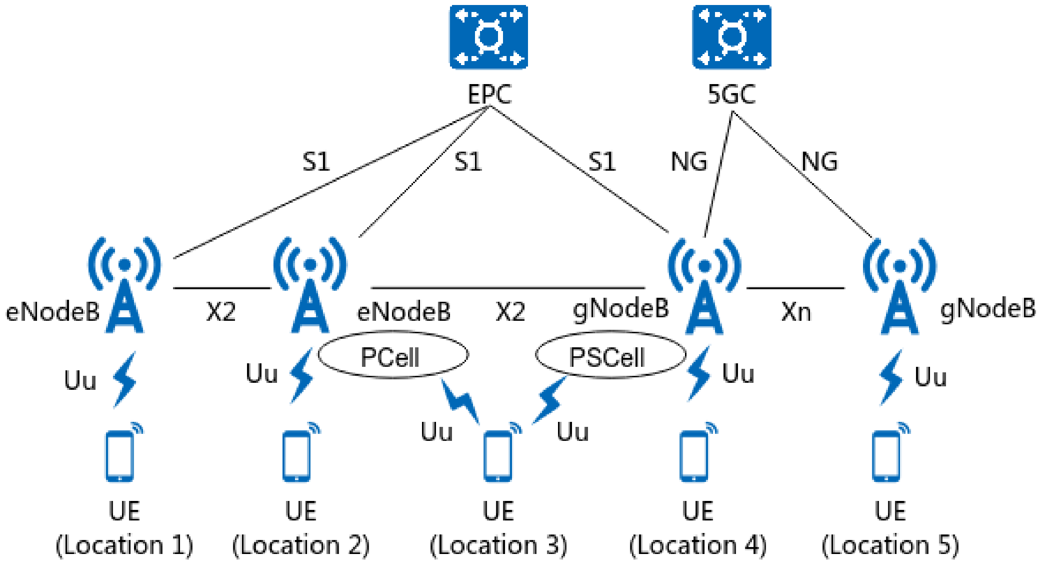
NSA networking does not support the RRC_INACTIVE state. Therefore, in the current version, NSA mobility management is categorized based on the UE's RRC state as follows:

- Mobility management in the RRC_CONNECTED state: UEs in this state are connected to both the PCell and PSCell. Therefore, mobility management is categorized as PCell change and PSCell change, as described in **5 PCell Change in NSA Networking** and **6 PSCell Change in NSA Networking**, respectively.
- Mobility management in the RRC_IDLE state: UEs in this state camp on LTE cells. For details about mobility management of UEs in this state, see cell reselection described in *Idle Mode Management* in eRAN feature documentation.

Mobility Management in NSA and SA Hybrid Networking

NSA and SA hybrid networking facilitates the evolution from NSA to SA. **Figure 3-3** shows the networking architecture.

Figure 3-3 NSA and SA hybrid networking architecture



As shown in **Figure 3-3**, the following mobility scenarios exist in NSA and SA hybrid networking:

- When a UE moves from location 1 to location 2: The UE camps on an LTE cell no matter it is at location 1 or location 2. In this scenario, intra-LTE mobility management is performed and the LTE mechanisms apply. For details, see *Idle Mode Management* and *Mobility Management in Connected Mode* in eRAN feature documentation.
- When a UE moves from location 2 to location 3: The UE camps on an LTE cell at location 2 and an LTE cell and NR cell simultaneously at location 3 (NSA networking). In this scenario, NSA mobility management as described in **NSA Mobility Management** is performed.
- When a UE moves from location 3 to location 4: The UE is connected to an LTE cell and NR cell simultaneously at location 3 (NSA networking) and an NR cell at location 4 (SA networking). In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.
- When a UE moves from location 4 to location 5: The UE camps on an NR cell in SA networking no matter it is at location 4 or location 5. In this scenario, SA mobility management is performed. For details, see *SA Mobility Management in Connected Mode*, *SA Mobility Management in Idle Mode*, and *SA Mobility Management in Inactive Mode* in 5G RAN feature documentation.

4 NSA Mobility Management in Connected Mode

Mobility management in connected mode is a process during which an eNodeB and a gNodeB monitor air interface status, determine serving cell change, and change serving cells for RRC_CONNECTED UEs. Generally termed as handover, this process ensures that the UEs receive uninterrupted services from networks while on the move. The serving cell change on the LTE side is referred to as PCell change, which is described in **5 PCell Change in NSA Networking**. The serving cell change on the NR side is referred to as PSCell change, which is described in **6 PSCell Change in NSA Networking**.

Figure 4-1 and **Table 4-1** provide an overview of NSA mobility management in connected mode and the corresponding signaling procedures, respectively. If LTE and NR have overlapping coverage areas, NR cells served by SgNB 1 and SgNB 2 must be configured as external cells of MeNB 1 and MeNB 2.

The SgNB is responsible for its mobility management, including measurement control and RRC connection reconfiguration.

Figure 4-1 Overview of NSA mobility management in connected mode

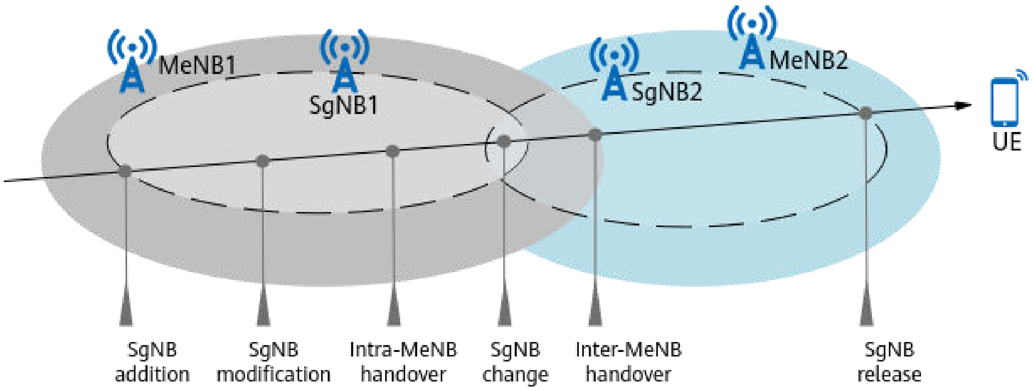


Table 4-1 Signaling procedures for NSA mobility management in connected mode

Mobility Scenario	Signaling Procedure
SgNB addition	8.1 MeNB-Initiated SgNB Addition
SgNB modification	• 8.8 MeNB-Initiated SgNB Modification • 8.9 SgNB-Initiated SgNB Modification
Intra-MeNB handover	• 8.2 MeNB-Initiated Intra-MeNB Handover Without an SgNB Change • 8.3 MeNB-Initiated Intra-MeNB Handover with an SgNB Change
SgNB change	8.10 SgNB-Initiated SgNB Change
Inter-MeNB handover	• 8.4 MeNB-Initiated Inter-MeNB Handover Without an SgNB Change • 8.5 MeNB-Initiated Inter-MeNB Handover with an SgNB Change • 8.6 MeNB-Initiated S1-based Inter-MeNB Handover Without an SgNB Change • 8.7 MeNB-Initiated S1-based Inter-MeNB Handover with an SgNB Change • 8.11 MeNB-to-eNodeB Handover
SgNB release	8.12 MeNB/SgNB-Initiated SgNB Release

This version supports data forwarding involved in SgNB modification, change, and release. For details, see chapter 10 "Multi-Connectivity operation related aspects" in 3GPP TS 37.340 V15.5.0.

5 PCell Change in NSA Networking

5.1 Principles

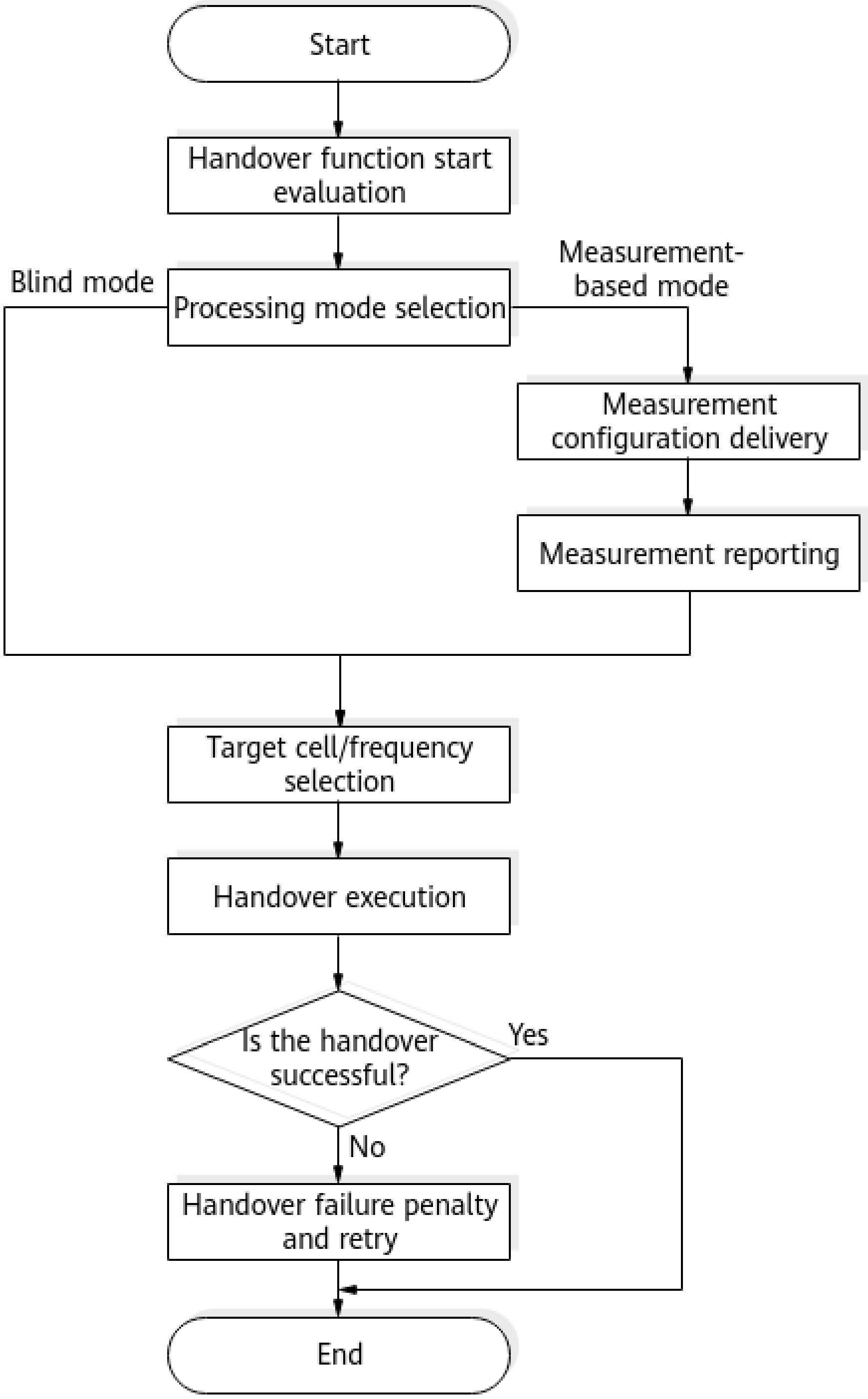
In NSA networking, PCell change refers to the change of the serving cell on the LTE side.

5.1.1 Basic Functions

Basic Procedure

Figure 5-1 shows the basic handover procedure on the LTE side in NSA networking.

Figure 5-1 Basic handover procedure on the LTE side



This procedure is implemented through the following steps:

1. Handover function start evaluation: The eNodeB evaluates the conditions for starting different handover functions.
2. Processing mode selection: The eNodeB selects the blind mode or measurement-based mode, which differ in whether to measure target cells.
3. Measurement configuration delivery: The eNodeB delivers measurement configurations. This step applies only to measurement-based mode.

4. Measurement reporting: The UE sends measurement reports to the eNodeB. This step applies only to measurement-based mode.
5. Target cell or frequency selection: The eNodeB selects a target cell or target frequency after a handover function is initiated. In this step, the eNodeB determines whether there is a suitable new serving cell for the handover. If there is, the eNodeB starts executing the handover.
6. Handover execution: The UE and eNodeB execute the handover based on handover policies.
7. Handover failure penalty and retry: Penalty and retry are triggered if the UE handover to the target cell fails.

Common Mobility Policies

The triggering and decision-making mechanisms of the MeNB-related handover for NSA UEs are the same as those for LTE-only UEs. For details, see *Mobility Management in Connected Mode* in eRAN feature documentation. In addition, thresholds dedicated to NSA UEs can be set for intra-frequency, inter-frequency, and inter-RAT handovers.

Dedicated inter-frequency handover thresholds for different QCIs can be set for NSA UEs in different handover policy groups specified by the **CellQciPara**.***NsaDcInterFreqHoGroupId*** parameter. This parameter is associated with the **InterFreqHoGroup**.***InterFreqHoGroupId*** parameter for which corresponding inter-frequency handover thresholds are set. For example, coverage-based inter-frequency handover thresholds (A2 and A4 thresholds) can be set for NSA UEs in different handover policy groups. Similarly, dedicated intra-frequency and inter-RAT handover thresholds can be configured for NSA UEs by setting handover policy groups different from LTE-only UEs.

Handover Policy Group on the LTE Side	QCI-specific Handover Policy Group ID for NSA UEs
Intra-frequency handover policy group	**CellQciPara**.***NsaDcIntraFreqHoGroupId***
Inter-frequency handover policy group	**CellQciPara**.***NsaDcInterFreqHoGroupId***
Inter-RAT common policy group	**CellQciPara**.***NsaDcInterRatHoCommGroupId***
LTE-to-UTRAN handover policy group	**CellQciPara**.***NsaDcUtranHoGroupId***
LTE-to-GERAN handover policy group	**CellQciPara**.***NsaDcGeranHoGroupId***

The eNodeB can include cell individual offsets (CIOs) independent of those for LTE-only UEs in measurement configurations delivered to NSA UEs. This NSA-UE-dedicated CIO function is controlled by the **CIO_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter. If this option is selected, the function is enabled.

- The CIO of an intra-frequency neighboring cell is specified by the **EutranIntraFreqNCell**.***NsaDcCellIndividualOffset*** parameter.
- The CIO of an inter-frequency neighboring cell is specified by the **EutranInterFreqNCell**.***NsaDcCellIndividualOffset*** parameter.

If this option is deselected, the eNodeB includes CIOs consistent with those for LTE-only UEs in measurement configurations delivered to NSA UEs.

- The CIO of an intra-frequency neighboring cell is specified by the **EutranIntraFreqNCell**.***CellIndividualOffset*** parameter.
- The CIO of an inter-frequency neighboring cell is specified by the **EutranInterFreqNCell**.***CellIndividualOffset*** parameter.

If measurement configurations for inter-frequency/inter-RAT MR/MDT and periodic measurement configurations for multi-carrier unified scheduling need to take effect for NSA UEs, the **MR_MDT_MC_SCH_MEAS_ENH_SW** option of the **EnodebAlgoExtSwitch**.***NsaProtocolCompatSw*** parameter must be selected. (MR stands for measurement report, and MDT stands for minimization of drive tests.)

For NSA UEs and NSA+SA UEs, the maximum number of NR frequencies delivered on the LTE and NR sides for measurement must meet the following conditions:

- Maximum number of NR frequencies delivered on the LTE side = min{**CellUeMeasControlCfg**.***MaxNrMeasFreqNum***, **CellUeMeasControlCfg**.***NsaMaxLteInterFMeasObjNum*** **–** Number of non-NR frequencies}
- The maximum number of NR frequencies delivered on the NR side is as follows:
  - 1 (default value) when **CellUeMeasControlCfg**.***NsaDcMaxInterFMeasObjNum*** is equal to **CellUeMeasControlCfg**.***NsaMaxLteInterFMeasObjNum***
  - **CellUeMeasControlCfg**.***NsaDcMaxInterFMeasObjNum*** – **CellUeMeasControlCfg**.***NsaMaxLteInterFMeasObjNum*** when **CellUeMeasControlCfg**.***NsaDcMaxInterFMeasObjNum*** is different from **CellUeMeasControlCfg**.***NsaMaxLteInterFMeasObjNum***
- The maximum total number of NR frequencies delivered on the LTE and NR sides is equal to **CellUeMeasControlCfg**.***NsaDcMaxInterFMeasObjNum***.

NOTE

In time synchronization scenarios, if the frame offset difference between a cell on an NR frequency to be measured and the serving LTE cell is not 0 and **NRDUCell**.***SmtcDuration*** is set to 2 ms, the gap delivered on the LTE side based on the SMTC configuration transferred from the NR side may not be long enough for measuring the SSBs of the frequency to be measured. As a result, the inter-frequency measurement is incomplete, affecting the handover success rate. Therefore, it is recommended that **NRDUCell**.***SmtcDuration*** be set to 3 ms or a larger value.

5.1.2 PCell Change Policies

The principles and configurations for the PCell change mechanisms in different scenarios are similar to those for LTE handover functions. Therefore, this section will only describe the principles and configurations that are unique in NSA networking. For details about the LTE handover functions, see *Mobility Management in Connected Mode* in eRAN feature documentation.

Coverage-based Inter-Frequency Handover

Coverage-based inter-frequency handover involves NR+LTE-coverage-based LTE inter-frequency handover for NSA DC and LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC.

- NR+LTE-coverage-based LTE inter-frequency handover for NSA DC is enabled when the **NsaDcUeCovBasedHoOptSw** option of the **CellHoParaCfg**.***CellHoAlgoSwitch*** parameter is selected. When an NSA UE is in an area without NR coverage, it is considered as an LTE-only UE, and the inter-frequency handover threshold for LTE-only UEs is used. When an NSA UE is in an area with NR coverage, the inter-frequency handover threshold for NSA UEs, which equals the inter-frequency handover threshold associated with the **CellQciPara**.***NsaDcInterFreqHoGroupId*** parameter, is used.
- LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC is enabled when the **NSA_DC_COV_HO_ENH_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected. When delivering measurement frequencies for inter-frequency handovers, the eNodeB selects the frequencies in descending order of the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter values. A larger value indicates a higher priority. The frequencies with non-zero NSA PCC anchoring priorities are selected preferentially.

If both NR+LTE-coverage-based LTE inter-frequency handover for NSA DC and LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC take effect, the eNodeB delivers measurement configurations to NSA UEs based on NR coverage. When an NSA UE is in an area without NR coverage, it is considered as an LTE-only UE, and the eNodeB selects measurement frequencies based on the frequency priorities for LTE-only UEs. When an NSA UE is in an area with NR coverage, the eNodeB selects measurement frequencies based on the non-zero NSA PCC anchoring priorities specified by the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter.

For NSA UEs, the serving cell's RSRP offset related to LTE-coverage-based LTE inter-frequency handover for NSA DC can be specified through the **EutranInterNFreq**.***NsaCovIfHoServCellRsrpOfs*** parameter. If this parameter is not set to **0**, the eNodeB triggers coverage-based inter-frequency handover only if any of the following conditions is met:

- The NSA DC UE reports an inter-frequency event A3, and the serving cell's RSRP is less than the sum of **InterFreqHoGroup**.***A3InterFreqHoA2ThdRsrp*** and **EutranInterNFreq**.***NsaCovIfHoServCellRsrpOfs***.
- The NSA DC UE reports an inter-frequency event A4, and the serving cell's RSRP is less than the sum of **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** and **EutranInterNFreq**.***NsaCovIfHoServCellRsrpOfs***.
- The NSA DC UE reports an inter-frequency event A5, and the serving cell's RSRP is less than the sum of **InterFreqHoGroup**.***InterFreqHoA5Thd1Rsrp*** and **EutranInterNFreq**.***NsaCovIfHoServCellRsrpOfs***.

If both NR+LTE-coverage-based LTE inter-frequency handover for NSA DC and the serving cell's RSRP offset related to LTE-coverage-based LTE inter-frequency handover for NSA DC take effect, the eNodeB selects target cells for NSA UEs based on NR coverage. When NSA UEs are in areas without NR coverage, the eNodeB does not consider the offset specified by the

EutranInterNFreq.***NsaCovIfHoServCellRsrpOfs*** parameter during target cell selection. When NSA UEs are in areas with NR coverage, the eNodeB considers this offset during target cell selection.

For a UE in the EN-DC state, if the UE supports ECGI measurement in the LTE-only state but not in the EN-DC state, the eNodeB is required to obtain the ECGIs of neighboring cells during a coverage-based intra-RAT handover, and the **IntraRatHoComm**.***EcgiMeasScgReaddWaitTmr*** parameter is set to a non-zero value, then the eNodeB removes the SCG of the UE, starts the timer indicated by the preceding parameter, and then instructs the UE to perform ECGI measurement. If the UE is not handed over finally, the eNodeB checks whether the UE meets SCG addition conditions only after the timer expires. If the UE meets the conditions, the eNodeB adds an SCG for the UE.

Frequency-Priority-based Inter-Frequency Handover

If the frequency priorities configured for frequency-priority-based inter-frequency handover differ from the NSA PCC anchoring priorities, ping-pong handovers may occur. Specifically, an NSA UE may be handed over from a frequency with low NSA PCC anchoring priority to one with high PCC anchoring priority based on NSA PCC anchoring, and then handed over to a frequency with high frequency priority but low NSA PCC anchoring priority based on frequency-priority-based inter-frequency handover. To prevent such ping-pong handovers, you can deselect the **FREQ_PRI_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter so that frequency-priority-based inter-frequency handover does not take effect for NSA UEs. For details about frequency-priority-based inter-frequency handover, see *Mobility Management in Connected Mode* in eRAN feature documentation.

Service-based Inter-Frequency Handover

When service-based inter-frequency handover is enabled on the LTE side but it is not required for NSA UEs with a specified QCI, deselect the **SERV_BASED_INTER_FREQ_HO_SW** option of the **CellQciPara**.***NsaDcOptSwitch*** parameter. Then, service-based inter-frequency handover will not be triggered for UEs with the specified QCI. This prevents, for example, NSA UEs from being handed over from anchor frequencies to non-anchor frequencies through service-based inter-frequency handover. If service-based inter-frequency handover is required for a specified QCI (for example, VoLTE services need to be carried only on specific frequencies), select this option.

For NSA UEs that are handed over from anchor frequencies to non-anchor frequencies, "PCC anchoring for NSA UEs when bearers of a specified QCI are released" can be enabled so that such UEs can return to anchor frequencies for NSA DC services after the bearers of the QCI are released. After this function is enabled, NSA DC PCC anchoring can be triggered when the bearers of the specified QCI are released. This function is controlled by the **QCI_REL_BASED_PCC_ANCHORING_SW** option of the **CellQciPara**.***NsaDcOptSwitch*** parameter.

For details about service-based inter-frequency handover, see *Mobility Management in Connected Mode* in eRAN feature documentation.

5.2 Network Analysis

5.2.1 Benefits

PCell change in NSA networking allows NSA UEs in connected mode to reselect cells with better signal quality for data transmission, ensuring uninterrupted service continuity for moving UEs.

For details about the benefits of different PCell change policies, see the corresponding "Benefits" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

For details about the impacts of different PCell change policies, see the corresponding "Impacts" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.

5.3 Requirements

5.3.1 Licenses

For details about the license requirements for NSA networking based on EPC, see the "Licenses" section in *EPC-based NSA Basics*.

For details about the license requirements for different PCell change policies, see the corresponding"Licenses" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	NSA networking based on EPC	LTE side: **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter NR side: **NRCellAlgoSwitch**.***NsaDcSwitch*** set to **ON**	*EPC-based NSA Basics*	PCell change in NSA networking requires NSA networking based on EPC to be enabled.
LTE FDD LTE TDD	NSA PCC anchoring based on NR coverage	**NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** set to **BASED_ON_NR_COVERAGE**	*EPC-based NSA Performance Enhancement*	NR+LTE-coverage-based LTE inter-frequency handover for NSA DC takes effect only when NSA PCC anchoring based on NR coverage is enabled.
For details about the prerequisite functions of different PCell change policies, see the corresponding "Prerequisite Functions" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.				

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	MRO	• **InterFreqMroSwitch** option of the **CellAlgoSwitch**.***MroSwitch*** parameter • **IntraFreqMroSwitch** option of the **CellAlgoSwitch**.***MroSwitch*** parameter • **UEMroSwitch** option of the **CellAlgoSwitch**.***MroSwitch*** parameter	*MRO* in eRAN feature documentation	The NSA-UE-dedicated CIO function is mutually exclusive with MRO.
LTE FDD LTE TDD	eICIC	**CellAlgoSwitch**.***EicicSwitch***	*TDM eICIC (FDD)* in eRAN feature documentation	The NSA-UE-dedicated CIO function is mutually exclusive with eICIC.
For details about the mutually exclusive functions of different PCell change policies, see the corresponding "Mutually Exclusive Functions" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.				

5.3.2.3 Function Impacts

For details about the function impacts of different PCell change policies, see the corresponding "Function Impacts" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For details, see "Base Station Models" under the "Hardware" section in *EPC-based NSA Basics*.

Boards

For details, see "Boards" under the "Hardware" section in *EPC-based NSA Basics*.

RF Modules

This function does not depend on RF modules.

5.3.4 Others

For other requirements, see the corresponding "Others" sections in *Mobility Management in Connected Mode* in eRAN feature documentation.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

For details about data preparation for neighbor relationships and mobility-related parameters on the LTE side, see the "Data Preparation" section under the "Basic Functions of Mobility Management in Connected Mode" chapter in *Mobility Management in Connected Mode* in eRAN feature documentation.

The following table describes the parameters to be set on the LTE side to configure the handover policy group for NSA UEs.

Parameter Name	Parameter ID	Setting Notes
NSA DC Intra-Frequency HO Group ID	**CellQciPara**.***NsaDcIntraFreqHoGroupId***	Set this parameter based on the network plan.
NSA DC Inter-Frequency HO Group ID	**CellQciPara**.***NsaDcInterFreqHoGroupId***	Set this parameter based on the network plan.
NSA DC Inter-RAT HO Common Group ID	**CellQciPara**.***NsaDcInterRatHoCommGroupId***	Set this parameter based on the network plan.
NSA DC UTRAN HO Group ID	**CellQciPara**.***NsaDcUtranHoGroupId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
NSA DC GERAN HO Group ID	**CellQciPara**.***NsaDcGeranHoGroupId***	Set this parameter based on the network plan.

The following table describes the parameters to be set on the LTE side to configure the NSA-UE-dedicated CIO function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	**NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch***	To enable the NSA-UE-dedicated CIO function, select the **CIO_SW** option.
NSA DC UE Cell Individual offset	**EutranIntraFreqNCell**.***NsaDcCellIndividualOffset***	Set this parameter based on the network plan.
NSA DC UE Cell Individual offset	**EutranInterFreqNCell**.***NsaDcCellIndividualOffset***	Set this parameter based on the network plan.
Cell individual offset	**EutranIntraFreqNCell**.***CellIndividualOffset***	Set this parameter based on the network plan.
Cell individual offset	**EutranInterFreqNCell**.***CellIndividualOffset***	Set this parameter based on the network plan.

The following table describes the parameter to be set on the LTE side to configure "inter-frequency/inter-RAT MR/MDT measurement and periodic measurement for multi-carrier unified scheduling".

Parameter Name	Parameter ID	Setting Notes
NSA Protocol Compatibility Switch	**EnodebAlgoExtSwitch**.***NsaProtocolCompatSw***	To enable "inter-frequency/inter-RAT MR/MDT measurement and periodic measurement for multi-carrier unified scheduling", select the **MR_MDT_MC_SCH_MEAS_ENH_SW** option.

The following table describes the optimization parameters to be set on the LTE side.

Parameter Name	Parameter ID	Setting Notes
Max NR Measure Freq Number	**CellUeMeasControlCfg**.*MaxNrMeasFreqNum*	Set this parameter based on the network plan.
NSA Max LTE Inter-Freq Measurement Object Num	**CellUeMeasControlCfg**.*NsaMaxLteInterFMeasObjNum*	Set this parameter based on the network plan.
NSA DC Max Inter-Freq Measurement Object Num	**CellUeMeasControlCfg**.*NsaDcMaxInterFMeasObjNum*	Set this parameter based on the network plan.

The following table describes the parameter to be set on the NR side to configure the SMTC duration.

Parameter Name	Parameter ID	Setting Notes
SMTC Duration	**NRDUCell**.*SmtcDuration*	Set this parameter based on the network plan.

The following table describes the parameters to be set on the LTE side to configure coverage-based inter-frequency handover.

Parameter Name	Parameter ID	Setting Notes
Handover Algo Switch	**CellHoParaCfg**.*CellHoAlgoSwitch*	To enable NR+LTE-coverage-based LTE inter-frequency handover for NSA DC, select the **NsaDcUeCovBased-HoOptSw** option.
NSA DC Algorithm Switch	**NsaDcMgmtConfig**.*NsaDcAlgoSwitch*	To enable LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC, select the **NSA_DC_COV_HO_ENH_SW** option.
NSA PCC Anchoring Priority	**PccFreqCfg**.*NsaPccAnchoringPriority*	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Cov-based IF HO Serving Cell RSRP Ofs for NSA DC	**EutranInterNFreq**.*NsaCovIfHoServCellRsrpOfs*	Set this parameter based on the network plan.
ECGI Meas SCG Readdition Waiting Timer	**IntraRatHoComm**.*EcgiMeasScgReaddWaitTmr*	Set this parameter based on the network plan.

The following table describes the parameter to be set on the LTE side to configure frequency-priority-based inter-frequency handover.

Parameter Name	Parameter ID	Setting Notes
NSA DC UE LTE Function Activation Switch	**NsaDcMgmtConfig**.*NsaDcUeLteFunActivationSw*	To enable frequency-priority-based inter-frequency handover for NSA UEs, select the **FREQ_PRI_HO_SW** option.

The following table describes the parameters to be set on the LTE side to configure service-based inter-frequency handover.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	**CellQciPara**.*NsaDcOptSwitch*	To enable service-based inter-frequency handover for NSA UEs, select the **SERV_BASED_INTER_FREQ_HO_SW** option.
NSA DC Optimization Switch	**CellQciPara**.*NsaDcOptSwitch*	To enable PCC anchoring for NSA UEs when bearers of a specified QCI are released, select the **QCI_REL_BASED_PCC_ANCHORING_SW** option.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, configure neighbor relationships and mobility-related parameters on the LTE side. For details, see the "Using MML Commands" section under the "Basic Functions of Mobility Management in Connected Mode" chapter in *Mobility Management in Connected Mode* in eRAN feature documentation.

Activation Command Examples

On the eNodeB side

```
//Configuring handover policy groups for NSA UEs
MOD CELLQCIPARA: LocalCellId=21, Qci=9, NsaDcGeranHoGroupId=0, NsaDcInterFreqHoGroupId=0, NsaDcInterRatHoCommGroupId=0, NsaDcIntraFreqHoGroupId=0, NsaDcUtranHoGroupId=0;
//(Optional) Enabling NSA-UE-dedicated CIO
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=CIO_SW-1;
//(Optional) Configuring the cell individual offset for NSA-UE-dedicated CIO (with MCC=460 and MNC=23 as an example)
MOD EUTRANINTRAFREQNCELL: LocalCellId=21, Mcc="460", Mnc="23", eNodeBId=0, CellId=0, CellIndividualOffset=dB0, NsaDcCellIndividualOffset=dB0;
MOD EUTRANINTERFREQNCELL: LocalCellId=21, Mcc="460", Mnc="23", eNodeBId=0, CellId=0, CellIndividualOffset=dB0, NsaDcCellIndividualOffset=dB0;
//(Optional) Configuring "inter-frequency/inter-RAT MR/MDT measurement and periodic measurement for multi-carrier unified scheduling"
MOD ENODEBALGOEXTSWITCH: NsaProtocolCompatSw=MR_MDT_MC_SCH_MEAS_ENH_SW-1;
//(Optional) Configuring the maximum number of NR frequencies to be delivered in measurement configurations
MOD CELLUEMEASCONTROLCFG: LocalCellId=21, MaxNrMeasFreqNum=4, NsaDcMaxInterFMeasObjNum=13, NsaMaxLteInterFMeasObjNum=7;
//(Optional) Enabling NR+LTE-coverage-based LTE inter-frequency handover for NSA DC
MOD CELLHOPARACFG: LocalCellId=21, CellHoAlgoSwitch=NsaDcUeCovBasedHoOptSw-1;
//(Optional) Enabling LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_COV_HO_ENH_SW-1;
//(Optional) Configuring the NSA PCC anchoring priority
MOD PCCFREQCFG: PccDlEarfcn=1300, NsaPccAnchoringPriority=2;
//(Optional) Configuring the serving cell's RSRP offset related to LTE-coverage-based LTE inter-frequency handover for NSA DC
MOD EUTRANINTERNFREQ: LocalCellId=21, DlEarfcn=1300, NsaCovIfHoServCellRsrpOfs=0;
//(Optional) Configuring EcgiMeasScgReaddWaitTmr
MOD INTRARATHOCOMM: EcgiMeasScgReaddWaitTmr=3;
//(Optional) Enabling frequency-priority-based inter-frequency handover
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcUeLteFunActivationSw=FREQ_PRI_HO_SW-1;
//(Optional) Enabling service-based inter-frequency handover
MOD CELLQCIPARA: LocalCellId=21, Qci=9, NsaDcOptSwitch=SERV_BASED_INTER_FREQ_HO_SW-1;
//(Optional) Enabling PCC anchoring for NSA UEs when bearers of a specified QCI are released
MOD CELLQCIPARA: LocalCellId=21, Qci=1, NsaDcOptSwitch=QCI_REL_BASED_PCC_ANCHORING_SW-1;
```

On the gNodeB side

```
//(Optional) Configuring the SMTC duration
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, SmtcDuration=MS2;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```
//Disabling NSA-UE-dedicated CIO
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=CIO_SW-0;
//Disabling "inter-frequency/inter-RAT MR/MDT measurement and periodic measurement for multi-carrier unified scheduling"
MOD ENODEBALGOEXTSWITCH: NsaProtocolCompatSw=MR_MDT_MC_SCH_MEAS_ENH_SW-0;
//Disabling NR+LTE-coverage-based LTE inter-frequency handover for NSA DC
MOD CELLHOPARACFG: LocalCellId=21, CellHoAlgoSwitch=NsaDcUeCovBasedHoOptSw-0;
//Disabling LTE-coverage-based LTE inter-frequency handover enhancement for NSA DC
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_COV_HO_ENH_SW-0;
//Disabling frequency-priority-based inter-frequency handover
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcUeLteFunActivationSw=FREQ_PRI_HO_SW-0;
//Disabling service-based inter-frequency handover
MOD CELLQCIPARA: LocalCellId=21, Qci=9, NsaDcOptSwitch=SERV_BASED_INTER_FREQ_HO_SW-0;
//Disabling PCC anchoring for NSA UEs when bearers of a specified QCI are released
MOD CELLQCIPARA: LocalCellId=21, Qci=1, NsaDcOptSwitch=QCI_REL_BASED_PCC_ANCHORING_SW-0;
```

5.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the following counters to obtain the number of handovers triggered for NSA UEs in a cell.

Counter ID	Counter Name	NE
1526755743	**L.NsaDc.HHO.PrepAttOut**	eNodeB
1526755744	**L.NsaDc.HHO.ExecAttOut**	eNodeB
1526755745	**L.NsaDc.HHO.ExecSuccOut**	eNodeB
1526759024	**L.NsaDc.HHO.IntraeNB.ExecSuccOut**	eNodeB
1526759025	**L.NsaDc.HHO.IntraeNB.ExecAttOut**	eNodeB
1526759026	**L.NsaDc.HHO.IntraeNB.PrepAttOut**	eNodeB

5.4.3 Network Monitoring

Monitor the counters listed in the following table and compare them against the network plan to evaluate the feature performance.

- On the eNodeB side

  After NSA DC is enabled, observe the following counters to determine whether the handover success rate and service drop rate of NSA UEs are different from those of LTE-only UEs:

  – Outgoing handover success rate of NSA UEs = **L.NsaDc.HHO.ExecSuccOut** / **L.NsaDc.HHO.ExecAttOut** x 100%
  – Incoming handover success rate of NSA UEs = **L.NsaDc.PCell.Change.Succ** / **L.NsaDc.PCell.Change.Exec** x 100%
  – Abnormal service drop rate of NSA UEs = **L.NsaDc.E-RAB.AbnormRel** / **L.NsaDc.E-RAB.NormRel** x 100%
- On the gNodeB side
  – Number of inter-site SgNB addition attempts in the LTE-NR NSA DC scenario (**N.NsaDc.InterSite.SgNB.Add.Att**)
  – Number of successful inter-site SgNB additions in the LTE-NR NSA DC scenario (**N.NsaDc.InterSite.SgNB.Add.Succ**)

6 PSCell Change in NSA Networking

In NSA networking, the gNodeB sends NR measurement configurations to the eNodeB through the X2 interface and the eNodeB then delivers these configurations to UEs.

NOTE

According to 3GPP specifications, EN-DC-capable UEs must support all the following types of inter-frequency handovers in NSA networking: intra-FR1 FDD inter-frequency handover, intra-FR1 TDD inter-frequency handover, intra-FR2 TDD inter-frequency handover, FDD-TDD inter-frequency handover, and FR1-FR2 inter-frequency handover. For details, see section 4.2.9 "MeasAndMobParameters" in 3GPP TS 38.306 V15.9.0. As such, the gNodeB considers by default that EN-DC-capable UEs support all the preceding types of inter-frequency handovers, regardless of the *UECapabilityInformation* IE reported by UEs over the Uu interface.

In NSA networking, measurement gap configuration varies with UE capabilities.

The per UE or per FR1 measurement gap is calculated on the LTE side based on the NR cell's SMTC information carried in the messages sent over the X2 interface. Therefore, SMTC parameters must be set to the same values for intra-frequency cells on the NR side to ensure that the NR cells can be measured. For details, see *EPC-based NSA Basics*.

The per FR2 measurement gap is calculated on the NR side and forwarded to the eNodeB over the X2 interface. Then, the eNodeB delivers it to UEs.

For details about UE measurement gap capabilities, see section 9 "Measurement Procedure" in 3GPP TS 38.133 V15.5.0.

Once the NSA DC SRB3 function is enabled in NSA networking, a UE that supports SRB3 can directly send measurement reports to the gNodeB on SRB3. This function is controlled by the **NSA_DC_SRB3_SWITCH** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter. For details about the NSA DC SRB3 function, see *EPC-based NSA Basics*.

The PSCell modification success rate and PSCell change success rate may decrease due to conflicts between the LTE RRC connection reestablishment procedure and the gNodeB-triggered SCG modification or SCG change procedure, following the initiation of reestablishment on the LTE side. To reduce such procedure conflicts, enabling the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs" is recommended. This function is controlled by the **NSA_REEST_FLOW_CONFLICT_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** parameter. For details about this function, see *EPC-based NSA Basics*.

6.1 Principles

In NSA networking, PSCell change refers to the change of the serving cell on the NR side.

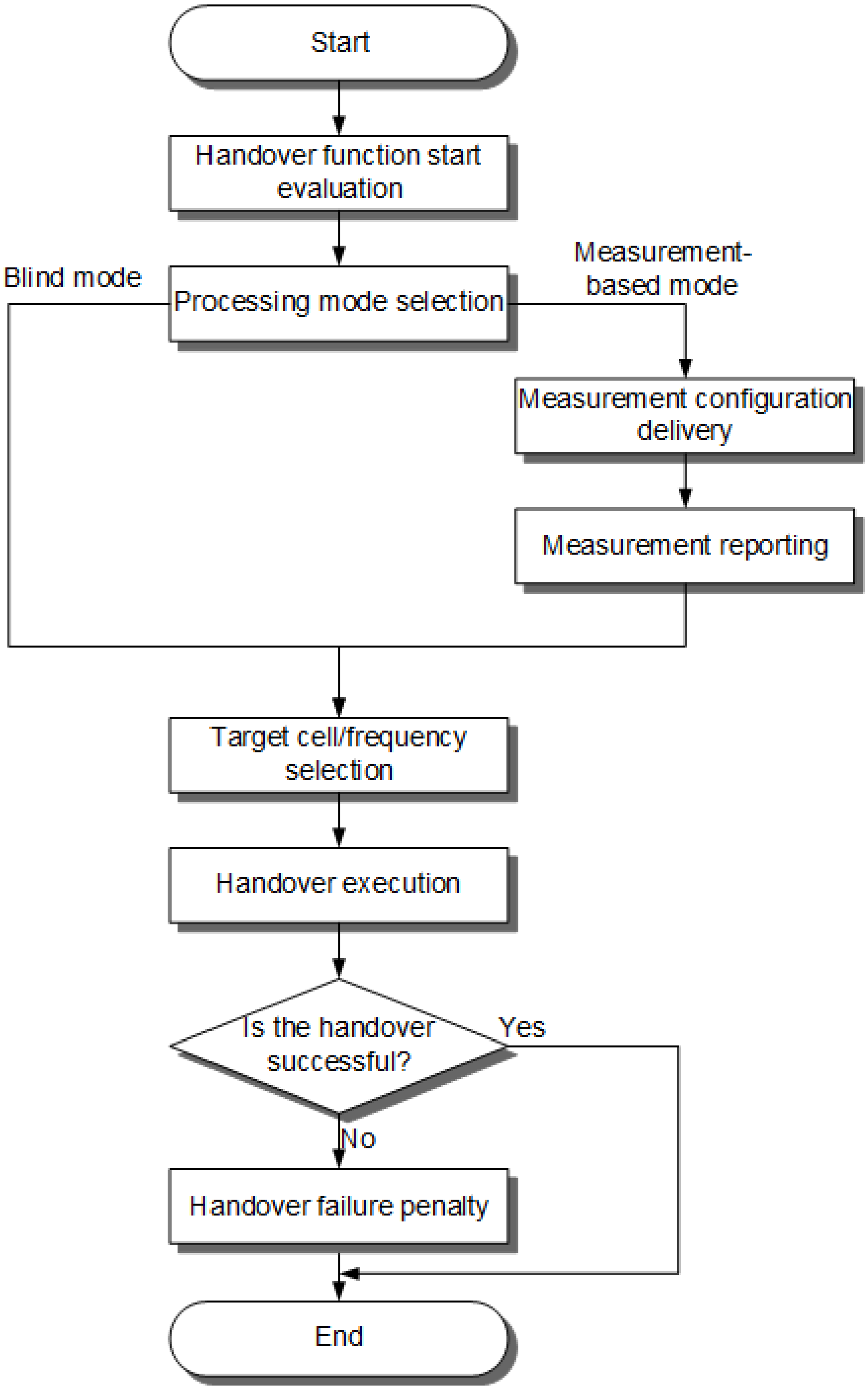
6.1.1 Basic Functions

This section describes the principles of PSCell change in NSA networking. The network analysis, requirements, and operation and maintenance of related functions are the same as those of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

6.1.1.1 Basic Procedure

Figure 6-1 shows the basic handover procedure on the NR side in NSA networking.

Figure 6-1 Basic handover procedure



1. **6.1.1.2 Handover Function Start Evaluation** describes the conditions for starting handover functions.
2. **6.1.1.3 Processing Mode Selection** describes processing mode selection (blind mode or measurement-based mode).
3. **6.1.1.4 Measurement Configuration Delivery** describes the method and contents of measurement configurations sent by the gNodeB to UEs.
4. **6.1.1.5 Measurement Reporting** describes the requirements for UEs to send measurement reports to the gNodeB.

5. **6.1.1.6 Target Cell or Frequency Evaluation** describes how the gNodeB selects a handover policy and generates target cells or frequencies for a handover.
6. **6.1.1.7 Handover Execution** describes how the gNodeB and UEs execute the handover based on the handover policy.
7. **6.1.1.8 Handover Failure Penalty** describes the penalty mechanism for UE handover failures.

6.1.1.2 Handover Function Start Evaluation

Handover function start evaluation is a process of determining whether to start a handover function based on its starting conditions. The process depends on whether:

- The switch of the handover function is turned on.
- The signal quality of the serving cell meets certain conditions.

6.1.1.3 Processing Mode Selection

After a handover function is started, the gNodeB selects a processing mode based on the signal quality of the serving cell and related events.

Depending on whether UEs measure the signal quality of candidate target cells before a handover, the processing mode is classified into measurement-based mode and blind mode.

- Measurement-based mode: UEs measure the signal quality of candidate target cells and report the results to the gNodeB based on the measurement configurations delivered from the gNodeB. The gNodeB then uses the measurement report to generate a target cell list.
- Blind mode: The gNodeB generates a target cell or frequency list based on priority parameter configurations without instructing UEs to measure the signal quality of candidate target cells.

NOTE

In blind mode, accessing a target cell has a high risk of failure due to the uncertain signal quality of the target cell. As such, blind mode is not recommended in most cases. This mode is only used when a quick handover is needed.

Blind mode is widely used in high-frequency scenarios. In these scenarios, mmWave signals attenuate quickly during transmission and provide a limited coverage range due to high frequencies and short wavelengths. When a UE moves at a high speed (for example, in high-speed railway scenarios), the UE may have moved out of the source cell before the measurement of the target cell or frequency is completed in measurement-based mode. Consequently, the service drop rate increases. To prevent this, blind mode can be used, in which case a handover is quickly implemented for the UE. However, blind mode has high requirements on neighboring cell planning. In the case of improper neighboring cell planning, the UE may experience an access failure.

6.1.1.4 Measurement Configuration Delivery

Measurement configuration delivery refers to the process by which the gNodeB sends measurement configurations to UEs.

The gNodeB delivers measurement configurations as follows:

- When a UE enters the RRC_CONNECTED state, the gNodeB includes the measurement configurations in an RRCReconfiguration message and sends it to the UE.
- After a UE enters the RRC_CONNECTED state or completes a handover, if the measurement configurations are updated, the gNodeB includes the updated configurations in an RRCReconfiguration message and sends it to the UE. This section describes measurement configuration delivery in this scenario.

Measurement configurations are configured by the gNodeB and then delivered to UEs. **Table 6-1** describes the information included in measurement configurations.

Table 6-1 Information included in measurement configurations

Measurement Configuration Information	Content	Function	Reference
Measurement object	It consists of attributes such as the RATs and frequencies (or cells) to be measured.	It specifies the frequencies or cells where a UE measures signal quality. The cells to be measured can be the serving cell or neighboring cells, for which the measurement procedures are the same. This section uses neighboring cell measurement as an example to describe measurement configuration delivery.	**6.1.1.4.1 Measurement Object**
Reporting configuration	It consists of attributes such as measurement event information and triggering quantities of the measurement reports.	It specifies the conditions and criteria for a UE to report measurement results.	**6.1.1.4.2 Reporting Configuration**
Measurement ID configuration	A measurement ID links a measurement object with a reporting configuration as a set.	Given a measurement ID, a UE measures the associated measurement object according to the reporting configuration.	**6.1.1.4.3 Measurement ID Configuration**
SSB-based measurement timing configuration (SMTC)	It consists of the SMTC period, SMTC duration, and SMTC offset.	It indicates the time window for a UE to perform measurements.	**6.1.1.4.4 SMTC Measurement Configuration**

Measurement Configuration Information	Content	Function	Reference
Measurement gap configuration	It consists of the measurement gap duration and measurement gap repetition period.	It indicates a period of time during which a UE tunes its receiver towards a target frequency to perform inter-frequency measurements.	**6.1.1.4.5 Measurement Gap Configuration**
Other configurations	They include optimized configuration for deriveSSB-IndexFromCell, configuration of SSB beam measurement result selection and combination, and measurement filtering configuration.	They specify auxiliary information used for a UE to perform measurements and process measurement results.	**6.1.1.4.6 Other Configurations**

6.1.1.4.1 Measurement Object

The attributes of measurement objects include RATs and frequencies (or cells) to be measured, and key attributes are different in different RATs. Inter-RAT measurement objects are described in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The key attributes of intra-RAT measurement objects include frequencies (or cells) to be measured.

The gNodeB determines an RAT for measurement first and then selects the frequencies from the cell-specific frequency relationships (or cells from the NRT) of this RAT.

RAT Selection for Measurement

In NSA networking, measurement events A1 to A6 are used, and the gNodeB selects NR as the RAT for measurement based on the measurement events.

For details about the events, see **Measurement Event**.

Frequency Selection for Measurement

5G UE handovers are based on SSB measurements performed by UEs. Measurement is classified into intra-frequency measurement and inter-frequency measurement, in terms of the frequencies to be measured.

- Intra-frequency measurement refers to measuring cells that have the same SSB frequency and SSB frequency subcarrier spacing (SCS) as the serving cell, regardless of whether their bandwidths are the same or not.
- Inter-frequency measurement refers to measuring cells that have different SSB frequencies from the serving cell, or those that have the same SSB frequency as the serving cell but different SSB frequency SCS.

The gNodeB obtains the information about the SSB frequencies to be measured from the cell configurations or cell-specific frequency relationships of the selected RAT.

- For intra-frequency measurement, the gNodeB sends the measurement configurations including the NR frequencies collectively determined by the **NRDUCell.*SsbDescMethod*** and **NRDUCell.*SsbFreqPos*** parameters to UEs.
- For inter-frequency measurement:

  The rules for selecting frequencies to be delivered vary depending on whether the RFSP-based inter-frequency measurement policy is configured. (RFSP is short for RAT/Frequency Selection Priority.)

  - Common policy for inter-frequency measurement: If no RFSP-based inter-frequency measurement policy is configured, the gNodeB selects the frequencies to be delivered for measurement based on the following common policy:

    For all UEs, the gNodeB selects the frequencies to be delivered for measurement based on the following factors, as described in **Table 6-2**.

    - Whether the operator-specific policy of an NR cell (**NRCellOpPolicy** MO) is configured.
    - Validity of the gNodeB frequency priority group ID (**NRCellOpPolicy**.***gNBFreqPriorityGroupId***) configured in the operator-specific policy of an NR cell: The value in the range of **0** to **254** indicates that the ID is valid. The value **255** indicates that the ID is invalid. The ID is mapped to the **gNBFreqPriorityGroup**.***gNBFreqPriorityGroupId*** parameter.
    - Whether the operator-specific neighboring frequency configuration function is enabled (**OPERATOR_NR_FREQ_CFG_SW** option of the **NRCellOpPolicy**.***FreqConfigPolicySwitch*** parameter).

Table 6-2 Common policy for inter-frequency measurement

<table>
<tr><th>Whether or Not the NRCellOpPolicy MO Is Configured</th><th>Whether or Not the NRCellOpPolicy.gNBFreqPriorityGroupId Parameter Value Is Valid</th><th>Operator-specific Neighboring Frequency Configuration</th><th>Frequency Delivery Policy</th></tr>
<tr><td>Yes</td><td>Valid</td><td>Enabled</td><td>The gNodeB delivers NR frequencies for measurement in the intersection of the following NR frequencies:
● NR frequencies determined by gNBFreqPriorityGroup.SsbDescMethod and gNBFreqPriorityGroup.SsbFreqPos
● NR frequencies determined by NRCellFreqRelation.SsbDescMethod and NRCellFreqRelation.SsbFreqPos</td></tr>
<tr><td>Yes</td><td>Valid</td><td>Disabled</td><td rowspan="3">The gNodeB delivers NR frequencies for measurement determined by NRCellFreqRelation.SsbDescMethod and NRCellFreqRelation.SsbFreqPos.</td></tr>
<tr><td>Yes</td><td>Invalid</td><td>Enabled or Disabled</td></tr>
<tr><td>No</td><td>Valid or Invalid</td><td>Enabled or Disabled</td></tr>
<tr><td colspan="4">Note: For unnecessary inter-frequency handovers except coverage-based inter-frequency handovers, the gNodeB also selects frequencies based on the frequency interference flag when delivering measurement configurations. If the NRCellFreqRelation.FreqIntrfFlag parameter is set to NO_INTRF for a frequency, the frequency can be selected as a to-be-measured frequency. Otherwise, the frequency cannot be selected as a to-be-measured frequency.</td></tr>
</table>

NOTE

For details about the **NRCellOpPolicy** MO and its configuration, see *Multi-Operator Sharing* in 5G RAN feature documentation.

When configuring the **gNBFreqPriorityGroup** MO associated with the serving PLMN of a UE, ensure that all neighboring frequencies of this operator are configured. Those unconfigured neighboring frequencies will not be delivered to the UE and the UE will not be handed over to cells working on them.

- RFSP-based inter-frequency measurement policy: If the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch**.***RfspAlgoSwitch*** parameter is selected, the frequencies to be delivered by the gNodeB for measurement are selected based on the RFSP-based inter-frequency measurement policy. For details about RFSP-based inter-frequency handover, see *Flexible User Steering* in 5G RAN feature documentation.

The gNodeB filters frequencies to be delivered as follows for all UEs regardless of which inter-frequency measurement policy is used:

- The gNodeB filters out neighboring frequencies not supported by the UE.
- The gNodeB filters out neighboring frequencies for which the **MOB_NEIGHBOR_FREQ_FILTER_SW** option of the **NRCellFreqRelation**.***FrequencyCtrlSwitch*** parameter is selected.

In NSA networking, the maximum number of non-serving NR frequencies to be delivered is indicated by the *maxMeasFreqsSCG* IE in an SgNB Addition Request message, which is sent by the eNodeB to the gNodeB over the X2 interface. If the IE does not indicate the maximum number, a maximum of seven non-serving NR frequencies can be delivered.

Cell Selection for Measurement

The cell individual offset (CIO) of a cell is specified by the **NRCellRelation.*CellIndividualOffset*** parameter. For details about how this parameter is used in measurement events, see the entering and leaving conditions of each measurement event in **Measurement Event**. The gNodeB determines whether to deliver the CIO information of a cell based on the setting of this parameter.

- If this parameter is set to a value other than **0**, the CIO information of the cell is included in the measurement configuration to be delivered for the measurement object.

  The CIO specification compatibility function controls whether to increase the maximum number of neighboring cells that can be delivered for each measurement object. This function is controlled by the **CIO_SPEC_COMPAT_SW** option of the **gNBMobilityCommParam**.***ProtocolCompatibilitySw*** parameter. When this function is enabled, the maximum number of neighboring cells that can be delivered for each measurement object is listed in **Table 6-3**. When this function is disabled, the maximum number of neighboring cells that can be delivered for each measurement object is listed in **Table 6-4**.

Table 6-3 Maximum number of neighboring cells that can be delivered for each measurement object when the CIO specification compatibility function is enabled

Neighboring Cell Type	Maximum Number of Neighboring Cells	IE
Intra-RAT intra-frequency neighboring cell	32	maxNrofCellMeas
Intra-RAT inter-frequency neighboring cell	32	maxNrofCellMeas
Neighboring E-UTRAN cell	32	maxCellMeasEUTRA

Table 6-4 Maximum number of neighboring cells that can be delivered for each measurement object when the CIO specification compatibility function is disabled

Neighboring Cell Type	Maximum Number of Neighboring Cells	IE
Intra-RAT intra-frequency neighboring cell	8	maxNrofCellMeas
Intra-RAT inter-frequency neighboring cell	8	maxNrofCellMeas
Neighboring E-UTRAN cell	32	maxCellMeasEUTRA

NOTE

If the number of cells that meet the requirements exceeds the delivered maximum allowed number, the gNodeB preferentially selects the cells with larger CIO values until the selected cells reach the maximum. For cells with the same CIO value, the cells with larger PCI values are preferentially selected.

For details about the maximum number of neighboring cells that can be delivered for each measurement object, see section 6.4 "RRC multiplicity and type constraint values" in 3GPP TS 38.331 V16.5.0.

- If this parameter is set to **0**, the CIO information of the cell is not included in the measurement configuration to be delivered for the measurement object.

6.1.1.4.2 Reporting Configuration

Reporting configurations include:

- Measurement event, which can be A1, A2, A3, A4, A5, A6, B1, or B2. Handover functions determine the measurement events to be used. For details, see the sections corresponding to each handover function. For details about the definition, entering conditions, and leaving conditions for each event, see **Measurement Event**.
- Triggering quantity, which specifies the policy used to trigger event reporting and can be reference signal received power (RSRP), reference signal received quality (RSRQ), or signal to interference plus noise ratio (SINR). This version supports SSB RSRP or RSRQ. For details, see **Triggering Quantity**.

Measurement Event

An event indicates the signal quality status of cells. **Table 6-5** lists the definition of each event.

Table 6-5 Event definition

Event Type	Event Definition
A1	The signal quality of the serving cell exceeds a specific threshold.
A2	The signal quality of the serving cell is below a specific threshold.
A3	The signal quality of a neighboring cell is higher than that of the serving cell by a certain offset.
A4	The signal quality of a neighboring cell exceeds a specific threshold.
A5	The signal quality of the serving cell is below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2.
A6	The signal quality of a neighboring cell is higher than that of the secondary cell by a certain offset.
B1	The signal quality of an inter-RAT neighboring cell exceeds a specific threshold.
B2	The signal quality of the serving cell is below threshold 1 and the signal quality of an inter-RAT neighboring cell exceeds threshold 2.

The mechanisms for measurement configuration delivery and measurement reporting are the same for all these events.

- Events A1 and A2 are used to evaluate the signal quality of the serving cell in the handover function start evaluation phase and determine whether to start or stop a handover function.
- The other events (A3, A4, A5, A6, B1, and B2) are used to evaluate the signal quality of neighboring cells in the target cell or frequency evaluation phase and determine whether to initiate handover execution.

Table 6-6 lists the entering and leaving conditions for these events. For details, see section 5.5.4 "Measurement report triggering" in 3GPP TS 38.331 V15.5.1.

Table 6-6 Entering and leaving conditions for events

Event Type	Entering Condition	Leaving Condition	Setting Notes
A1	(Ms – Hys > Thresh) is true during the time specified by *TimeToTrig*.	(Ms + Hys < Thresh) is true during the time specified by *TimeToTrig*.	See **Table 6-7**.
A2	(Ms + Hys < Thresh) is true during the time specified by *TimeToTrig*.	(Ms – Hys > Thresh) is true during the time specified by *TimeToTrig*.	See **Table 6-8**.
A3	(Mn + Ofn + Ocn – Hys > Ms + Ofs + Ocs + Off) is true during the time specified by *TimeToTrig*.	(Mn + Ofn + Ocn + Hys < Ms + Ofs + Ocs + Off) is true during the time specified by *TimeToTrig*.	See **Table 6-9**.
A4	(Mn + Ofn + Ocn – Hys > Thresh) is true during the time specified by *TimeToTrig*.	(Mn + Ofn + Ocn + Hys < Thresh) is true during the time specified by *TimeToTrig*.	See **Table 6-10**.
A5	(Ms + Hys < Thresh1) and (Mn + Ofn + Ocn – Hys > Thresh2) are true during the time specified by *TimeToTrig*.	(Ms – Hys > Thresh1) or (Mn + Ofn + Ocn + Hys < Thresh2) is true during the time specified by *TimeToTrig*.	See **Table 6-11**.
A6	(Mn + Ocn – Hys > Ms + Ocs + Off) is true during the time specified by *TimeToTrig*.	(Mn + Ocn + Hys < Ms + Ocs + Off) is true during the time specified by *TimeToTrig*.	Applicable to carrier aggregation. For details, see *Carrier Aggregation* in 5G RAN feature documentation.
B1	(Mn + Ofn + Ocn – Hys > Thresh) is true during the time specified by *TimeToTrig*.	(Mn + Ofn + Ocn + Hys < Thresh) is true during the time specified by *TimeToTrig*.	Applicable to inter-RAT scenarios. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Event Type	Entering Condition	Leaving Condition	Setting Notes
B2	(Ms + Hys < Thresh1) and (Mn + Ofn + Ocn – Hys > Thresh2) are true during the time specified by *TimeToTrig*.	(Ms – Hys > Thresh1) or (Mn + Ofn + Ocn + Hys < Thresh2) is true during the time specified by *TimeToTrig*.	Applicable to inter-RAT scenarios. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

The variables in the preceding formulas are defined as follows:

- Ms and Mn: measurement results of the serving cell and a neighboring cell, respectively
- Hys: hysteresis for an event
- TimeToTrig: duration during which a condition is continuously met before the event can be triggered
- Thresh, Thresh1, and Thresh2: thresholds
- Ofs and Ofn: frequency-specific offsets for the serving cell and a neighboring cell, respectively
- Ocs and Ocn: cell individual offset (CIO) for the serving cell and a neighboring cell, respectively
- Off: offset for a measurement result

Variable configuration is event specific. The following describes the entering conditions and parameters for variables of each event.

Entering condition for event A1: (*Ms* – *Hys* > *Thresh*) is true throughout the duration specified by *TimeToTrig*. **Table 6-7** describes the parameters for the variables.

Table 6-7 Variable configuration of event A1

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-frequency A1A2 Hysteresis	**NRCellInterFHoMeaGrp.*InterFreqA1A2Hyst***	This variable represents the hysteresis for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Cov Inter-freq Handover A1 RSRP Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA1RsrpThld***	This variable represents the RSRP threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter. This threshold can be set to different values for different frequencies using the **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter. The threshold equals the sum of the **NRCellInterFHoMeaGrp.*CovInterFreqA1RsrpThld*** and **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter values.
Thresh	Cov Inter-freq HO A1 RSRQ Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA1RsrqThld***	This variable represents the RSRQ threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter.
Thresh	SSB SINR Inter-freq HO A1 Threshold	**NRCellInterFHoMeaGrp**.***SsbSinrInterFreqHoA1Thld***	This variable represents the SINR threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Freq Priority Inter-freq A1 RSRP Threshold	**NRCellInterFHoMeaGrp**.***FreqPriInterFA1RsrpThld***	• If the threshold for RFSP-specific frequency-priority-based event A1 is not customized for a UE, the threshold for frequency-priority-based event A1 equals the value of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA1RsrpThld*** parameter. • If the threshold for RFSP-specific frequency-priority-based event A1 is customized for a UE, the threshold for frequency-priority-based event A1 equals the sum of the parameter values of **NRCellInterFHoMeaGrp**.***FreqPriInterFA1RsrpThld*** and **gNBRfspConfig**.***FreqPriHoA1A2RsrpThldOfs***.
Thresh	A3 Inter-freq Handover A1 RSRP Threshold	**NRCellInterFHoMeaGrp**.***A3InterFreqHoA1RsrpThld***	This variable represents the RSRP threshold for event A1 related to event A3-based coverage-based inter-frequency handover and is configured through this parameter.
TimeToTrig	Inter-frequency A1A2 Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA1A2TimeToTrig***	This variable represents the time-to-trigger for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

Entering condition for event A2: (*Ms* + *Hys* < *Thresh*) is true throughout the duration specified by *TimeToTrig*. **Table 6-8** describes the parameters for the variables.

Table 6-8 Variable configuration of event A2

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-frequency A1A2 Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA1A2Hyst***	This variable represents the hysteresis for events A1/A2 related to inter-frequency measurements and is configured through this parameter.
Thresh	Cov Inter-freq Handover A2 RSRP Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld***	This variable represents the RSRP threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter. This threshold can be set to different values for different frequencies using the **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter. The threshold equals the sum of the **NRCellInterFHoMeaGrp.*CovInterFreqA2RsrpThld*** and **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter values.
Thresh	Cov Inter-freq HO A2 RSRQ Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrqThld***	This variable represents the RSRQ threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter.
Thresh	SSB SINR Inter-freq HO A2 Threshold	**NRCellInterFHoMeaGrp**.***SsbSinrInterFreqHoA2Thld***	This variable represents the SINR threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Cov HO to E-UTRAN Blind A2 RSRP Thld	**NRCellInterRHoMeaGrp**.***CovHoToEutranBlindA2Thld***	This variable represents the RSRP threshold for event A2 related to inter-frequency blind redirection and is configured through this parameter (also used for inter-RAT blind redirection).
Thresh	Cov HO to E-UTRAN Blind A2 RSRQ Thld	**NRCellInterRHoMeaGrp**.***CovHoToLteBlindA2RsrqThld***	This variable represents the RSRQ threshold for event A2 related to inter-frequency blind redirection and is configured through this parameter (also used for inter-RAT blind redirection).
Thresh	Freq Priority Inter-freq A2 RSRP Threshold	**NRCellInterFHoMeaGrp**.***FreqPriInterFA2RsrpThld***	• If the threshold for RFSP-specific frequency-priority-based event A2 is not customized for a UE, the threshold for frequency-priority-based event A2 equals the value of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA2RsrpThld*** parameter. • If the threshold for RFSP-specific frequency-priority-based event A2 is customized for a UE, the threshold for frequency-priority-based event A2 equals the sum of the parameter values of **NRCellInterFHoMeaGrp**.***FreqPriInterFA2RsrpThld*** and **gNBRfspConfig**.***FreqPriHoA1A2RsrpThldOfs***.
Thresh	Service-based Inter-Freq HO A2 RSRP Thld	**NRCellInterFHoMeaGrp**.***InterFreqServHoA2ThldRsrp***	This variable represents the RSRP threshold for event A2 related to service-based inter-frequency handover and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	A3 Inter-freq Handover A2 RSRP Threshold	**NRCellInterFHoMeaGrp**.***A3InterFreqHoA2RsrpThld***	This variable represents the RSRP threshold for event A2 related to event A3-based coverage-based inter-frequency handover and is configured through this parameter.
TimeToTrig	Inter-frequency A1A2 Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA1A2TimeToTrig***	This variable represents the time-to-trigger for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

Entering condition for event A3: (*Mn* + *Ofn* + *Ocn* – *Hys* > *Ms* + *Ofs* + *Ocs* + *Off*) is true throughout the duration specified by *TimeToTrig*. **Table 6-9** describes the parameters for the variables.

Table 6-9 Variable configuration of event A3

Variable	Parameter Name	Parameter ID	Remarks
Ofn	None	None	This variable is fixed at **0**.
Ocn	Cell Individual Offset	**NRCellRelation**.***CellIndividualOff-set***	This variable represents the CIO of a neighboring cell. ● If the value is not **DB0**, this parameter is included in measurement configurations. ● If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
Hys	Intra-frequency Handover A3 Hysteresis	**NRCellIntraFHoMeaGrp**.***IntraFreqHoA3Hyst***	This variable represents the hysteresis for event A3 related to intra-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst***	This variable represents the hysteresis for event A3 related to inter-frequency handover and is configured through this parameter.
Hys	None	None	In the case of uplink-interference-based inter-frequency handovers, the hysteresis for event A3 related to inter-frequency measurements is fixed at 1 dB.
Ofs	None	None	This variable is fixed at **0**.
Ocs	None	None	This variable is fixed at **0**.
Off	Intra-frequency Handover A3 Offset	**NRCellIntraFHoMeaGrp**.***IntraFreqHoA3Offset***	This variable represents the offset for event A3 related to intra-frequency measurements and is configured through this parameter.
Off	Intrf-based Inter-Freq HO A3 RSRP Offset	**NRCellInterFHoMeaGrp**.***IntrfInterFreqHoA3RsrpOfs***	This variable represents the RSRP offset for event A3 related to uplink-interference-based inter-frequency handover and is configured through this parameter. If the RSRP of a neighboring cell minus the RSRP of the serving cell is greater than the offset, event A3 will be reported.
Off	Inter-frequency Handover A3 Offset	**NRCellInterFHoMeaGrp**.***InterFreqHoA3Offset***	This variable represents the offset for event A3 related to inter-frequency handover and is configured through this parameter.
TimeToTrig	Intra-frequency Handover A3 Time To Trigger	**NRCellIntraFHoMeaGrp**.***IntraFreqHoA3TimeToTrig***	This variable represents the time-to-trigger for event A3 related to intra-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
TimeToTrig	Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig***	This variable represents the time-to-trigger for event A3 related to inter-frequency handover and is configured through this parameter.
TimeToTrig	None	None	In the case of uplink-interference-based inter-frequency handovers, the time-to-trigger for event A3 related to inter-frequency measurements is fixed at 320 ms.

Entering condition for event A4: (*Mn* + *Ofn* + *Ocn* – *Hys* > *Thresh*) is true throughout the duration specified by *TimeToTrig*. **Table 6-10** describes the parameters for the variables.

Table 6-10 Variable configuration of event A4

Variable	Parameter Name	Parameter ID	Remarks
Ofn	Frequency Offset for Connected Mode	**NRCellFreqRelation**.***ConnFreqOffset***	This variable represents the connected-mode frequency-specific offset of a non-serving frequency and is configured through this parameter.
Ocn	Cell Individual Offset	**NRCellRelation**.***CellIndividualOff-set***	This variable represents the CIO of a neighboring cell. • If the value is not **DB0**, this parameter is included in measurement configurations. • If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp**.***InterFreqA4A5Hyst***	This variable represents the hysteresis for event A4 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Freq Priority Inter-freq A4 RSRP Threshold	**NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld***	• If the threshold for event A4 related to RFSP-specific frequency-priority-based measurements is not customized for a UE, the RSRP threshold for event A4 related to frequency-priority-based inter-frequency measurements equals the value of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld*** parameter. • If the threshold for event A4 related to RFSP-specific frequency-priority-based measurements is customized for a UE, the RSRP threshold for event A4 related to frequency-priority-based inter-frequency measurements equals the sum of the values of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld*** and **gNBRfspConfig**.***FreqPriIFHoA4RsrpThldOfs*** parameters. The threshold for event A4 related to RFSP-specific frequency-priority-based measurements can be customized for a UE only when RFSP-specific frequency-priority-based inter-frequency handover is enabled. For details about RFSP-specific frequency-priority-based inter-frequency handover, see *Flexible User Steering* in 5G RAN feature documentation.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Operator Ded Pri Inter-freq HO A4 RSRP Thld	**NRCellInterFHoMeaGrp.*OpDedPriHoA4RsrpThld***	This variable represents the RSRP threshold for event A4 related to operator-specific-priority-based inter-frequency measurements and is configured through this parameter.
Thresh	Srv-Based Inter-freq A4 RSRP Threshold	**NRCellInterFHoMeaGrp.*SrvInterFreqA4RsrpThld***	This variable represents the RSRP threshold for event A4 related to service-based inter-frequency measurements and is configured through this parameter.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp.*InterFreqA4A5TimeToTrig***	This variable represents the time-to-trigger for event A4 related to inter-frequency measurements and is configured through this parameter.

Entering condition for event A5: (*Ms* + *Hys* < *Thresh1*) and (*Mn* + *Ofn* + *Ocn* – *Hys* > *Thresh2*) are true throughout the duration specified by *TimeToTrig*. **Table 6-11** describes the parameters for the variables.

Table 6-11 Variable configuration of event A5

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-freq Measurement Event Hysteresis	**NRCellInterFHoMeaGrp.*InterFreqA4A5Hyst***	This variable represents the hysteresis for event A5 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh1	Cov Inter-freq Handover A5 RSRP Threshold1	**NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld1***	● RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements is specified by this parameter when **NRCellMobilityConfig**.***MeasTrigQuantity*** is set to **RSRP**. This threshold can be set to different values for different frequencies using the **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter. The threshold equals the sum of the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld1*** and **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter values. ● RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements has a fixed value of -43 dBm and the value of this parameter is invalid when **NRCellMobilityConfig**.***MeasTrigQuantity*** is not set to **RSRP**.
Thresh1	RSRP threshold 1 for event A5 related to frequency-priority-based inter-frequency measurements	None	This variable is fixed at –43 dBm.

Variable	Parameter Name	Parameter ID	Remarks
Thresh1	RSRP threshold 1 for event A5 related to operator-specific-priority-based inter-frequency measurements	None	This variable is fixed at –43 dBm.
Thresh1	RSRP threshold 1 for event A5 related to service-based inter-frequency measurements	**NRCellInterFHoMeaGrp**.***InterFreqServHoA2ThldRsrp***	This variable represents RSRP threshold 1 for event A5 related to service-based inter-frequency handover and is configured through this parameter. This variable reuses the specific parameter of the RSRP threshold for event A2.
Thresh2	Cov Inter-freq Handover A5 RSRP Threshold2	**NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrpThld2***	This variable represents RSRP threshold 2 for event A5 related to coverage-based inter-frequency measurements and is configured through this parameter.
Thresh2	Cov Inter-freq HO RSRQ Threshold	**NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrqThld2***	This variable represents the RSRQ threshold related to coverage-based inter-frequency measurements and is configured through this parameter, when RSRP combined with RSRQ-based filtering is used as the triggering quantity (by setting **NRCellFreqRelation**.***FreqIntrfFlag*** to **INTRF**) for event A5 related to coverage-based inter-frequency handover.

Variable	Parameter Name	Parameter ID	Remarks
Thresh2	Cov Inter-Freq HO A5 SINR Threshold2	**NRCellInterFHoMeaGrp**.***CovInterFreqA5SinrThld2***	This variable represents the SINR threshold related to coverage-based inter-frequency measurements and is configured through this parameter, when RSRP combined with SINR-based filtering is used as the triggering quantity (by setting **NRCellFreqRelation**.***FreqIntrfFlag*** to **INTRF_SINR**) for event A5 related to coverage-based inter-frequency handover.

Variable	Parameter Name	Parameter ID	Remarks
Thresh2	RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements	**NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld***	• If threshold 2 for event A5 related to RFSP-specific frequency-priority-based measurements is not customized for a UE, RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements equals the value of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld*** parameter. • If threshold 2 for event A5 related to RFSP-specific frequency-priority-based measurements is customized for a UE, RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements equals the sum of the values of the **NRCellInterFHoMeaGrp**.***FreqPriInterFA4RsrpThld*** and **gNBRfspConfig**.***FreqPriIFHoA4RsrpThldOfs*** parameters. Threshold 2 for event A5 related to RFSP-specific frequency-priority-based measurements can be customized for a UE only when RFSP-specific frequency-priority-based inter-frequency handover is enabled. For details about RFSP-specific frequency-priority-based inter-frequency handover, see *Flexible User Steering* in 5G RAN feature documentation.

Variable	Parameter Name	Parameter ID	Remarks
Thresh2	RSRP threshold 2 for event A5 related to operator-specific-priority-based inter-frequency measurements	**NRCellInterFHoMeaGrp**.***OpDedPriHoA4RsrpThld***	This variable represents RSRP threshold 2 for event A5 related to operator-specific-priority-based inter-frequency measurements and is configured through this parameter.
Thresh2	RSRP threshold 2 for event A5 related to service-based inter-frequency measurements	**NRCellInterFHoMeaGrp**.***SrvInterFreqA4RsrpThld***	This variable represents RSRP threshold 2 for event A5 related to service-based inter-frequency measurements and is configured through this parameter.
Ofn	Frequency Offset for Connected Mode	**NRCellFreqRelation**.***ConnFreqOffset***	This variable represents the connected-mode frequency-specific offset of a non-serving frequency and is configured through this parameter.
Ocn	Cell Individual Offset	**NRCellRelation**.***CellIndividualOff-set***	This variable represents the CIO of a neighboring cell. ● If the value is not **DB0**, this parameter is included in measurement configurations. ● If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	**NRCellInterFHoMeaGrp**.***InterFreqA4A5TimeToTrig***	This variable represents the time-to-trigger for event A5 related to inter-frequency measurements and is configured through this parameter.

QCI-specific Measurement Parameter Configuration

Quality of service (QoS) is a measure that helps define the level of service and provides an indication of customer satisfaction. The QoS class identifier (QCI) is used to identify the QoS class of a data radio bearer (DRB).

A measurement parameter group can be configured separately for each QCI. If a UE has multiple radio bearers (in the case of combined services), each of the bearers has a QCI and measurement parameters can be configured separately for each QCI. Further details are as follows:

- For intra-frequency measurement, the QCI-specific measurement parameters for NSA UEs can be configured as follows: Use the **NRCellIntraFHoMeaGrp** MO with **NRCellIntraFHoMeaGrp**.***IntraFreqHoMeasGroupId*** specified to set a measurement parameter group related to intra-frequency handovers for an NR cell. Then, bind this group with a QCI using the **NRCellQciBearer**.***IntraFreqHoMeasGroupId*** and **NRCellQciBearer**.***Qci*** parameters. Through such configuration, intra-frequency measurement parameters are configured separately for services with different QCIs.
- For inter-frequency measurement, the QCI-specific measurement parameters for NSA UEs can be configured as follows: Use the **NRCellInterFHoMeaGrp** MO with **NRCellInterFHoMeaGrp**.***InterFreqHoMeasGroupId*** specified to set a measurement parameter group related to inter-frequency handovers for an NR cell. Then, bind this group with a QCI using the **NRCellQciBearer**.***NsaInterFreqHoMeasGroupId*** and **NRCellQciBearer**.***Qci*** parameters. Through such configuration, inter-frequency measurement parameters are configured separately for services with different QCIs.

NOTE

If the ID of a measurement parameter group related to inter-frequency handovers of NSA UEs (specified by the **NRCellQciBearer**.***NsaInterFreqHoMeasGroupId*** parameter) is not configured in the **NRCellInterFHoMeaGrp** MO, the gNodeB does not deliver measurement configurations to the UEs with the QCI bound to the **NRCellQciBearer**.***NsaInterFreqHoMeasGroupId*** parameter. As such, service drops may occur.

In the case of combined services, the gNodeB sends a group of measurement parameters configured for the QCI that has the highest priority for handovers. The priority of a QCI for handovers is configured by the **gNBQciBearer**.***QciPriorityForHo*** parameter and a smaller value of this parameter indicates a higher QCI priority for handovers. When **gNBQciBearer**.***QciPriorityForHo*** is set to **255**, the priority of a QCI for handovers equals the value of **gNBQciBearer**.***PriorityLevel*** and a smaller value indicates a higher QCI priority for handovers.

If the priorities of different QCIs for handovers obtained based on the preceding principle are the same, the gNodeB sorts the QCIs and sends a group of measurement parameters configured for the smallest QCI. For details, see section 6.1.7 "Standardized QoS characteristics" in 3GPP TS 23.203 V15.4.0.

Triggering Quantity

A triggering quantity refers to the policy that triggers event reporting. This policy uses concepts of RSRP, RSRQ, and SINR. **Table 6-12** lists the triggering quantities supported by the current version.

Table 6-12 Triggering quantities

Triggering Quantity	Definition	Description
Only RSRP	Measurement is triggered and stopped based only on RSRP.	1. The gNodeB sends only the RSRP measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP of a cell meets the entering conditions for the event, the UE sends a measurement report.
RSRP or RSRQ	Measurement is triggered and stopped based on both RSRP and RSRQ, and the RSRP-based process and the RSRQ-based process are independent of each other.	1. The gNodeB sends both RSRP and RSRQ measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP or RSRQ of a cell meets the entering conditions for the event, the UE sends a measurement report.

Triggering Quantity	Definition	Description
RSRP or SINR	Measurement is triggered and stopped based on both RSRP and SINR, and the RSRP-based process and the SINR-based process are independent of each other.	1. The gNodeB sends both RSRP and SINR measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP or SINR of a cell meets the entering conditions for the event, the UE sends a measurement report.
RSRP combined with RSRQ-based filtering	Measurement is triggered and stopped based only on RSRP, but a measurement report includes both the measured RSRP and RSRQ, and specific neighboring cells are filtered out based on the measured RSRQ.	1. The gNodeB sends both RSRP and RSRQ measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP of a cell meets the entering conditions for the event, the UE sends a measurement report including the measured RSRP and RSRQ. 3. The gNodeB generates a candidate cell list based on the measured RSRP, and filters out specific candidate cells based on the measured RSRQ. For details about RSRQ-based filtering, see **6.1.1.6 Target Cell or Frequency Evaluation**.

Triggering Quantity	Definition	Description
RSRP combined with SINR-based filtering	Measurement is triggered and stopped based only on RSRP, but a measurement report includes both the measured RSRP and SINR, and specific neighboring cells are filtered out based on the measured SINR.	1. The gNodeB sends both RSRP and SINR measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP of a cell meets the entering conditions for the event, the UE sends a measurement report including the measured RSRP and SINR. 3. The gNodeB generates a candidate cell list based on the measured RSRP, and filters out specific candidate cells based on the measured SINR. For details about SINR-based filtering, see **6.1.1.6 Target Cell or Frequency Evaluation**.
Only SINR	Measurement is triggered and stopped based only on SINR.	1. The gNodeB sends only the SINR measurement configuration related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured SINR of a cell meets the entering conditions for the event, the UE sends a measurement report.

Table 6-13 lists the triggering quantities specific to handover functions. For details, see the sections corresponding to each handover function.

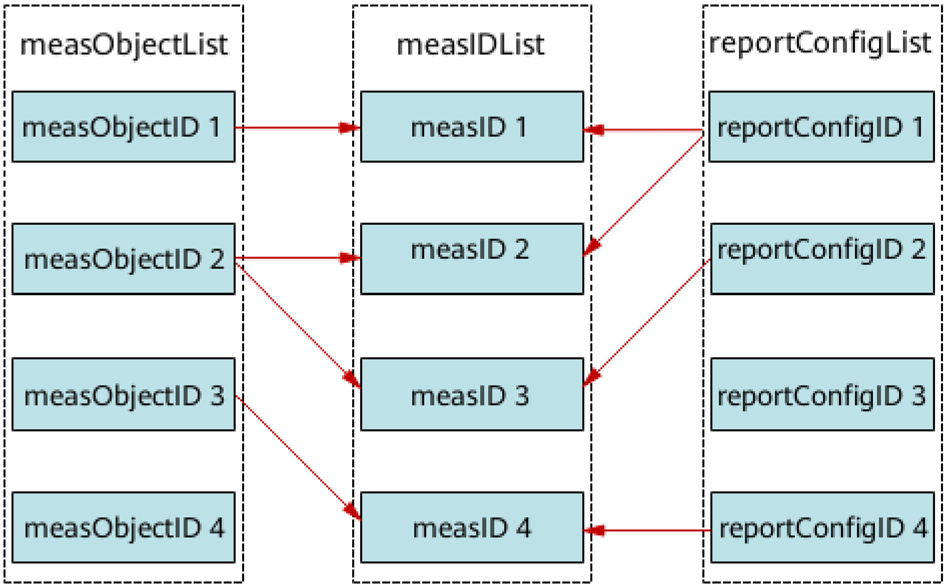
Table 6-13 Event triggering quantities specific to handover functions

Handover Function	Event That Triggers the Start of a Handover Function	Event That Stops a Handover Function	Event That Triggers the Execution of a Handover
Coverage-based intra-frequency handover	N/A	N/A	Event A3: only RSRP
Coverage-based inter-frequency handover	Event A2: • Only RSRP • RSRP or RSRQ • RSRP or SINR	Event A1: • Only RSRP • RSRP or RSRQ • RSRP or SINR	Event A5: • Only RSRP • RSRP combined with RSRQ-based filtering • RSRP combined with SINR-based filtering Event A3: only RSRP
Frequency-priority-based inter-frequency handover	Event A1: only RSRP	Event A2: only RSRP	Event A4: only RSRP Event A5: only RSRP
Operator-specific-priority-based inter-frequency handover	N/A	N/A	Event A4: only RSRP Event A5: only RSRP
Service-based inter-frequency handover	N/A	N/A	Event A4: only RSRP Event A5: only RSRP
Uplink-interference-based inter-frequency handover	N/A	N/A	Event A3: only RSRP

6.1.1.4.3 Measurement ID Configuration

A measurement ID links a measurement object with a reporting configuration as a set. Given a measurement ID, a UE measures the associated measurement object according to the reporting configuration. With multiple measurement IDs configured, it is possible to link multiple measurement objects to the same reporting configuration, as well as to link multiple reporting configurations to the same measurement object. **Figure 6-2** provides an example, in which both measObjectID 1 and measObjectID 2 are linked to reportConfigID 1, and measObjectID 2 is linked to both reportConfigID 1 and reportConfigID 2. A UE includes a measurement ID in a measurement report sent to the base station, which then identifies the corresponding measurement object and reporting configuration based on this measurement ID.

Figure 6-2 Measurement ID



The base station sends measurement configurations to UEs on a per measurement ID basis and in sequence based on the triggering time until the number of sent measurement IDs reaches the upper limit. Once this limit is reached, the base station cannot send new measurement configurations to UEs. Consequently, some high-priority measurement configurations (such as those for coverage-based handovers) may fail to be delivered, leading to service drops or handover failures. To prevent this, measurement ID preemption is introduced. This function is controlled by the **MEAS_ID_PREEMPTION_SW** option of the **gNBMobilityCommParam.*MeasAlgoSwitch*** parameter.

- If this option is selected, measurement ID preemption is enabled. If the number of sent measurement IDs has reached the upper limit, a UE's high-priority measurement IDs can preempt the measurement resources of its low-priority measurement IDs. This mechanism ensures that high-priority measurement configurations, such as mobility-related measurement configurations, can be successfully sent. If the number of sent measurement IDs has not reached the upper limit, measurement IDs are not preempted and measurement configurations are sent to UEs in sequence based on the triggering time.
- If this option is deselected, measurement ID preemption is disabled. Measurement IDs are not preempted and measurement configurations are sent to UEs in sequence based on the triggering time.

6.1.1.4.4 SMTC Measurement Configuration

An SMTC is a measurement configuration that the gNodeB sends to a UE, which includes an SMTC period, SMTC duration, and SMTC offset. These serve as the instructions for the UE to perform SSB measurements in a cell. For details, see section 5.5.2.10 "Reference signal measurement timing configuration" and section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.5.1.

- The SMTC period determines the interval at which the UE starts SSB measurements. It is configured by the **NRDUCell.*SmtcPeriod*** parameter for intra-frequency measurements and by the **NRCellFreqRelation**.***SmtcPeriod*** parameter for inter-frequency measurements.

NOTE

SMTC is specific to frequencies. If the cells on the same frequency have different SSB periods (specified by the **NRDUCell**.***SsbPeriod*** parameter), SSB measurements may be inaccurate for the cells whose SSB periods are longer than the corresponding SMTC period. To prevent this, ensure that the cells on the same frequency use the same SSB period (specified by the **NRDUCell**.***SsbPeriod*** parameter) according to the network plan.

- The SMTC duration determines the time during which the UE performs SSB measurements in each SMTC period. It is configured by the **NRDUCell**.***SmtcDuration*** parameter for intra-frequency measurements and by the **NRCellFreqRelation**.***SmtcDuration*** parameter for inter-frequency measurements.

  The **NRDUCell**.***SmtcDuration*** and **NRCellFreqRelation**.***SmtcDuration*** parameters take effect only for low frequency bands. The system automatically calculates the SMTC duration for high frequency bands.

- The SMTC offset determines the time offset for the UE to start SSB measurements in each SMTC period. The offset is calculated by the gNodeB to ensure accurate SSB measurements by the UE. The following configurations affect the SMTC offset calculation. If they do not meet the corresponding requirements, the SSB measurement result may be inaccurate.

  The frame offsets (specified by the **gNBFreqBandConfig**.***FrameOffset*** parameter) of frequency bands must meet all of the following conditions:

  – The frame offset of a frequency band must be the same for all base stations on the entire network.
  – If the frame offsets for the to-be-measured frequency bands differ, the maximum difference must not exceed 3 ms.
  – If the frame offsets for the serving and to-be-measured frequency bands differ, the maximum difference must not exceed 4 ms.
  – If the frame offset of a frequency band is configured on a base station on the network, the frame offset of this band must also be specified on neighboring base stations, even if these neighboring base stations do not serve any cell in this band. This ensures that this frequency band is accurately measured by UEs served by these neighboring base stations.

NOTE

If frame offsets are different between to-be-measured frequency bands, the measurement gap configured for a UE must be extended to include the SMTC duration of cells in different frequency bands, reducing the throughput during inter-frequency measurements. If the maximum difference between said frame offsets exceeds 3 ms, the measurement gap is insufficient to include the SMTC duration of all cells. As a result, inter-frequency measurement is incomplete, affecting the handover success rate.

In cases where the frame offsets for the serving and to-be-measured frequency bands differ and the maximum difference exceeds 4 ms, inter-frequency measurement may be incomplete, affecting the number of handover attempts. For details, see **6.1.1.4.5 Measurement Gap Configuration**.

For details about frame offsets of frequency bands, see *Standards Compliance* in 5G RAN feature documentation.

6.1.1.4.5 Measurement Gap Configuration

A measurement gap is a period during which a UE performs measurements on a non-serving frequency, and is only used for inter-frequency measurements.

Measurement Gap Configuration on the eNodeB

The eNodeB derives the signal quality of NR cells from the measurement results of synchronization signal and PBCH block (SSB) signals from the gNodeB. To this end, SSB signals from NR cells must be transmitted within measurement gaps on the E-UTRAN side. Otherwise, the eNodeB fails to initiate handovers. The start position of a gap for NR measurement calculated by the eNodeB is relevant to the SSB period and SSB offset. The recommended settings of the SSB period and SSB offset vary depending on the synchronization mode used by the eNodeB and gNodeB. It is recommended that time synchronization be used.

If time synchronization is used:

- SSB period

  The eNodeB parameter **NrNFreq**.***SsbPeriod*** must be set to the same value as the gNodeB parameter **NRDUCell**.***SsbPeriod***. If multiple NR cells operate on the same frequency on the NR network and have different SSB period settings, it is recommended that, on the E-UTRAN side, the SSB period for that frequency be set to the maximum value among these SSB period settings.

- SSB offset

  On the gNodeB side, the frame offset is specified by the **gNodeBParam**.***FrameOffset*** or **gNBFreqBandConfig**.***FrameOffset*** parameter. The value ranges of the two parameters are the same. The former applies to a gNodeB, and the latter applies to a frequency band. If both parameters are configured, the latter takes effect.

  On the eNodeB side, the frame offset is specified by the **ENodeBFrameOffset**.***TddFrameOffset***, **ENodeBFrameOffset**.***FddFrameOffset***, or **CellFrameOffset**.***FrameOffset*** parameter. The first two parameters apply to an eNodeB, and the last parameter applies to a cell. If all these parameters are configured, the **CellFrameOffset**.***FrameOffset*** parameter takes effect.

  The following uses the gNodeB frame offset specified by the **gNBFreqBandConfig**.***FrameOffset*** parameter and the eNodeB frame offset specified by the **CellFrameOffset**.***FrameOffset*** parameter as an example.

  a. The time difference between the start time of the subframe for the NR SSB and the global navigation satellite system (GNSS) standard time, also known as the offset of SSB to the clock source, is calculated based on the NG-RAN frame offset.

     - If the value of the **gNBFreqBandConfig**.***FrameOffset*** parameter is less than or equal to 261120, the offset of SSB to the clock source is calculated as follows: (**gNBFreqBandConfig**.***FrameOffset***/30720) MOD **NrNFreq**.***SsbPeriod***.

     - If the value of the **gNBFreqBandConfig**.***FrameOffset*** parameter is greater than or equal to 275943, the offset of SSB to the clock source is calculated as follows: (**gNBFreqBandConfig**.***FrameOffset***/30720 – 10 + **NrNFreq**.***SsbPeriod***) MOD **NrNFreq**.***SsbPeriod***

If the NR cells operate on different frequencies, the **gNBFreqBandConfig**.***FrameOffset*** parameter must be set to an identical value across the frequencies.

b. The SSB offset is corrected based on the eNodeB frame offset.

The offset of SSB to the clock source calculated in the previous step is used as the SSB offset before correction. Perform the following steps to correct the SSB offset:

i. The NG-RAN frame offset remainder and E-UTRAN frame offset remainder are calculated as follows:

- NG-RAN frame offset remainder = (**gNBFreqBandConfig**.***FrameOffset*** MOD 30720)/30720
- E-UTRAN frame offset remainder = (**CellFrameOffset**.***FrameOffset*** MOD 30720)/30720

ii. If the E-UTRAN frame offset remainder is greater than the NG-RAN frame offset remainder and is less than 0.5 ms, the corrected SSB offset is calculated as follows: (SSB offset before correction – 0.5 + **NrNFreq**.***SsbPeriod***) MOD **NrNFreq**.***SsbPeriod***.

iii. If the preceding condition is not met, the corrected SSB offset is equal to the SSB offset before correction.

c. The corrected SSB offset is rounded down to its nearest integer, and the integer is used as the value of the **NrNFreq**.***SsbOffset*** parameter.

If frequency synchronization is used:

- SSB period

  It is recommended that the SSB period (specified by the eNodeB parameter **NrNFreq**.***SsbPeriod*** and the gNodeB parameter **NRDUCell**.***SsbPeriod***) be set to **5MS**.

- SSB offset

  Set the eNodeB parameter **NrNFreq**.***SsbOffset*** to **0**.

NOTE

The unit of the frame offset is Ts, which equals 1/30720 ms. Frame offset can be expressed in the unit of ms as long as the value in Ts is divided by 30720.

When the eNodeB initiates gap-assisted measurements of NR frequencies, it does not deliver B1- or B2-related measurement gap configurations to the UE if the UE is capable of non-gap-assisted measurements of NR frequencies and the **NR_B1_NO_GAP_SW** option of the **EnodeBAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter is selected.

Measurement Gap Configuration on the gNodeB

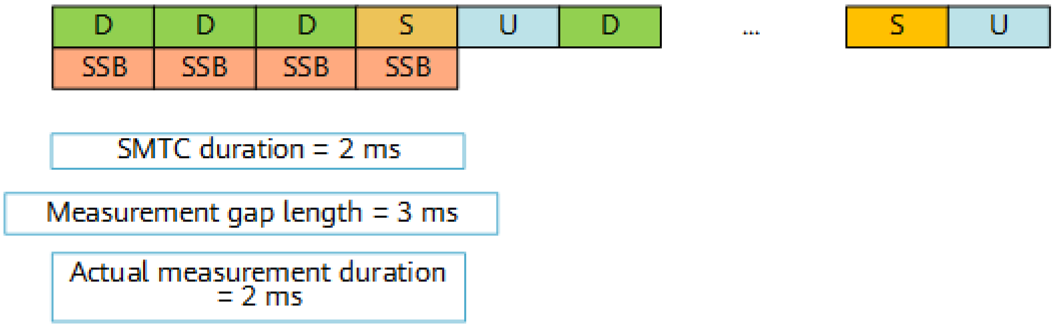
There are 24 protocol-defined gap patterns, with each specific to a measurement gap length and a measurement gap repetition period. Measurement gap configurations are sent to UEs through measurement configurations. In the current version, the gNodeB supports gap pattern adaptation based on UE capabilities and SMTC to minimize the overhead of gap-assisted measurements. For details about measurement gaps, see section 9.1.2 "Measurement gap" in 3GPP TS 38.133 V15.5.0.

When an inter-frequency measurement is required, the gNodeB determines whether to send a measurement gap as follows:

- If a measurement gap has been configured, a new measurement gap is not sent.
- If no measurement gap has been configured, a measurement gap is sent.

To ensure the complete measurement of a non-serving frequency, the actual measurement duration within a measurement gap must cover the entire SMTC duration of this frequency. The actual measurement duration equals the measurement gap length minus the time required for a frequency switch before and after the measurement. **Figure 6-3** shows an example for TDD.

Figure 6-3 Example of measurement gap length and SMTC duration configuration



If a UE needs to start both intra-frequency and inter-frequency measurements, and the SMTC period for the intra-frequency measurement is greater than or equal to the measurement gap repetition period, the SMTC duration for the intra-frequency measurement is entirely included in the inter-frequency measurement gap. As a result, the intra-frequency measurement cannot be performed. In this case, the gNodeB configures equal-opportunity intra-frequency and inter-frequency measurements, allowing the UE to alternatively perform such measurements through gap sharing.

If a UE needs to initiate inter-frequency measurements, frame offsets must be configured in accordance with the following requirements for the serving and to-be-measured frequency bands:

- The difference in the frame offsets (**gNBFreqBandConfig**.***FrameOffset***) between the serving and to-be-measured frequency bands cannot exceed 4 ms. If it exceeds 4 ms, inter-frequency measurement may be incomplete, affecting the number of handover attempts.
- If the UE needs to measure multiple frequency bands with different frame offsets, the gNodeB selects the maximum gap to cover the SMTC durations of all the frequency bands to be measured. The protocol-defined maximum gap is 6 ms, but in cases where the frame offsets for multiple to-be-measured frequency bands differ by more than 3 ms, the maximum gap may not cover the SMTC durations of all the to-be-measured frequency bands. As a result, inter-frequency measurement is incomplete, affecting the handover success rate. As such, it is recommended that the frame offsets (configured by the **gNBFreqBandConfig**.***FrameOffset*** parameter) remain within a margin of 3 ms between different frequency bands.

Optimized Non-Overlapping Between Gap and NR Resources

UEs cannot transmit uplink signals or receive downlink signals on the serving frequency throughout the measurement gap for inter-frequency measurements, which reduces the uplink and downlink throughput of those UEs. In addition, if the time-domain positions of NR channel resources conflict with those of the measurement gaps, user experience is affected. For example, if the SRS period is shorter than the gap period, the time-domain positions of SRSs and gap-assisted measurements will conflict, increasing uplink IBLER and RBLER and causing significant signal fluctuation. This affects the timeliness of measurement reporting and the service experience of moving UEs.

To address this issue, the "optimized non-overlapping between gap and NR resources" function is introduced. This function staggers the channel resources of SRS, PUCCH SR, CSI-RS for CM, and CSI-RS for CM measurement reporting signals in low frequency bands from measurement gaps in the time domain.

In NSA networking, the "optimized non-overlapping between gap and NR resources" function is controlled by the **GAP_AVOID_NR_RES_OPT_SW** option of the **NRDUCellMeas**.***MeasAlgoSwitch*** parameter and is supported only in low frequency bands. It is recommended that this function be enabled when signal fluctuations are large.

- If this option is deselected, the channel resources of the said signals are not staggered from the measurement gap in the time domain. In this case, the measurement gap preferentially takes effect on UEs, which do not transmit or receive signals during the measurement gap.
- If this option is selected, the channel resources of the said signals are staggered from the measurement gap in the time domain. This reduces uplink IBLER and RBLER and decreases the number of delayed measurement reports, but it may increase the number of reconfiguration signaling messages.

For more details about the "optimized non-overlapping between gap and NR resources" function, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

Non-Gap-Assisted Measurement for Active BWP

The UEs capable of non-gap-assisted measurements support inter-frequency measurements (within the frequency-domain range of the active BWP) without stopping downlink scheduling. Accordingly, the non-gap-assisted measurement for active BWP function is developed to reduce data transmission interruptions during inter-frequency measurements for the preceding UEs in a cell. In addition, this function reduces the service drops caused by time-domain conflicts between PUCCH SRs and gap-assisted measurements, improving user experience. Non-gap-assisted measurement for active BWP is supported only in low frequency bands, and it is controlled by the **SSB_MEAS_WITHOUT_GAP_SW** option of the **NRDUCellMeas**.***SsbMeasPolicySwitch*** parameter. When this function is enabled, it takes effect for the UE that meets all of the following conditions:

- The UE supports non-gap-assisted measurements. For details, see the *interFrequencyMeas-NoGap-r16* IE in the UECapabilityInformation message.
- At least one of the non-serving frequencies to be measured is within the frequency-domain range of the UE's active BWP.

After non-gap-assisted measurement for active BWP takes effect, the gNodeB sends an RRCReconfiguration message containing the *interFrequencyConfig-NoGap-r16* IE with a value of **TRUE** to the UE that meets all of the preceding conditions. For non-serving frequencies included in the frequency-domain range of the UE's active BWP, the gNodeB does not calculate or deliver measurement gaps, and the UE can still perform inter-frequency measurements.

NOTE

In carrier aggregation scenarios, non-gap-assisted measurement for active BWP takes effect only for the non-serving frequencies within the frequency-domain range of the active BWP for the PCC of a UE enabled with carrier aggregation.

Gap-assisted Scheduling Compatibility

In low-frequency NR TDD scenarios, if the measurement gap timing advance (MGTA) value delivered by the gNodeB is not 0, UEs with compatibility issues fail to decode information in the first slot after the measurement gap. This triggers RLC retransmissions, resulting in increased delay for these UEs. To mitigate this, the gap-assisted scheduling compatibility function is introduced. This function is controlled by the **GAP_SCH_COMP_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter. When this option is selected, the function takes effect, and no service or signaling is transmitted in the first slot after the measurement gap, improving the performance and experience of UEs with compatibility issues.

For details about the gap-assisted scheduling compatibility function, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

NOTE

For details about MGTA, see section 6.3.5 "Measurement information elements" in 3GPP TS 38.331 V16.7.0.

6.1.1.4.6 Other Configurations

SSB Time-Domain Position Delivery

SSB patterns in the time domain are defined in section 4.1 "Cell Search" in 3GPP TS 38.213 V15.12.0. For high-frequency cells, the number of SSBs in a certain pattern can be changed to adjust cell capacity and coverage capability. For details, see *mmWave Beam Management (High-Frequency TDD)* in 5G RAN feature documentation. After SSB beam adjustment, SSB beam time-domain positions are sent to UEs through the *ssb-ToMeasure* IE, and the UEs then perform SSB measurement at the corresponding positions.

For UEs to perform neighboring cell measurement, the gNodeB not only sends the SSB time-domain positions of the serving cell, but may also send the SSB time-domain positions of intra-frequency neighboring cells, depending on the **SSB_MEAS_POS_POLICY_SW** option of the **NRCellAlgoSwitch**.***MeasPolicySwitch*** parameter.

- If this option is deselected, the gNodeB sends only the SSB time-domain positions of the serving cell to UEs. The gNodeB performs downlink data transmission in time-domain positions that differ from those for SSB and remaining minimum system information (RMSI) transmission in the serving cell.

If the time-domain positions for SSB or RMSI transmission in intra-frequency neighboring cells conflict with those for downlink data transmission in the serving cell, UEs may encounter interference during intra-frequency neighboring cell measurement or experience downlink data transmission failure in the serving cell.

- If this option is selected, the gNodeB obtains the SSB time-domain positions of intra-frequency neighboring cells and sends the SSB time-domain positions of both the serving cell and intra-frequency neighboring cells to UEs. After obtaining the SSB time-domain positions of intra-frequency neighboring cells, the gNodeB can deduce the RMSI time-domain positions of these cells based on the relationship between SSB and RMSI time-domain positions. The gNodeB performs downlink data transmission in time-domain positions that differ from those for SSB and RMSI transmission in both the serving cell and intra-frequency neighboring cells.

  The gNodeB obtains the SSB time-domain positions of intra-frequency neighboring cells as follows:

  – If an Xn interface is available between base stations, the SSB time-domain positions of intra-frequency neighboring cells can be obtained over this interface.
  – If no Xn interface is available between base stations, an Xn interface self-setup is triggered and then the SSB time-domain positions of intra-frequency neighboring cells can be obtained over this interface.

NOTE

This function applies only to high-frequency cells. It is recommended that this function be enabled if the serving cell has intra-frequency neighboring cells and its SSB time-domain positions differ from those of intra-frequency neighboring cells (for example, if the **NRDUCellTrpMmwavBeam**.***CoverageScenario*** parameter is set to **SCENARIO_103** for both the serving cell and intra-frequency neighboring cells).

If there is no intra-frequency neighboring cell or if the SSB time-domain positions of intra-frequency neighboring cells cannot be obtained due to reasons such as the use of devices from other vendors, the gNodeB will send only the SSB time-domain positions of the serving cell.

This function may have the following network impacts:

- As the time-domain positions for downlink data transmission in the serving cell differ from those for SSB transmission in intra-frequency neighboring cells, the serving cell has the following changes: the downlink IBLER decreases, the number of downlink scheduling times decreases, the average downlink cell throughput decreases, and the intra-frequency handover success rate increases.
- As the time-domain positions for downlink data transmission in the serving cell differ from those for RMSI transmission in intra-frequency neighboring cells, the interference to RMSI sent in intra-frequency neighboring cells decreases, the success rate of accessing intra-frequency neighboring cells increases, but the average downlink throughput of the serving cell slightly decreases.

Optimization of *deriveSSB-IndexFromCell*

The gNodeB includes the *deriveSSB-IndexFromCell* IE (either set to **TRUE** or **FALSE**) in measurement configurations. For intra-frequency neighboring cell measurement, the value of this IE indicates whether UEs derive the SSB indexes of all intra-frequency neighboring cells based on the timing information of the serving cell. For inter-frequency neighboring cell measurement, it indicates whether UEs derive the SSB indexes of all neighboring cells on the frequency of any detected inter-frequency cell based on the timing information of that cell. If all cells on a frequency are time-synchronized, you can improve the efficiency and accuracy of detecting neighboring cells' SSB indexes by setting this IE to **TRUE**.

Optimization of the *deriveSSB-IndexFromCell* IE has been introduced, and this function is enabled by selecting the **PROTOCOL_IE_OPT_SW** option the **NRCellAlgoSwitch**.***HoCompatibilitySwitch*** parameter. When this function is enabled, the value of the *deriveSSB-IndexFromCell* IE in the measurement configurations sent from the gNodeB varies with the RATs corresponding to the frequencies that are to be measured. If an NR TDD frequency is to be measured, the value of the *deriveSSB-IndexFromCell* IE is **TRUE**.

SSB Beam Measurement Result Selection and Combination

A UE needs to combine the SSB beam-level measurement results of a cell when calculating the cell-level SSB measurement result. These beam-level results must meet the SSB combination threshold requirement. The SSB combination threshold is specified by the **NRCellMobilityConfig**.***SsbConsolidationThld*** parameter. Upon detecting that multiple SSB beams in a cell meet the combination threshold requirement, the UE combines the measurement results of these SSB beams into a cell-level result. The maximum number of SSB beams whose measurement results can be combined is specified by the **NRCellMobilityConfig**.***SsbNumToAverage*** parameter.

Measurement Filtering

Before evaluating the conditions for entering or leaving an event and sending measurement reports, the UE filters measurement results based on filtering coefficients as follows:

- Beam-level filtering, in which the filtering coefficient is specified by the **NRCellMobilityConfig.*BeamRsrpFilterCoeff*** parameter.
- Cell-level filtering, in which the filtering coefficient is specified by the **NRCellMobilityConfig.*CellRsrpFilterCoeff*** parameter.

6.1.1.5 Measurement Reporting

A UE performs measurements based on the measurement configuration received from the gNodeB, and filters the measurement results based on filtering coefficients. When determining that the entering conditions for the corresponding event are met, the UE sends measurement reports to the gNodeB.

- The maximum number of cells that can be reported by a UE is indicated by the *maxReportCells* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.
- The reporting type of a measurement report is indicated by the *reportType* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.
- The reporting interval of a UE is indicated by the *reportInterval* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.

- The number of reporting times allowed for a UE is indicated by the *reportAmount* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.

6.1.1.6 Target Cell or Frequency Evaluation

The process includes:

- Measurement report processing (needed only in measurement-based mode)
- Determination of a handover policy
- Generation of a target cell or frequency

Measurement Report Processing

The gNodeB processes measurement reports in the order they arrive and generates candidate cells or frequencies based on the analysis results.

Determination of a Handover Policy

The gNodeB selects a handover policy for a UE based on information about candidate cells or frequencies, the UE's current service type, and UE capabilities.

Generation of a Target Cell or Frequency List

A target cell or frequency list is generated based on the processing mode, measurement report, and neighboring cell filtering rule. The process is as follows:

1. Candidate cells or frequencies are generated according to different processing modes (measurement-based mode and blind mode):
   - Measurement-based mode: The gNodeB selects candidate cells based on measurement reports and then filters the candidate cells according to rules in 2 to determine the target cell.
   - Blind mode:

     The gNodeB filters frequencies based on factors such as UE capabilities and the frequencies associated with each operator, and then generates a candidate frequency list based on related priorities.

     The frequency filtering rules are as follows:

     - For all UEs, the gNodeB excludes non-serving frequencies that are unsupported by the UE's capabilities.
     - For all UEs, whether the gNodeB filters frequencies based on the frequencies associated with a specific operator is related to the factors listed in **Table 6-14**.
       - Whether the operator-specific policy of an NR cell (**NRCellOpPolicy** MO) is configured.
       - Validity of the gNodeB frequency priority group ID (**NRCellOpPolicy**.***gNBFreqPriorityGroupId***) configured in the operator-specific policy of an NR cell: The value in the range of **0** to **254** indicates that the ID is valid. The value **255** indicates that the ID is invalid. The ID is mapped to the **gNBFreqPriorityGroup**.***gNBFreqPriorityGroupId*** parameter.

- ○ Whether the operator-specific neighboring frequency configuration function is enabled (**OPERATOR_NR_FREQ_CFG_SW** option of the **NRCellOpPolicy**.***FreqConfigPolicySwitch*** parameter).

Table 6-14 Frequency filtering based on the frequencies associated with a specific operator

<table>
<tr><th>Whether or Not the NRCellOpPolicy MO Is Configured</th><th>Whether or Not the NRCellOpPolicy.gNBFreqPriorityGroupId Parameter Value Is Valid</th><th>Operator-specific Neighboring Frequency Configuration</th><th>Frequency Filtering Mode</th></tr>
<tr><td>Yes</td><td>Valid</td><td>Enabled</td><td>The gNodeB excludes NR frequencies that fall outside the intersection of the following NR frequency sets:
● NR frequencies determined by gNBFreqPriorityGroup.SsbDescMethod and gNBFreqPriorityGroup.SsbFreqPos
● NR frequencies determined by NRCellFreqRelation.SsbDescMethod and NRCellFreqRelation.SsbFreqPos</td></tr>
<tr><td>Yes</td><td>Valid</td><td>Disabled</td><td rowspan="3">The gNodeB excludes frequencies that are not in the range specified by the NRCellFreqRelation.SsbDescMethod and NRCellFreqRelation.SsbFreqPos parameters.</td></tr>
<tr><td>Yes</td><td>Invalid</td><td>Enabled or Disabled</td></tr>
<tr><td>No</td><td>Valid or Invalid</td><td>Enabled or Disabled</td></tr>
<tr><td colspan="4">Note: For unnecessary inter-frequency handovers excluding those triggered by coverage, the gNodeB also selects candidate frequencies based on the frequency interference flag in blind mode. If the NRCellFreqRelation.FreqIntrfFlag parameter is set to NO_INTRF for a given frequency, that frequency is eligible for selection as a candidate. Otherwise, it is excluded from candidate selection.</td></tr>
</table>

Once the candidate frequency list is generated, the gNodeB selects the highest-priority frequency as the target frequency. It then filters cells on the target frequency according to the rules in 2 to generate the target cell.

2. The target cell is generated in accordance with the following neighboring cell filtering rules:
   – The gNodeB excludes neighboring cells for which the **NRCellRelation**.***NoHoFlag*** parameter is set to **FORBID_HO**.
   – The gNodeB excludes cells that are not supported by the current UE, whether NSA or SA.

     For an NSA UE, candidate neighboring cells using the **SA** networking option are excluded, whereas those using the **NSA** or **SA_NSA** option are retained.

     Networking option of a neighboring cell:
     - The networking option of a neighboring cell is specified by **NRExternalNCell.*NrNetworkingOption*** in non-multi-operator sharing scenarios, or by **NRExternalNCellPlmn.*NrNetworkingOption*** in multi-operator sharing scenarios.
     - The networking options of neighboring cells can be automatically managed through ANR when the **NR_NR_ANR_SW** option of the **NRCellAlgoSwitch.*AnrSwitch*** parameter is selected. For details about ANR, see *ANR* in 5G RAN feature documentation.
   – For the coverage-based inter-frequency handover function:
     - If **NRCellFreqRelation**.***FreqIntrfFlag*** is set to **INTRF** for a given frequency, the gNodeB filters the cells on this frequency based on the following rules:

       The gNodeB filters candidate cells based on the measured RSRQ. That is, the gNodeB determines whether the RSRQ of a cell meets the following condition. If the condition is met, the gNodeB excludes the cell. Otherwise, the gNodeB retains the cell.

       $Mn + Ofn + Ocn + Hys < Thresh2$

       In the above formula:
       - *Mn* is the measured RSRQ of the neighboring cell.
       - *Ofn* is the frequency-specific offset for RRC_CONNECTED UEs, and is specified by the **NRCellFreqRelation.*ConnFreqOffset*** parameter.
       - *Ocn* is the CIO, and is specified by the **NRCellRelation**.***CellIndividualOffset*** parameter. If the value is not **DB0**, this parameter is included in measurement configurations. If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
       - *Hys* is the hysteresis for events related to inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.*InterFreqA4A5Hyst*** parameter.

- *Thresh2* is the RSRQ threshold for events related to coverage-based inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp**.***CovInterFreqA5RsrqThld2*** parameter.

- If **NRCellFreqRelation**.***FreqIntrfFlag*** is set to **INTRF_SINR** for a frequency, the gNodeB filters the cells on this frequency based on the following rules:

  The gNodeB filters candidate cells based on the measured SINR. That is, the gNodeB determines whether the SINR of a cell meets the following condition. If the condition is met, the gNodeB excludes the cell. Otherwise, the gNodeB retains the cell.

  Mn + *Ofn* + *Ocn* + *Hys* < *Thresh2*

  In the above formula:

  - *Mn* is the measured SINR of the neighboring cell.
  - *Ofn* is the frequency-specific offset for RRC_CONNECTED UEs, and is specified by the **NRCellFreqRelation.*ConnFreqOffset*** parameter.
  - *Ocn* is the CIO, and is specified by the **NRCellRelation**.***CellIndividualOffset*** parameter. If the value is not **DB0**, this parameter is included in measurement configurations. If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
  - *Hys* is the hysteresis for events related to inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.*InterFreqA4A5Hyst*** parameter.
  - *Thresh2* is the SINR threshold for events related to coverage-based inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp**.***CovInterFreqA5SinrThld2*** parameter.

– In low-frequency scenarios, the gNodeB excludes terrestrial neighboring cells whose SINRs fall below a defined threshold for UEs with specific bearers.

  In specific scenarios, if the RSRP- and SINR-based joint handover function is enabled for a cell and the bearer-level switch for this function is turned on for any bearer of a UE, this function will take effect for the UE. That is, the gNodeB will exclude terrestrial neighboring cells whose SINRs fall below a defined SINR threshold based on SINR measurements reported by the UE. The specific scenarios and related parameters are as follows:

  - Specific scenarios: Handover or redirection scenarios except the following:
    - Blind handover
    - Blind redirection
    - Handovers in which the gNodeB performs evaluation upon receiving measurement results. For example, the gNodeB evaluates user experience after receiving measurement results during experience-based smart carrier selection.
    - Grid-based handover

- Related parameters and options:
  - The RSRP- and SINR-based joint handover function is controlled by the **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter.
  - The bearer-level switch for RSRP- and SINR-based joint handover is controlled by the **RSRP_SINR_HO_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter.
  - A terrestrial neighboring cell is defined as a cell for which the **NRExternalNCell**.***AerialCellFlag*** parameter is set to **GROUND_CELL**.
  - The SINR threshold corresponds to the bearer with the QCI that has the highest handover priority. For details about the priority of a QCI for handovers, see **QCI-specific Measurement Parameter Configuration** in **6.1.1.4.2 Reporting Configuration**.

    The SINR threshold for necessary handovers is specified by the **NRCellInterFHoMeaGrp**.***RsrpSinrBasedNecSinrThld*** parameter.

    The SINR threshold for unnecessary handovers is specified by the **NRCellInterFHoMeaGrp**.***RsrpSinrBasedUnecSinrThld*** parameter.

– When the **MOBILITY_RESTRICTION_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter is selected, the gNodeB excludes any cells subject to handover restrictions, as specified in the *Mobility Restriction List* IE within signaling messages such as the SgNB Addition Request.

– The gNodeB excludes neighboring cells whose PLMNs are not listed among those permitted for UE handovers. If the target cell supports multiple PLMNs (including the current serving PLMN), the source gNodeB preferentially selects the current serving PLMN for use after the handover.

– The gNodeB excludes the neighboring cells whose frequency bands are incompatible with the UE's supported frequency bands and whose uplink and downlink bandwidths fall outside the UE's bandwidth capabilities. The details are as follows:

- For intra-base-station handovers, the gNodeB excludes neighboring cells as follows:
  - If an intra-base-station neighboring cell is configured with frequency bands incompatible with the UE's supported frequency bands, the gNodeB excludes that cell.
  - If the UE's minimum supported bandwidth exceeds both the downlink and uplink bandwidths of an intra-base-station neighboring cell, the gNodeB excludes that cell.
- For inter-base-station handovers, the gNodeB excludes inter-base-station neighboring cells configured with frequency bands that are incompatible with those supported by the UE.
- For inter-base-station handovers, the gNodeB does not filter inter-base-station neighboring cells based on the UE's supported frequency bands or bandwidths if those cells are not configured with any frequency bands.

In the preceding information:

- The frequency band of an intra-base-station neighboring cell is specified by the **NRCell**.***FrequencyBand*** parameter, which is mandatory.
- The frequency band of an inter-base-station neighboring cell is specified by the **NRExternalNCell**.***FrequencyBand*** parameter, which is optional.
- The frequency bands supported by a UE can be queried from the *supportedBandListNR* IE in the *UECapabilityInformation* IE signaled over the Uu interface.
- The minimum bandwidth supported by a UE can be queried from the *channelBWs-UL*, *channelBWs-DL*, *channelBWs-UL-v1590*, and *channelBWs-DL-v1590* IEs in the *UECapabilityInformation* IE signaled over the Uu interface.

For details, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V16.5.0.

– The gNodeB excludes NR cells served by blacklisted gNodeBs. The blacklist of gNodeBs is configured using the **gNBNRBlacklist** MO. For details, see *ANR* in 5G RAN feature documentation.

– The gNodeB excludes cells based on gNodeB mobility control lists defined using **gNBMobilityCtrlList** MOs. A maximum of 16 such lists can be configured, each of which is identified by the **gNBMobilityCtrlList**.***MobilityCtrlListId*** parameter. The following operations need to be performed for each control list:

- Set the **gNBMobilityCtrlList**.***gNodeBStartId*** and **gNBMobilityCtrlList**.***gNodeBEndId*** parameters to respectively specify the start and end gNodeB IDs of the control list.
- Set the **gNBMobilityCtrlList**.***Mcc*** and **gNBMobilityCtrlList**.***Mnc*** parameters to respectively specify the MCC and MNC of the primary operator of external NR cells served by gNodeBs indicated in the control list. The MCC and MNC together indicate the PLMN of the primary operator.

After the configuration is complete, the gNodeB prohibits handovers of NSA UEs to a target cell that meets all the following conditions:

- The ID of the gNodeB serving the target cell is in the range of [start gNodeB ID, end gNodeB ID] configured for the control list.
- The PLMN to which the target cell belongs is the same as the PLMN of the primary operator of external NR cells served by gNodeBs indicated in the control list.

6.1.1.7 Handover Execution

After determining the target cell or frequency list, the gNodeB executes a handover based on the selected handover policy.

During the inter-gNodeB PSCell change procedure, the reconfiguration message sent to the UE carries only incremental configurations, which may lead to compatibility issues of defective UEs and result in degraded user experience, including increased service drop rates and reduced traffic throughput. In this context, the "full configuration during inter-gNodeB handovers" function is introduced. Once this function is enabled, the target gNodeB includes full configurations and the *fullConfig* IE in the SGNB ADDITION REQUEST ACKNOWLEDGE message, and delivers an RRC reconfiguration message containing such information to the UE through the eNodeB during the inter-gNodeB PSCell change procedure. If this function is disabled, the eNodeB sends the UE an RRC reconfiguration message that carries only incremental configurations. When the **INTER_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch**.***HoCompatibilitySwitch*** parameter is selected for the target cell of an inter-gNodeB PSCell change, the "full configuration during inter-gNodeB handovers" function takes effect.

This function supports differentiated application across UEs. That is, this function can be enabled only for specific UEs. **Table 6-15** describes how it is implemented.

Table 6-15 Function effectiveness of "full configuration during inter-gNodeB handovers"

Setting of the HO_FULL_CONFIG_SW_ON Option of the gNBUeCompatSw.*WhitelistCtrlSwitch* Parameter	Setting of the INTER_GNB_HO_FULL_CONFIG_SW Option of the NRCellAlgoSwitch.*HoCompatibilitySwitch* Parameter	Function Effectiveness for UEs
Deselected	Deselected	Full configuration during inter-gNodeB handovers does not take effect for any UE.
Deselected	Selected	Full configuration during inter-gNodeB handovers takes effect for all UEs.
Selected	Deselected	Full configuration during inter-gNodeB handovers does not take effect for any UE.

Setting of the HO_FULL_CONFIG_SW_ON Option of the gNBUeCompatSw.*WhitelistCtrlSwitch* Parameter	Setting of the INTER_GNB_HO_FULL_CONFIG_SW Option of the NRCellAlgoSwitch.*HoCompatibilitySwitch* Parameter	Function Effectiveness for UEs
Selected	Selected	● Full configuration during inter-gNodeB handovers takes effect only for specific UEs that correspond to the UE information index for which the **HO_FULL_CONFIG_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlSwitch*** parameter is selected. ● Full configuration during inter-gNodeB handovers does not take effect for other UEs.

6.1.1.8 Handover Failure Penalty

The handover failure penalty mechanism aims to reduce the number of ineffective retries that follow a handover failure. In NSA networking, handover failure penalty is the penalty for handover preparation failures.

Handover Preparation Failure Penalty

Depending on the causes, handover preparation failures are categorized as resource-caused handover preparation failures and non-resource-caused handover preparation failures.

- For resource-caused handover preparation failures, a penalty timer is configured to impose penalties on the target cell. A retry cannot be initiated to the cell for the UE until the penalty timer expires.
  - For coverage-based intra-frequency and inter-frequency handovers, the penalty timer is specified by the **gNBMobilityCommParam**.***NecHoPrepFailPunishTimer*** parameter.
  - For other handover functions, the penalty timer is specified by the **gNBMobilityCommParam**.***OptHoPrepFailPunishTimer*** parameter.

NOTE

For handover preparation failures with the cause value "Transport resource unavailable", the following applies within the penalty time:

- If the UE involved is an NSA DC UE, the NSA DC UEs that are connected to the same eNodeB and served by the same PLMN as this NSA DC UE cannot initiate handover attempts to the target cell.
- If the UE involved is an SA UE, the SA UEs that are served by the same PLMN as this SA UE cannot initiate handover attempts to the target cell.

- For non-resource-caused handover preparation failures, the number of penalty times is configured to impose penalties on the target cell. A retry cannot be initiated to the cell for the UE until the number of penalty times is reached. If the handover preparation continues to fail after the retry, the UE can no longer initiate a handover to the cell.

  The number of penalty times is specified by the **gNBMobilityCommParam**.***NonResHoPrepFailPunishTimes*** parameter.

NOTE

- After cell A enters the penalty state, the number of penalty times is incremented by one each time the measurement result indicates that cell A meets the handover condition.
- Retry means that a request for a handover to cell A is sent again when this cell meets the handover condition after the specified penalties imposed on it have ended.
- Resource-caused handover preparation failures are only those whose failure causes are as follows:
  - No radio resources available in target cell
  - Transport resource unavailable
  - Not enough User Plane Processing Resources
  - Radio resources not available
  - Control Processing Overload

6.1.2 PSCell Change Policies

NSA networking supports the following types of PSCell change:

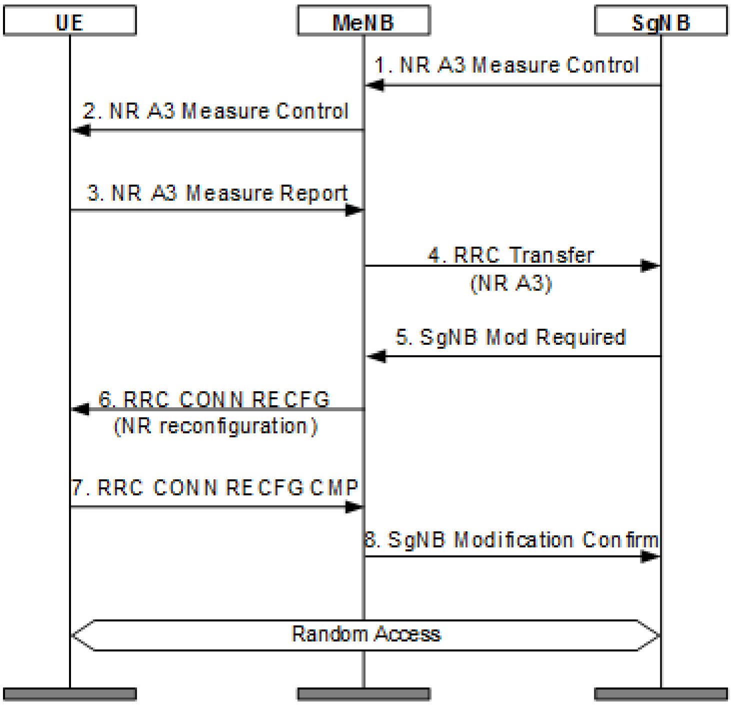
- Coverage-based intra-frequency PSCell change
- Coverage-based inter-frequency PSCell change
- Frequency-priority-based inter-frequency PSCell change
- Operator-specific-priority-based inter-frequency PSCell change
- Service-based inter-frequency PSCell change
- Uplink-interference-based inter-frequency PSCell change

The principles and configurations for the PSCell change mechanisms in different scenarios are similar to those in SA networking. Therefore, this section will only describe the principles and configurations that are different. Inter-frequency PSCell change must be performed within the EN-DC band combinations supported by the UE's EN-DC capability.

Coverage-based Intra-Frequency PSCell Change

The principles, network analysis, requirements, and operation and maintenance for coverage-based intra-frequency PSCell change in NSA networking are the same as those for coverage-based intra-frequency handover of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. The SgNB delivers intra-frequency A3 measurement configurations and triggers an intra-frequency cell change based on A3 measurement reports. It sends an SgNB Modification Required (for intra-base-station cell change) or SgNB Change Required (for inter-base-station cell change) message to the MeNB to trigger a cell change. **Figure 6-4** shows an intra-base-station cell change procedure.

Figure 6-4 Intra-base-station cell change procedure



Coverage-based Inter-Frequency PSCell Change

The principles of coverage-based inter-frequency PSCell change are similar to those of coverage-based inter-frequency handover in SA networking. This section describes only the unique aspects of the PSCell change principles. For details about coverage-based inter-frequency handover in SA networking, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

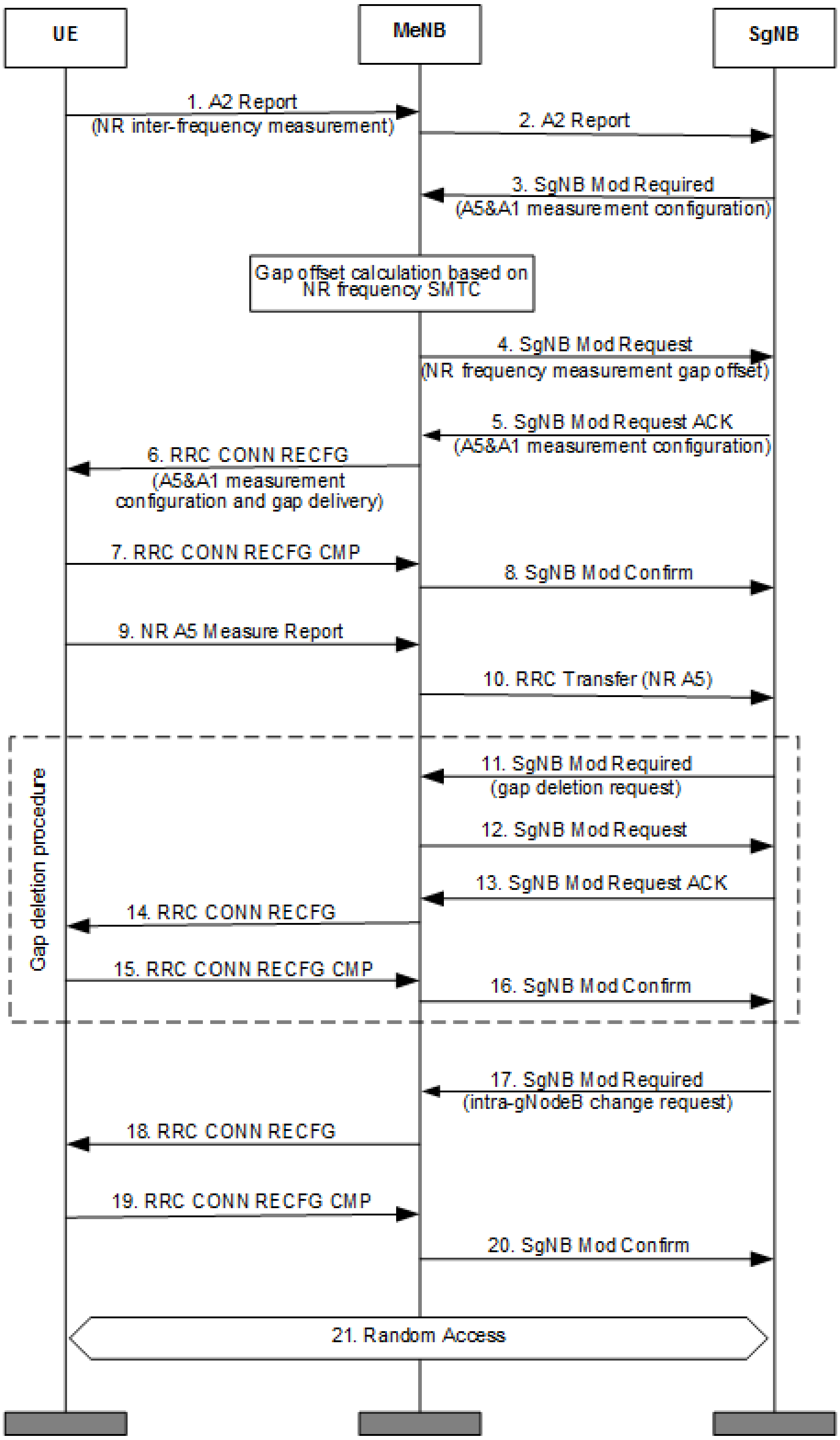
The SgNB, after receiving an inter-frequency event A2 report, triggers an inter-frequency handover procedure, where it delivers events A5 and A1 measurement configurations and configures an NR inter-frequency measurement gap. Event A5 is used to measure neighboring NR frequencies, and event A1 is used to cancel inter-frequency measurement. After receiving an event A1 report, the SgNB deletes the NR inter-frequency measurement gap. In NSA networking, gaps for NR inter-frequency measurement (except for per-FR2 measurement) are calculated and delivered by the eNodeB, according to section 7.2 "Measurements" in 3GPP TS 37.340 V15.5.0.

- If the **NsaDcPerFr1GapSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter is selected, the eNodeB checks whether an NSA UE supports independent gaps during gap configuration.

  - If the *MeasAndMobParametersMRDC* IE in the MR-DC capability information contains an *independentGapConfig* IE, the NSA UE supports independent gaps. In this case, per-FR1 gaps are configured.
  - If the *MeasAndMobParametersMRDC* IE in the MR-DC capability information does not contain an *independentGapConfig* IE, the NSA UE does not support independent gaps. In this case, per-UE gaps are configured.
- If the **NsaDcPerFr1GapSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter is deselected, the eNodeB does not check whether an NSA UE supports independent gaps during gap configuration. In this case, per-UE gaps are configured.

Figure 6-5 shows a PSCell change procedure, using an intra-base-station inter-frequency cell change as an example.

Figure 6-5 Intra-base-station inter-frequency cell change procedure



NOTE

LTE and NR time synchronization is required in NSA scenarios for NR inter-frequency measurement in per-UE or per-FR1 gaps. If LTE and NR time synchronization is not achieved, NR inter-frequency SSBs may not be measured in NR inter-frequency measurement gaps configured on the LTE side; in this case, the corresponding frequencies cannot be measured.

In NSA networking, coverage-based inter-frequency PSCell change can be implemented in measurement-based mode or blind mode, but redirection is not supported. Measurement-based inter-frequency PSCell change can be implemented between low frequency bands as well as between high and low frequency bands. Blind inter-frequency PSCell change can only be implemented from high frequency bands to low frequency bands.

Measurement-based inter-frequency PSCell change is the same as measurement-based coverage-based inter-frequency handover in SA networking in the following aspects: principles, network analysis, requirements, and operation and maintenance. For details about measurement-based coverage-based inter-frequency handover in SA networking, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

The following only describes the aspects of blind inter-frequency PSCell change that differ from the basic functions of mobility management in connected mode. For details about the basic functions of mobility management in connected mode, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

The principles of blind inter-frequency PSCell change in NSA networking are as follows:

- Handover function start evaluation

  Blind inter-frequency PSCell change starts when all of the following conditions are met:

  – The **COVERAGE_BASED_BLIND_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter is selected for the serving cell.

  – There are low-frequency neighboring cells with the **COV_BASED_BLIND_HO_FLAG** option of the **NRCellRelation**.***BlindHoFlag*** parameter selected.

  – The gNodeB receives a measurement report indicating that the involved frequency (or cell) meets the entering conditions for event A2 related to the blind mode. The threshold configuration for the preceding event A2 is related to its triggering quantity. The triggering quantity for event A2 can be "only RSRP", "RSRP or RSRQ", or "RSRP or SINR", which is specified by the **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter. For details, see **Triggering Quantity**.

    ▪ If this parameter is set to **RSRP**, only RSRP is used as the triggering quantity for event A2 related to the blind mode.

      The gNodeB receives a measurement report indicating that the measured RSRP of a frequency (or cell) meets the entering condition for event A2 related to the blind mode. That is, the signal quality of the serving cell drops below the threshold specified by the **NRCellInterFHoMeaGrp.*CovInterFreqA2RsrpThld*** parameter.

- If this parameter is set to **RSRP_OR_RSRQ**, RSRP or RSRQ is used as the triggering quantity for event A2 related to the blind mode.

  The gNodeB receives a measurement report indicating that the measured RSRP of a frequency (or cell) meets the entering condition for event A2 related to the blind mode. That is, the signal quality of the serving cell drops below the threshold specified by the **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** parameter. Alternatively, it receives a measurement report indicating that the measured RSRQ of a frequency (or cell) meets the entering condition for event A2 related to the blind mode. That is, the signal quality of the serving cell drops below the threshold specified by the **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrqThld*** parameter.

- If this parameter is set to **RSRP_OR_SINR**, RSRP or SINR is used as the triggering quantity for event A2 related to the blind mode.

  The gNodeB receives a measurement report indicating that the measured RSRP of a frequency (or cell) meets the entering condition for event A2 related to the blind mode. That is, the signal quality of the serving cell drops below the threshold specified by the **NRCellInterFHoMeaGrp.*CovInterFreqA2RsrpThld*** parameter. Alternatively, it receives a measurement report indicating that the measured SINR of a frequency (or cell) meets the entering condition for event A2 related to the blind mode. That is, the signal quality of the serving cell drops below the threshold specified by the **NRCellInterFHoMeaGrp**.***SsbSinrInterFreqHoA2Thld*** parameter.

NOTE

Blind inter-frequency PSCell change can be implemented only from high frequency bands to low frequency bands.

Blind inter-frequency PSCell change and measurement-based inter-frequency PSCell change share the A2 thresholds specified by **NRCellInterFHoMeaGrp.*CovInterFreqA2RsrpThld***, **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrqThld***, and **NRCellInterFHoMeaGrp**.***SsbSinrInterFreqHoA2Thld***. The RSRP threshold for event A2 can be set to different values for different frequencies using the **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter. This threshold equals the sum of the **NRCellInterFHoMeaGrp.*CovInterFreqA2RsrpThld*** and **NRCellFreqRelation**.***InterFreqHoA1A2A51FreqOfs*** parameter values

When both blind inter-frequency PSCell change and measurement-based inter-frequency PSCell change are enabled and the UE reports event A2:

- If there is a neighboring cell for blind handover (that is, the **COV_BASED_BLIND_HO_FLAG** option of the **NRCellRelation**.***BlindHoFlag*** parameter is selected for the neighboring cell), blind inter-frequency PSCell change instead of measurement-based inter-frequency PSCell change is triggered.
- If there are no neighboring cells for blind handovers (that is, the **COV_BASED_BLIND_HO_FLAG** option of the **NRCellRelation**.***BlindHoFlag*** parameter is deselected for all neighboring cells), measurement-based inter-frequency PSCell change instead of blind inter-frequency PSCell change is triggered.

– UEs support inter-frequency measurement and inter-frequency handover.

- Target cell or frequency evaluation

  After receiving an A2 measurement report related to the blind mode, the gNodeB selects a low-frequency neighboring cell for which the

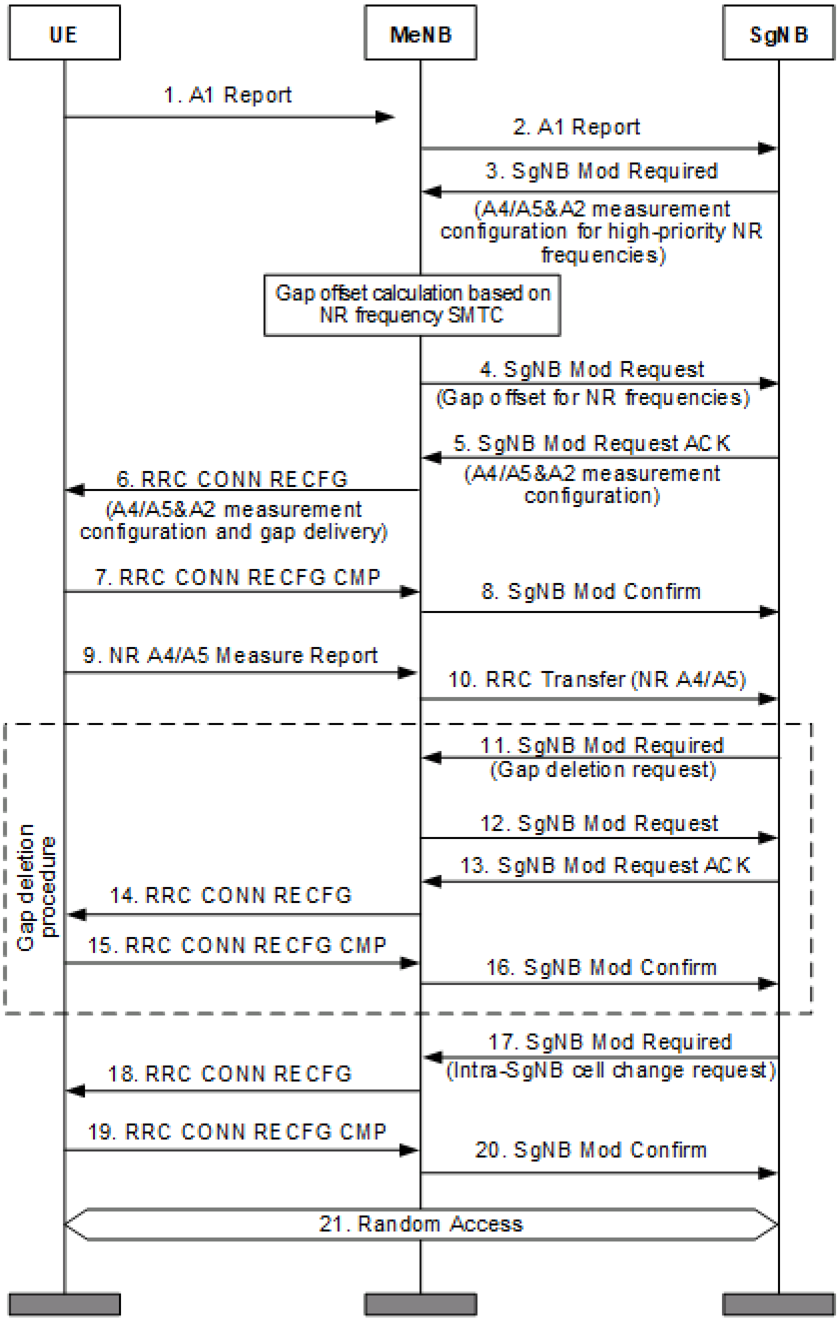
COV_BASED_BLIND_HO_FLAG option of the **NRCellRelation**.***BlindHoFlag*** parameter is selected to perform an inter-frequency PSCell change.

- If there are multiple such cells, the cell on the frequency with a higher connected-mode priority (specified by the **NRCellFreqRelation**.***ConnFreqPriority*** parameter) is preferentially selected.
- If there are no such cells (that is, the **COV_BASED_BLIND_HO_FLAG** option of the **NRCellRelation**.***BlindHoFlag*** parameter is deselected for all low-frequency neighboring cells), then measurement-based inter-frequency PSCell change instead of blind inter-frequency PSCell change is performed.

Frequency-Priority-based Inter-Frequency PSCell Change

The principles, network analysis, requirements, and operation and maintenance for frequency-priority-based inter-frequency PSCell change in NSA networking are the same as those for frequency-priority-based inter-frequency handover of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. After the PSCell is configured, event A1 or A2 can be used to determine whether to start a frequency-priority-based inter-frequency PSCell change procedure. After the procedure is started, if the UE reports event A4 or A5 and the frequency corresponding to the PSCell does not have the highest priority, frequency-priority-based inter-frequency PSCell change enables the UE to be handed over to a higher-priority non-serving frequency. **Figure 6-6** shows a frequency-priority-based inter-frequency PSCell change procedure where event A1 is used to determine whether to start the procedure. Frequency-priority-based inter-frequency PSCell change can be implemented between high frequency bands, between low frequency bands, and between high and low frequency bands.

Figure 6-6 Procedure for frequency-priority-based inter-frequency PSCell change (event A1 used to determine whether to start the procedure)



Operator-specific-Priority-based Inter-Frequency PSCell Change

The principles, network analysis, requirements, and operation and maintenance for operator-specific-priority-based inter-frequency PSCell change in NSA networking are the same as those for operator-specific-priority-based inter-frequency handover of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. Operator-specific-priority-based inter-frequency PSCell change can be implemented between high frequency bands, between low frequency bands, and between high and low frequency bands.

Service-based Inter-Frequency PSCell Change

The principles, network analysis, requirements, and operation and maintenance for service-based inter-frequency PSCell change in NSA networking are the same as those for service-based inter-frequency handover of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. Service-based inter-frequency PSCell change can be implemented between high frequency bands, between low frequency bands, and between high and low frequency bands.

Uplink-Interference-based Inter-Frequency PSCell Change

The principles (except the starting conditions), network analysis, requirements, and operation and maintenance for uplink-interference-based inter-frequency PSCell change in NSA networking are the same as those for uplink-interference-based inter-frequency handover of SA mobility management in connected mode. For details, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation. The conditions for starting uplink-interference-based inter-frequency PSCell change in NSA networking are listed below. Uplink-interference-based inter-frequency PSCell change can be implemented only between low frequency bands.

- The **UL_INTRF_BASED_HO** and **NSA_UL_INTRF_BASED_HO** options of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter are both selected.
- The strength of uplink interference in the serving cell is greater than the uplink interference threshold (specified by the **NRDUCellIntrfIdent**.***UlIntrfThld*** parameter).

6.2 Network Analysis

6.2.1 Benefits

PSCell change in NSA networking allows NSA UEs in connected mode to reselect PSCells with better signal quality for data transmission, ensuring service continuity for moving UEs.

For details about the benefits of different PSCell change policies, see the corresponding "Benefits" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

After the "full configuration during inter-gNodeB handovers" function is enabled, the user experience improves and the service drop rate decreases for defective UEs with compatibility issues. However, for UEs without compatibility issues, the size of the RRC reconfiguration message increases, leading to higher base station resource consumption and greater UE processing complexity, and potentially a decreased inter-gNodeB handover success rate.

For details about the impacts of different PSCell change policies, see the corresponding "Impacts" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

6.3 Requirements

6.3.1 Licenses

For details about the license requirements, see the "Licenses" section in *EPC-based NSA Basics*.

There are no license requirements for PSCell change in NSA networking.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	NSA networking based on EPC	LTE side: **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter NR side: **NRCellAlgoSwitch**.***NsaDcSwitch*** set to **ON**	*EPC-based NSA Basics*	PSCell change in NSA networking requires NSA networking based on EPC to be enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	X2 self-setup	• LTE: **LTE_NR_X2_SON_SETUP_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter • NR: **X2SON_SETUP_SWITCH** option of the **gNBX2SonConfig**.***X2SonConfigSwitch*** parameter	*X2 and S1 Self-Management in NSA Networking*	EN-DC services can be performed only after an X2 link has been set up between the source/target MeNB and the source/target SgNB in scenarios with SgNB addition, SgNB-initiated SgNB change, or MeNB-initiated inter-MeNB handover without an SgNB change.
For details about the prerequisite functions of different PSCell change policies, see the corresponding "Prerequisite Functions" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.				

6.3.2.2 Mutually Exclusive Functions

For details about the mutually exclusive functions of different PSCell change policies, see the corresponding "Mutually Exclusive Functions" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	User-experience-based NSA SCG coverage expansion	Both the **SCG_ADAPT_COV_THLD_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter and the **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter selected	*Multi-Frequency Smart Selection* in 5G RAN feature documentation	User-experience-based NSA SCG coverage expansion modifies the threshold for event A2, threshold 2 for event A5, and threshold for event A1 related to coverage-based inter-frequency handovers of NSA UEs only when coverage-based inter-frequency PSCell change is enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	User-experience-based NSA SCG coverage expansion	Both the **SCG_ADAPT_COV_THLD_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter and the **NSA_COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter selected	*Multi-Frequency Smart Selection* in 5G RAN feature documentation	When user-experience-based NSA SCG coverage expansion is enabled, the RSRP threshold used in coverage threshold adaptation (specified by **gNBMobilityComm-Param**.***AdaptCovRsrpThld***) must be greater than or equal to the threshold for event A2 related to SCG deletion plus 4 dB. The purpose is to prevent a significant increase in the number of SCG releases (indicated by **L.NsaDc.SgnbTrig.SgNB.NormRel**). ● Threshold for event A2 related to SCG deletion = **NRCellNsaDcConfig**.***PscellA2RsrpThld*** + **NRCellNsaDcConfigGrp**.***PscellA2RsrpThldOffset*** ● It is recommended that the threshold for event A2 related to SCG deletion be set to the recommended value. – A higher threshold for event A2 related to SCG deletion results in a smaller SCG coverage expansion area. To expand the SCG coverage

RAT	Function Name	Function Switch	Reference	Description
				area, the threshold for event A2 related to SCG deletion can be decreased. – Decreasing the threshold for event A2 related to SCG deletion may increase the number of intra-frequency PSCell changes (indicated by **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att** and **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**).
For details about the function impacts of different PSCell change policies, see the corresponding "Function Impacts" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.				

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For details, see "Base Station Models" under the "Hardware" section in *EPC-based NSA Basics*.

Boards

For details, see "Boards" under the "Hardware" section in *EPC-based NSA Basics*.

RF Modules

This function does not depend on RF modules.

6.3.4 Others

For other requirements, see the corresponding "Others" sections in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

For details about data preparation for neighbor relationships and mobility-related parameters on the NR side, see the "Data Preparation" section under the "Basic Functions of Mobility Management in Connected Mode" chapter in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

The following table describes the parameters to be set on the NR side for SSB, SMTC, frame offset, and more.

Parameter Name	Parameter ID	Setting Notes
SSB Frequency Position	**NRCellFreqRelation.*SsbFreqPos***	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	**NRCellFreqRelation.*SsbDescMethod***	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	**gNBFreqPriorityGroup**.***SsbDescMethod***	Set this parameter based on the network plan.
SSB Frequency Position	**gNBFreqPriorityGroup**.***SsbFreqPos***	Set this parameter based on the network plan.
Frequency Interference Flag	**NRCellFreqRelation**.***FreqIntrfFlag***	Set this parameter based on the network plan.
SMTC Period	**NRDUCell.*SmtcPeriod***	Use the recommended value.
SMTC Duration	**NRDUCell.*SmtcDuration***	Use the recommended value.
SMTC Period	**NRCellFreqRelation.*SmtcPeriod***	Use the recommended value.
SMTC Duration	**NRCellFreqRelation.*SmtcDuration***	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Frame Offset	**gNBFreqBandConfig**.*Frame Offset*	Set this parameter based on the network plan.

The following table describes the parameters to be set on the NR side to configure blind inter-frequency PSCell change.

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	**NRCellRelation**.*NrCellId*	Set this parameter based on the network plan.
Mobile Network Code	**NRCellRelation**.*Mnc*	Set this parameter based on the network plan.
Mobile Country Code	**NRCellRelation**.*Mcc*	Set this parameter based on the network plan.
gNodeB ID	**NRCellRelation**.*gNBId*	Set this parameter based on the network plan.
Cell ID	**NRCellRelation**.*CellId*	Set this parameter based on the network plan.
Blind Handover Flag	**NRCellRelation**.*BlindHoFlag*	It is recommended that this parameter be set only for neighboring cells whose coverage includes that of the serving cell.
Inter-Frequency Handover Switch	**NRCellAlgoSwitch**.*InterFreqHoSwitch*	When you enable coverage-based inter-frequency blind handover, select the **COV_BASED_BLIND_HO_FLAG** option for low-frequency neighboring cells to configure them as neighboring cells for blind handover. It is recommended that this option be selected only for the neighboring cells whose coverage includes that of the serving cell.

The following table describes the parameters to be set on the NR side to configure QCI-specific measurement.

Parameter Name	Parameter ID	Setting Notes
Intra-freq Handover Measurement Group ID	**NRCellIntraFHoMeaGrp**.***IntraFreqHoMeasGroupId***	Set this parameter based on the network plan.
Intra-freq Handover Measurement Group ID	**NRCellQciBearer**.***IntraFreqHoMeasGroupId***	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	**NRCellInterFHoMeaGrp**.***InterFreqHoMeasGroupId***	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	**NRCellQciBearer**.***NsaInterFreqHoMeasGroupId***	Set this parameter based on the network plan.
QCI Priority for Handover	**gNBQciBearer**.***QciPriorityForHo***	Set this parameter based on the network plan.
Priority Level	**gNBQciBearer**.***PriorityLevel***	Set this parameter based on the network plan.

The following table describes the parameters to be set on the LTE side to configure gap-assisted measurement.

Parameter Name	Parameter ID	Setting Notes
SSB Period	**NrNFreq**.***SsbPeriod***	Set this parameter to the same value as the **NRDUCell**.***SsbPeriod*** parameter.
TDD Frame Offset	**ENodeBFrameOffset**.***TddFrameOffset***	Set this parameter based on the network plan.
FDD Frame Offset	**ENodeBFrameOffset**.***FddFrameOffset***	Set this parameter based on the network plan.
Frame Offset	**CellFrameOffset**.***FrameOffset***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	**EnodeBAlgoExtSwitch**.***MultiNetworkingOptionOptSw***	To prevent the eNodeB from delivering gap configurations for events B1 and B2, select the **NR_B1_NO_GAP_SW** option.

The following table describes the parameters that must be set on the NR side to configure a gNodeB mobility control list.

Parameter Name	Parameter ID	Setting Notes
Mobility Control List ID	**gNBMobilityCtrlList**.***MobilityCtrlListId***	Set this parameter based on the network plan.
gNodeB Start ID	**gNBMobilityCtrlList**.***gNodeBStartId***	Set this parameter based on the network plan.
gNodeB End ID	**gNBMobilityCtrlList**.***gNodeBEndId***	Set this parameter based on the network plan.
Mobile Country Code	**gNBMobilityCtrlList**.***Mcc***	Set this parameter based on the network plan.
Mobile Network Code	**gNBMobilityCtrlList**.***Mnc***	Set this parameter based on the network plan.

The following table describes the optimization parameters to be set on the NR side.

Parameter Name	Parameter ID	Setting Notes
Necessary HO Prepare Fail Punish Timer	**gNBMobilityCommParam**.***NecHoPrepFailPunishTimer***	Use the recommended value.
Optimized HO Prepare Fail Punish Timer	**gNBMobilityCommParam**.***OptHoPrepFailPunishTimer***	Use the recommended value.
Non-Res HO Prepare Fail Punish Times	**gNBMobilityCommParam**.***NonResHoPrepFailPunishTimes***	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Measurement Algorithm Switch	**NRDUCellMeas**.***MeasAlgoSwitch***	It is recommended that the **GAP_AVOID_NR_RES_OPT_SW** option be selected in scenarios with significant channel quality variation to enable the "optimized non-overlapping between gap and NR resources" function.
Compatibility Algorithm Switch	**gNodeBParam**.***CompatibilityAlgoSwitch***	It is recommended that the **GAP_SCH_COMP_SW** option be selected in scenarios where there are UEs with compatibility issues to enable gap-assisted scheduling compatibility.
SSB Measurement Policy Switch	**NRDUCellMeas**.***SsbMeasPolicySwitch***	It is recommended that the **SSB_MEAS_WITHOUT_GAP_SW** option be selected to enable non-gap-assisted measurements for active BWP in multi-frequency coverage scenarios where there is a large proportion of CA UEs capable of non-gap-assisted measurements.
Measurement Algorithm Switch	**gNBMobilityCommParam**.***MeasAlgoSwitch***	Select the **MEAS_ID_PREEMPTION_SW** option to enable measurement ID preemption.
Handover Compatibility Switch	**NRCellAlgoSwitch**.***HoCompatibilitySwitch***	It is recommended that the **PROTOCOL_IE_OPT_SW** option be selected to enable optimization of the *deriveSSB-IndexFromCell* IE.
SSB Consolidation Threshold	**NRCellMobilityConfig**.***SsbConsolidationThld***	Use the recommended value.
SSB Number To Average	**NRCellMobilityConfig**.***SsbNumToAverage***	Use the recommended value.
Beam RSRP Filter Coefficient	**NRCellMobilityConfig**.***BeamRsrpFilterCoeff***	Use the recommended value.
Cell RSRP Filter Coefficient	**NRCellMobilityConfig**.***CellRsrpFilterCoeff***	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
No Handover Flag	**NRCellRelation**.*NoHoFlag*	The gNodeB filters out neighboring cells for which this parameter is set to **FORBID_HO**.
Mobility Algorithm Switch	**gNBMobilityCommParam**.*MobilityAlgoSwitch*	It is recommended that the **MOBILITY_RESTRICTION_SW** option be selected if the INITIAL CONTEXT SETUP REQUEST message sent from the core network includes the *Mobility Restriction List* IE.
Frequency Interference Flag	**NRCellFreqRelation**.*FreqIntrfFlag*	Set this parameter based on the network plan.
Frequency Control Switch	**NRCellFreqRelation**.*FrequencyCtrlSwitch*	To prevent the base station from delivering measurement configurations related to a non-serving frequency, select the **MOB_NEIGHBOR_FREQ_FILTER_SW** option of this parameter for the frequency.
Protocol Compatibility Switch	**gNBMobilityCommParam**.*ProtocolCompatibilitySw*	For a cell with the **NRCellRelation.***CellIndividualOffset* parameter set to a value other than **0**, it is recommended that the **CIO_SPEC_COMPAT_SW** option be selected if a measurement object contains more than eight intra-RAT intra-frequency or inter-frequency neighboring cells whose CIO information needs to be delivered.

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Extend Switch	**NRCellMobilityConfig**.***InterFreqHoExtSwitch***	In certain scenarios, to enable the gNodeB to exclude ground neighboring cells with SINRs below a specific threshold for specific UEs, you are advised to select the **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig**.***InterFreqHoExtSwitch*** parameter, and also select the **RSRP_SINR_HO_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter for any bearer of the UEs.
Service Differentiation Algorithm Switch	**gNBQciBearer**.***ServiceDiffAlgoSwitch***	
RSRP and SINR-based Nec HO SINR Thld	**NRCellInterFHoMeaGrp**.***RsrpSinrBasedNecSinrThld***	Set this parameter based on the network plan. For necessary handovers, a larger value of this parameter results in a lower probability of initiating handovers, while a smaller value results in a higher probability of initiating handovers.
RSRP and SINR-based Unec HO SINR Thld	**NRCellInterFHoMeaGrp**.***RsrpSinrBasedUnecSinrThld***	Set this parameter based on the network plan. For unnecessary handovers, a larger value of this parameter results in a lower probability of initiating handovers, while a smaller value results in a higher probability of initiating handovers.
Handover Compatibility Switch	**NRCellAlgoSwitch**.***HoCompatibilitySwitch***	Select the **INTER_GNB_HO_FULL_CONFIG_SW** option to enable full configuration during inter-gNodeB handovers.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, configure neighbor relationships and mobility-related parameters on the NR side. For details, see the "Using MML Commands" section under the "Basic Functions of Mobility Management in Connected Mode" chapter in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

Activation Command Examples

On the eNodeB side

```
//Adding a neighboring NR frequency
ADD NRNFREQ: LocalCellId=21, DlArfcn=636666, UlArfcnConfigInd=NOT_CFG, SsbOffset=0,
SsbPeriod=20MS, SubcarrierSpacing=30KHZ;
//Specifying a frequency band and an additional frequency band for the neighboring NR frequency whose
NR-ARFCN falls within the range of 158200-160600, 285400-303400, 386000-398000, 402000-404000,
422000-434000, 514000-537999, 620000-653333, or 2054166-2084999.
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=636666, FrequencyBand=N77, AdditionalFrequencyBand=N78;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="262", Mnc="01", GnodebId=255, CellId=1, DlArfcn=636666,
UlArfcnConfigInd=NOT_CFG, PhyCellId=1, Tac=1, NrNetworkingOption=NSA, FrequencyBand=N77,
AdditionalFrequencyBand=N78, MasterPlmnReservedFlag=TRUE;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=21, Mcc="262", Mnc="01", GnodebId=255, CellId=1;
//Adding a PCC frequency
ADD PCCFREQCFG: PccDlEarfcn=1500, NsaPccAnchoringPriority=1;
//Adding an NR SCG frequency
ADD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=636666, ScgDlArfcnPriority=6,
NsaDcB1ThldRsrp=-105, NrB1TimeToTrigger=40MS;
```

On the gNodeB side

```
//Setting the blind handover flag for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
BlindHoFlag=COV_BASED_BLIND_HO_FLAG-1;
//Enabling coverage-based inter-frequency blind handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=COVERAGE_BASED_BLIND_HO-1;
//Configuring a measurement parameter group for intra-frequency handovers
MOD NRCELLINTRAFHOMEAGRP: NrCellId=0, IntraFreqHoMeasGroupId=0, IntraFreqHoA3Offset=2,
IntraFreqHoA3Hyst=2, IntraFreqHoA3TimeToTrig=320MS;
//Configuring a measurement parameter group for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, CovInterFreqA5RsrpThld1=-110, CovInterFreqA5RsrpThld2=-106,
CovInterFreqA2RsrpThld=-110, CovInterFreqA2RsrqThld=-30, SsbSinrInterFreqHoA2Thld=-10,
CovInterFreqA1RsrpThld=-106, CovInterFreqA1RsrqThld=-26, CovInterFreqA5RsrqThld2=-24,
CovInterFreqA5SinrThld2=-4, SsbSinrInterFreqHoA1Thld=-6, A3InterFreqHoA1RsrpThld=-96,
A3InterFreqHoA2RsrpThld=-100, InterFreqHoA3Offset=2;
//Configuring a QCI-specific measurement parameter group
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, IntraFreqHoMeasGroupId=0, InterFreqHoMeasGroupId=0;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=9, PriorityLevel=7, QciPriorityForHo=7;
//(Optional) Adding a gNodeB mobility control list
ADD GNBMOBILITYCTRLLIST: MobilityCtrlListId=2, gNodeBStartId=3, gNodeBEndId=4, Mcc="302",
Mnc="220";
//Configuring penalty parameters for handover preparation failures
```



```
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=0, OptHoPrepFailPunishTimer=0,
NonResHoPrepFailPunishTimes=10;
//(Optional) Selecting the GAP_AVOID_NR_RES_OPT_SW option in scenarios with significant channel
quality variation
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=GAP_AVOID_NR_RES_OPT_SW-1;
//(Optional) Enabling gap-assisted scheduling compatibility
MOD GNODEBPARAM: CompatibilityAlgoSwitch=GAP_SCH_COMP_SW-1;
//(Optional) Enabling non-gap-assisted measurements for active BWP
MOD NRDUCELLMEAS: NrDuCellId=0, SsbMeasPolicySwitch=SSB_MEAS_WITHOUT_GAP_SW-1;
//(Optional) Selecting the MEAS_ID_PREEMPTION_SW option
MOD GNBMOBILITYCOMMPARAM: MeasAlgoSwitch=MEAS_ID_PREEMPTION_SW-1;
//(Optional) Selecting the PROTOCOL_IE_OPT_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=PROTOCOL_IE_OPT_SW-1;
//(Optional) Modifying the configuration for SSB beam measurement result selection and combination
MOD NRCELLMOBILITYCONFIG: NrCellId=0, SsbConsolidationThld=70, SsbNumToAverage=8;
//(Optional) Modifying the measurement filtering configuration
MOD NRCELLMOBILITYCONFIG: NrCellId=0, BeamRsrpFilterCoeff=FC4, CellRsrpFilterCoeff=FC4;
//(Optional) Selecting the MOBILITY_RESTRICTION_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=MOBILITY_RESTRICTION_SW-1;
//(Optional) Enabling non-serving frequency filtering
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77,
FrequencyCtrlSwitch=MOB_NEIGHBOR_FREQ_FILTER_SW-1;
//(Optional) Enabling CIO specification compatibility
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=CIO_SPEC_COMPAT_SW-1;
//(Optional) Enabling RSRP- and SINR-based joint handover
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=RSRP_SINR_HO_SW-1;
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=RSRP_SINR_HO_SW-1;
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, RsrpSinrBasedNecSinrThld=-6,
RsrpSinrBasedUnecSinrThld=-6;
//(Optional) Enabling full configuration during inter-gNodeB handovers
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=INTER_GNB_HO_FULL_CONFIG_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the gNodeB side

```
//Disabling coverage-based inter-frequency blind handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=COVERAGE_BASED_BLIND_HO-0;
//Deselecting the GAP_AVOID_NR_RES_OPT_SW option
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=GAP_AVOID_NR_RES_OPT_SW-0;
//Disabling gap-assisted scheduling compatibility
MOD GNODEBPARAM: CompatibilityAlgoSwitch=GAP_SCH_COMP_SW-0;
//Disabling non-gap-assisted measurements for active BWP
MOD NRDUCELLMEAS: NrDuCellId=0, SsbMeasPolicySwitch=SSB_MEAS_WITHOUT_GAP_SW-0;
//Deselecting the MEAS_ID_PREEMPTION_SW option
MOD GNBMOBILITYCOMMPARAM: MeasAlgoSwitch=MEAS_ID_PREEMPTION_SW-0;
//Deselecting the PROTOCOL_IE_OPT_SW option
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=PROTOCOL_IE_OPT_SW-0;
//Deselecting the MOBILITY_RESTRICTION_SW option
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=MOBILITY_RESTRICTION_SW-0;
//Disabling non-serving frequency filtering
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77,
FrequencyCtrlSwitch=MOB_NEIGHBOR_FREQ_FILTER_SW-0;
//Disabling CIO specification compatibility
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=CIO_SPEC_COMPAT_SW-0;
//Disabling RSRP- and SINR-based joint handover
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=RSRP_SINR_HO_SW-0;
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=RSRP_SINR_HO_SW-0;
//Disabling full configuration during inter-gNodeB handovers
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=INTER_GNB_HO_FULL_CONFIG_SW-0;
```

6.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Coverage-based Intra-Frequency PSCell Change

Coverage-based intra-frequency PSCell change has taken effect if any of the following counters have a value other than 0:

- **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att**, which indicates the number of coverage-based inter-gNodeB intra-frequency PSCell change attempts
- **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Succ**, which indicates the number of successful coverage-based inter-gNodeB intra-frequency PSCell changes
- **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**, which indicates the number of coverage-based intra-gNodeB intra-frequency PSCell change attempts
- **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Succ**, which indicates the number of successful coverage-based intra-gNodeB intra-frequency PSCell changes

Coverage-based Inter-Frequency PSCell Change

Coverage-based inter-frequency PSCell change has taken effect if any of the following counters have a value other than 0:

- **N.NsaDc.InterFreq.Coverage.PSCell.Change.Att**, which indicates the number of coverage-based inter-frequency PSCell change attempts
- **N.NsaDc.InterFreq.Coverage.PSCell.Change.Succ**, which indicates the number of successful coverage-based inter-frequency PSCell changes

Blind inter-frequency PSCell change has taken effect if any of the following counters have a value other than 0:

- **N.NsaDc.InterFreq.Blind.Coverage.PSCell.Change.Att**, which indicates the number of blind inter-frequency PSCell change attempts

- **N.NsaDc.InterFreq.Blind.Coverage.PSCell.Change.Succ**, which indicates the number of successful blind inter-frequency PSCell changes

Currently, no counter can directly measure the number of measurement-based inter-frequency PSCell changes, but the number can be calculated using the following formulas. If any of the results of these formulas is not 0, measurement-based inter-frequency PSCell change has taken effect:

- Number of measurement-based inter-frequency PSCell change attempts = Number of coverage-based inter-frequency PSCell change attempts – Number of blind inter-frequency PSCell change attempts
- Number of successful measurement-based inter-frequency PSCell changes = Number of successful coverage-based inter-frequency PSCell changes – Number of successful blind inter-frequency PSCell changes

Frequency-Priority-based Inter-Frequency PSCell Change

Frequency-priority-based inter-frequency PSCell change has taken effect if any of the following counters have a value other than 0:

- **N.NsaDc.InterFreq.FreqPri.PSCell.Change.Att**, which indicates the number of frequency-priority-based inter-frequency PSCell change attempts
- **N.NsaDc.InterFreq.FreqPri.PSCell.Change.Succ**, which indicates the number of successful frequency-priority-based inter-frequency PSCell changes

Operator-specific-Priority-based Inter-Frequency PSCell Change

After operator-specific-priority-based inter-frequency PSCell change is enabled alone, this function has taken effect if the value of any of the following counters increases:

- **N.NsaDc.InterSgNB.PSCell.Change.Att**, which indicates the total number of inter-gNodeB PSCell change attempts
- **N.NsaDc.IntraSgNB.PSCell.Change.Att**, which indicates the total number of intra-gNodeB PSCell change attempts

Service-based Inter-Frequency PSCell Change

Service-based inter-frequency PSCell change has taken effect if any of the following counters have a value other than 0:

- **N.NsaDc.InterFreq.Service.PSCell.Change.Att**, which indicates the number of service-based inter-frequency PSCell change attempts
- **N.NsaDc.InterFreq.Service.PSCell.Change.Succ**, which indicates the number of successful service-based inter-frequency PSCell changes

Uplink-Interference-based Inter-Frequency PSCell Change

After the uplink-interference-based inter-frequency PSCell change function is enabled, if the value of **HoCause** in the external CHR event PRIVATE_INTRA_RAT_HO_OUT is **INTRF_BASED**, this function has taken effect.

6.4.3 Network Monitoring

Monitor the counters listed in the following table and compare them against the network plan to evaluate the feature performance.

- On the eNodeB side

  After NSA DC is enabled, observe the following counters to determine whether the handover success rate and service drop rate of NSA UEs are different from those of LTE-only UEs:

  - Outgoing handover success rate of NSA UEs = **L.NsaDc.HHO.ExecSuccOut** / **L.NsaDc.HHO.ExecAttOut** x 100%
  - Incoming handover success rate of NSA UEs = **L.NsaDc.PCell.Change.Succ** / **L.NsaDc.PCell.Change.Exec** x 100%
  - Abnormal service drop rate of NSA UEs = **L.NsaDc.E-RAB.AbnormRel** / **L.NsaDc.E-RAB.NormRel** x 100%
- On the gNodeB side
  - Number of inter-site SgNB addition attempts in the LTE-NR NSA DC scenario (**N.NsaDc.InterSite.SgNB.Add.Att**)
  - Number of successful inter-site SgNB additions in the LTE-NR NSA DC scenario (**N.NsaDc.InterSite.SgNB.Add.Succ**)

7 Functions Related to NSA Mobility Management

7.1 Principles

7.1.1 Non-Gap-Assisted B1 Measurement

When an NSA UE performs inter-RAT measurement in NSA scenarios, it requires a gap for NR frequency measurement. As a result, the user-perceived rate decreases.

To reduce the impact of gap-assisted measurement on data rates, non-gap-assisted measurement of NR frequencies is allowed for specific UEs in LTE-only mode. This function is controlled by the **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter.

- If this option is selected, non-gap-assisted B1 measurement of a frequency not in blacklisted combinations takes effect for specific UEs.
- If this option is deselected, non-gap-assisted B1 measurement does not take effect.

Blacklist for Non-Gap-Assisted Measurement

After the **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected, a blacklist can be specified by the **B1_NO_GAP_FREQ_COMB_ID_*X*** option of the **UeCompat**.***LnrNoGapFreqCombBlacklist*** parameter. ***X*** equals the value of an **LnrFreqComb**.***FreqCombId*** parameter, which indicates the ID of an LTE and NR frequency combination. When the option is selected for an ID specified by the **LnrFreqComb**.***FreqCombId*** parameter, non-gap-assisted B1 measurements cannot be performed. Under this setting, an NSA UE cannot perform non-gap-assisted B1 measurement of the NR frequency specified by the corresponding **LnrFreqComb**.***DlNrArfcn*** parameter when the UE is working on the LTE frequency specified by the corresponding **LnrFreqComb**.***DlEarfcn*** parameter.

Assume that an NSA UE does not support non-gap-assisted measurement of NR-ARFCN 636666 when it is working on EARFCN 1500. Then, this frequency combination needs to be blacklisted. Specifically:

- Select the **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter.
- Select the **B1_NO_GAP_FREQ_COMB_ID_0** option of the **UeCompat**.***LnrNoGapFreqCombBlacklist*** parameter.
- Set the **LnrFreqComb**.***FreqCombId*** parameter to **0**.
- Set the **LnrFreqComb**.***DlEarfcn*** parameter to **1500**.
- Set the **LnrFreqComb**.***DlNrArfcn*** parameter to **636666**.

Whitelist for Non-Gap-Assisted Measurement

An LTE-only or EN-DC band combination can be whitelisted in NSA networking for specific UEs so that these UEs can perform non-gap-assisted measurement of the whitelisted band combination.

A whitelist is specified when the **ENDC_GAP_FREE_COMB_ID_*X*** option of the **UeCompat**.***WhitelistGapFreeCombSwitch*** parameter is selected. ***X*** equals the value of an **EndcBandGapFreeCfg**.***EndcGapFreeCombId*** parameter, which indicates the ID of an EN-DC band combination for which non-gap-assisted measurements can be performed.

A maximum of 32 band combinations supporting non-gap-assisted measurement can be configured and take effect accordingly. There are four scenarios as follows:

- Non-gap-assisted measurement of an LTE band by UEs in LTE-only mode
- Non-gap-assisted measurement of an NR band by UEs in LTE-only mode
- Non-gap-assisted measurement of an LTE band by UEs in EN-DC mode
- Non-gap-assisted measurement of an NR band by UEs in EN-DC mode

After setting the **EndcBandGapFreeCfg**.***EndcGapFreeCombId*** parameter, configure each band in the band combination using the following parameters:

- **EndcBandGapFreeCfg**.***BandType***, which specifies the band type. It can be an LTE band, NR band, LTE band of which a UE can perform non-gap-assisted measurement, or NR band of which a UE can perform non-gap-assisted measurement.
- **EndcBandGapFreeCfg**.***BandIndex***, which specifies the index of a frequency band. The maximum number of indexes that can be configured for LTE bands, NR bands, LTE bands of which a UE can perform non-gap-assisted measurement, and NR bands of which a UE can perform non-gap-assisted measurement is 8, 4, 1 and 1, respectively.
- **EndcBandGapFreeCfg**.***FrequencyBandId***, which specifies the ID of a frequency band. For details about the IDs of LTE and NR frequency bands, see 3GPP TS 36.101 and 3GPP TS 38.101.

Assume that an NSA UE supports an EN-DC band combination including LTE band 3, LTE band 10, NR band 14, and NR band 41 of which the UE can perform non-gap-assisted measurement. In this case, set the **EndcBandGapFreeCfg**.***EndcGapFreeCombId*** parameter for each band to **0** first and then set other parameters by referring to **Table 7-1**:

Table 7-1 Example of configuring a band combination of which a UE can perform non-gap-assisted measurement

EndcBandGapFreeCfg.*EndcGapFreeCombId*	EndcBandGapFreeCfg.*BandType*	EndcBandGapFreeCfg.*BandIndex*	EndcBandGapFreeCfg.*FrequencyBandId*
0	LTE	1	3
0	LTE	2	10
0	NR	9	14
0	GAP_FREE_NR_BAND	14	41

After the preceding configurations are completed, select the **ENDC_GAP_FREE_COMB_ID_0** option of the **UeCompat**.***WhitelistGapFreeCombSwitch*** parameter. Then the UE can perform non-gap-assisted measurement of this band combination.

7.1.2 Gap Sharing

For an NSA UE in the DC state, if LTE inter-frequency, LTE inter-RAT, or NR inter-frequency measurement gaps are configured on the LTE side, inter-frequency measurement gaps are also used on the NR side. In this case, the UE performs inter-frequency measurement and NR intra-frequency measurement. If the SMTC period for NR intra-frequency measurement is longer than or equal to the inter-frequency measurement gap, NR intra-frequency measurement cannot be performed during LTE inter-frequency or inter-RAT measurement. To enable the NSA UE to perform NR intra-frequency measurement during LTE inter-frequency or inter-RAT measurement, the gap sharing function is introduced, as described in section 7.2 "Measurements" in 3GPP TS 37.340 V15.4.0.

- When the **NsaDcMgmtConfig**.***MeasGapSharingScheme*** parameter is set to **EQUAL_SPLITTING**, **25_PERCENT**, **50_PERCENT**, or **75_PERCENT**, the eNodeB configures the specified gap sharing proportion for the UE and the UE performs gap sharing based on this proportion.
- When the **NsaDcMgmtConfig**.***MeasGapSharingScheme*** parameter is set to **NO_SHARING**, the eNodeB does not configure a gap sharing proportion for the UE.

For details about gap-assisted measurement, see **6.1.1.4.5 Measurement Gap Configuration** and *Mobility Management in Connected Mode* in eRAN feature documentation.

7.1.3 Measurement-based Simultaneous LTE and NR Cell Changes

This function allows a cell change on the LTE side and an intra-frequency cell change on the NR side to be performed simultaneously and instructs the UE through an RRC reconfiguration message. In LTE and NR co-coverage scenarios, this function prevents an immediate SCG change on the NR side after an LTE handover, thereby preventing NR data transmission from being interrupted again.

The MeNB determines whether to trigger a coverage-based intra- or inter-frequency handover on the LTE side based on the UE-reported event A3, A4, or A5 measurement results on the LTE side. If the **NsaDcLteMeasCtrlwithNbrMeasSw** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter is selected, the MeNB includes a *reportAddNeighMeas* IE in the delivered LTE measurement configurations and the UE reports the measurement results of the serving NR cell and intra-frequency neighboring NR cells when reporting the LTE measurement results. Otherwise, the UE does not do so. The MeNB determines whether to trigger an intra-frequency neighboring NR cell change based on the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter and the measurement results of the serving NR cell and intra-frequency neighboring NR cells, which are reported along with the LTE measurement results. If the source serving NR cell and intra-frequency neighboring NR cells are intra-gNodeB cells, the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side needs to be selected. If the option is not selected, the source serving NR cell will not change during the MeNB cell change.

- If the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option is selected, an intra-frequency neighboring NR cell can be selected as a target NR cell during the MeNB cell change when the measurement result of the neighboring NR cell minus that of the source serving NR cell is greater than the threshold specified by the **NrScgFreqConfig**.***NsaDcLteNrSimulHoThld*** parameter.
- If the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option is deselected, the MeNB does not check the difference between the measurement results of an intra-frequency neighboring NR cell and the source serving NR cell. Instead, it selects an intra-frequency neighboring NR cell with the highest RSRP as a target NR cell. If the cell with the highest RSRP is the source serving NR cell, this cell remains unchanged during the MeNB cell change.

Measurement-based simultaneous LTE and NR cell changes include intra- or inter-eNodeB cell changes and intra- or inter-gNodeB cell changes. Table 7-2 describes the supported changes and the conditions for the changes to take effect.

Table 7-2 Supported measurement-based simultaneous LTE and NR cell changes and conditions for the changes to take effect

Scenario	Intra-gNodeB Cell Change	Inter-gNodeB Cell Change
Intra-eNodeB Cell Change	Supported. The configuration requirements on both the source MeNB and the SgNB must be met.	Supported. The configuration requirements on only the source MeNB need to be met.

Scenario	Intra-gNodeB Cell Change	Inter-gNodeB Cell Change
Inter-eNodeB Cell Change	Supported. The configuration requirements on all the source MeNB, the target MeNB, and the SgNB must be met.	Supported. The configuration requirements on both the source MeNB and the target MeNB must be met.

If measurement-based simultaneous LTE and NR cell changes need to be supported in an area and the target neighboring NR cells meet configuration requirements, it is recommended that all base stations in this area meet the following configuration requirements:

- On the source MeNB, both the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter and the **NsaDcLteMeasCtrlwithNbrMeasSw** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter are selected.
- On the target MeNB, the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is selected.
- On the NR side, the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter is selected.

NOTE

- In the current version, a target NR cell can be selected based only on RSRP measurement results during an LTE cell handover with an NR cell change. This requires that the **NRCellMobilityConfig**.***MeasTrigQuantity*** parameter be set to **RSRP** in NSA networking. Otherwise, if the UE reports RSRQ-based event A2 immediately after the NR cell change, the SgNB will immediately trigger a coverage-based inter-frequency handover.
- Assume that there are blind-configurable intra-frequency neighboring NR cells that are co-sited with the source serving NR cell. If the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter on the MeNB side is deselected, the MeNB filters out these blind-configurable cells. If this option is selected, the MeNB does not filter out these blind-configurable cells.

7.1.4 Band Combination Change Triggered by NR Inter-Frequency Handover

Band combination change triggered by NR inter-frequency handover is supported in the following three NSA scenarios:

Scenario 1: If the signal quality of the PSCell on the NR side is poor and the SgNB determines that the UE does not support EN-DC between a neighboring NR frequency and the current NSA anchor frequency, band combination change triggered by NR inter-frequency handover is controlled by the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter on the NR side. The implementation is as follows:

- If this option is selected, the SgNB delivers coverage-based A2 measurement configurations. After receiving A2 measurement results from the UE, the SgNB sends a proprietary X2 message to the MeNB, and the MeNB triggers NSA PCC anchoring based on NR coverage or NSA carrier combination selection based on user experience (the latter is preferentially triggered if both are enabled).

- If this option is deselected, the SgNB does not deliver coverage-based A2 measurement configurations.

Scenario 2: Assume that the frequency combinations formed by the anchor frequency and the neighboring NR frequencies configured for the PSCell include the EN-DC frequency combinations supported by the UE and those not supported by the UE. If the downlink signal quality of the PSCell is poor and the UE is beyond the coverage area of the neighboring NR frequencies in the EN-DC frequency combinations supported by the UE, then band combination change triggered by NR inter-frequency handover is controlled by the **gNBMobilityCommParam**.***NsaInterFreqMeasWaitTimer*** parameter, the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter, and the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter. The implementation is as follows:

- If the **gNBMobilityCommParam**.***NsaInterFreqMeasWaitTimer*** parameter is set to a non-zero value and both the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter and the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter are selected, the SgNB delivers coverage-based A2 measurement configurations.

  After receiving A2 measurement results from the UE, the SgNB delivers coverage-based A5 measurement configurations for the configured neighboring NR frequencies and starts the timer specified by the **gNBMobilityCommParam**.***NsaInterFreqMeasWaitTimer*** parameter. After receiving A5 measurement results from the UE:

  - If EN-DC between the neighboring NR frequencies indicated in the A5 measurement results and the current anchor frequency is allowed, the SgNB triggers an intra-NR inter-frequency handover. For details about this procedure, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.
  - If EN-DC between the neighboring NR frequencies indicated in the A5 measurement results and the current anchor frequency is not allowed, the SgNB checks whether the timer specified by **gNBMobilityCommParam**.***NsaInterFreqMeasWaitTimer*** has expired. If the timer has expired, the SgNB deletes the measurement configurations of the neighboring NR frequencies that cannot be used with the current anchor frequency for EN-DC, and sends a proprietary X2 message to the MeNB. Then, the MeNB triggers NSA PCC anchoring based on NR coverage or NSA carrier combination selection based on user experience (the latter is preferentially triggered if both are enabled). If the timer does not expire, the SgNB discards the A5 measurement results.

  To prevent gap-assisted NR inter-frequency measurements from affecting the user-perceived rate, the non-gap-assisted coverage-based inter-frequency PSCell change function can be enabled so that the SgNB includes only NR SCell frequencies in coverage-based A5 measurement configurations delivered for neighboring NR frequencies and does not start gap-assisted NR inter-frequency measurements. For details about the non-gap-assisted coverage-based inter-frequency PSCell change function, see **7.1.14 Non-Gap-Assisted Coverage-based Inter-Frequency PSCell Change**.

- If the **gNBMobilityCommParam**.***NsaInterFreqMeasWaitTimer*** parameter is set to **0** or either the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter or the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter is deselected, this function does not take effect.

Scenario 3: Assume that the frequency combinations formed by the anchor frequency and the neighboring NR frequencies configured for the PSCell include the EN-DC frequency combinations supported by the UE and those not supported by the UE. If the uplink signal quality of the PSCell is poor and the UE is beyond the coverage area of the neighboring NR frequencies in the EN-DC frequency combinations supported by the UE, then band combination change triggered by NR inter-frequency handover is controlled by the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter, **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter, and the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter. The implementation is as follows:

- When the **UL_WEAK_COV_UE_PROTECT_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter, the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter, and the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter are all selected, the SgNB delivers A5 measurement configurations to UEs under weak coverage in the uplink and starts a 2s timer. Threshold 1 and threshold 2 for event A5 are specified by the **NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1*** and **NRCellInterFHOMeaGrp**.***CovInterFreqA5RsrpThld2*** parameters, respectively.

  Assume that the SgNB receives A5 measurement results from the UE before the timer expires.

  – If EN-DC between the neighboring NR frequencies indicated in the measurement results and the current anchor frequency is not allowed:

    After the timer expires, the SgNB sends a proprietary X2 message to the MeNB, and the MeNB triggers NSA PCC anchoring based on NR coverage or NSA carrier combination selection based on user experience (the latter is preferentially triggered if both are enabled).

  – If EN-DC between the neighboring NR frequencies indicated in the measurement results and the current anchor frequency is allowed:

    The SgNB triggers an intra-NR inter-frequency handover. For details about this procedure, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.

  If the SgNB does not receive any A5 measurement results before the timer expires, it deletes the A5 measurement configurations.

- This function does not take effect when the **UL_WEAK_COV_UE_PROTECT_SW** option of the

NRCellAlgoSwitch.***MultiFreqAlgoSwitch*** parameter, the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter, or the **NO_NR_COV_EN_DC_RESEL_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter is deselected.

For details about NSA PCC anchoring based on NR coverage and NSA carrier combination selection based on user experience in the preceding three scenarios, see *EPC-based NSA Performance Enhancement* and *EPC-based NSA Experience Improvement*, respectively.

NOTE

The values of the **N.NsaDc.IntraSgNB.PSCell.Change.Fail.Conflict** and **N.NsaDc.InterSgNB.PSCell.Change.Fail.Conflict** counters increase when the MeNB responds to the SgNB with the SGNB MODIFICATION REFUSE and SGNB CHANGE REFUSE messages upon reception of the SGNB MODIFICATION REQUIRED and SGNB CHANGE REQUIRED messages from the SgNB for a band combination change triggered by NR inter-frequency handover during LTE PCC anchoring.

It is recommended that the **NrScgFreqConfig**.***NsaDcB1ThldRsrp*** parameter be set to a value greater than **NRCellInterFHoMeaGrp**.***CovInterFreqA2RsrpThld*** by over 3 dB. Otherwise, the UE will measure and send measurement results of the SCG frequency that is selected for DC with the current anchor in scenarios 1 and 2, and the eNodeB will then select the current EN-DC combination as the target EN-DC combination. As a result, ping-pong switching of band combinations will occur.

It is recommended that the **NrScgFreqConfig**.***NsaDcB1ThldRsrp*** parameter be set to a value greater than **NRCellInterFHoMeaGrp**.***UlCovInterFreqA5RsrpThld1*** by over 3 dB. Otherwise, the UE will measure and send measurement results of the SCG frequency that is selected for DC with the current anchor in scenario 3, and the eNodeB will then select the current EN-DC combination as the target EN-DC combination. As a result, ping-pong switching of band combinations will occur.

7.1.5 User-Experience-based and UE-Traffic-Model-based SCG Releases

In NSA networking, SCG release can be triggered on the NR side to improve user experience when specific conditions are met. Three functions are involved: user-experience-based SCG release, user-experience-based SCG release enhancement, and UE-traffic-model-based SCG release.

The **NsaDcMgmtConfig**.***ScgAddPenaltyDuration*** parameter can be set to a non-zero value on the LTE side to prevent an SCG from being added within the penalty duration after the SCG is released on the NR side. This prevents ping-pong SCG addition and release.

The following provides more details on the SCG addition penalty duration:

- The SCG addition penalty duration does not take effect if RRC connection reestablishment or handover occurs on the LTE side after the SCG is released.
- For user-experience-based SCG release, if the software version of the LTE side is earlier than eRAN18.1 or the SCG addition penalty duration is set to 0, ping-pong SCG addition and release may occur. To reduce this risk, it is recommended that the **NsaDcMgmtConfig**.***ScgAdditionInterval*** parameter be set to a large value.

User-Experience-based SCG Release

User-experience-based SCG release is enabled when the **USER_EXP_SCG_REL_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter is selected. The NR side triggers an SCG release when any of the following conditions is met:

- The uplink SRS SINR of a UE is less than the threshold specified by **NRDUCellNsaDcConfig**.***ScgRelUlSrsSinrThld*** for 200 ms a consecutive of *X* times (*X* is specified by **NRDUCellNsaDcConfig**.***UserExpScgRelMeasNum***).
- The downlink spectral efficiency calculated based on the CQI and RI of a UE is less than the spectral efficiency corresponding to the threshold specified by **NRDUCellNsaDcConfig**.***ScgRelDlCqiThld*** for 200 ms a consecutive of *X* times (*X* is specified by **NRDUCellNsaDcConfig**.***UserExpScgRelMeasNum***). For the mapping between CQIs and spectral efficiencies, see "Table 5.2.2.1-3: 4-bit CQI Table 2" in 3GPP TS 38.214.

User-Experience-based SCG Release Enhancement

User-experience-based SCG release enhancement is enabled when the **UL_DL_USER_EXP_SCG_REL_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter is selected. When the downlink SSB RSRP of a UE is less than the threshold specified by **NRCellNsaDcConfig**.***UserExpScgRelA2RsrpThld***, the NR side will trigger an SCG release when any of the following conditions is met:

- The uplink SRS SINR of the UE is less than the threshold specified by **NRDUCellNsaDcConfig**.***ScgRelUlSrsSinrThld*** for 200 ms a consecutive of *X* times (*X* is specified by **NRDUCellNsaDcConfig**.***UserExpScgRelMeasNum***).
- The downlink spectral efficiency calculated based on the CQI and RI of the UE is less than the spectral efficiency corresponding to the threshold specified by **NRDUCellNsaDcConfig**.***ScgRelDlCqiThld*** for 200 ms a consecutive of *X* times (*X* is specified by **NRDUCellNsaDcConfig**.***UserExpScgRelMeasNum***). For the mapping between CQIs and spectral efficiencies, see "Table 5.2.2.1-3: 4-bit CQI Table 2" in 3GPP TS 38.214.

During the preceding evaluation, if the downlink SSB RSRP of the UE is greater than the threshold specified by **NRCellNsaDcConfig**.***UserExpScgRelA1RsrpThld***, the NR side stops evaluating the uplink SRS SINR and downlink spectral efficiency. It restarts the evaluation when the downlink SSB RSRP of the UE is less than the **NRCellNsaDcConfig**.***UserExpScgRelA2RsrpThld*** parameter value again.

NOTE

- User-experience-based SCG release and user-experience-based SCG release enhancement are applicable when there is only one frequency on the NR side. If there are multiple frequencies on the NR side, the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter needs to be selected. This ensures that inter-frequency SCG changes have been attempted on the NR side before the SCG is released.
- Uplink fallback to LTE does not take effect when the value of **NRDUCellNsaDcConfig**.***ScgRelUlSrsSinrThld*** is greater than the value of **NRDUCellSrsMeas**.***NsaUlFackToLteSinrThld***.

UE-Traffic-Model-based SCG Release

For a UE in weak NR coverage areas, if the uplink transmit power on the LTE side in NSA scenarios with an SCG added is lower than that in the LTE-only state, the

NSA UE may have a lower uplink user-perceived rate than in the LTE-only state. UE-traffic-model-based SCG release is recommended to prevent the preceding issue.

UE-traffic-model-based SCG release is enabled when all the following functions are enabled:

- The **uplink large-packet service detection** function is enabled on the LTE side.
- The function of **notification of the uplink large-packet status of uplink fallback UEs to NR** is enabled on the LTE side. This function enables the eNodeB to notify the NR side of the uplink large-packet status of UEs involved in uplink fallback to LTE.
- The **traffic-volume-based NR SCG release** function is enabled on the NR side.

After UE-traffic-model-based SCG release is enabled, the SgNB directly initiates an SCG release for an NSA UE performing uplink large-packet services and downlink small-packet services when the uplink SINR on the NR side deteriorates. The uplink SINR on the NR side is detected at a certain **interval** and is considered as deteriorating if it is less than the **traffic-volume-based SCG release UL SINR threshold** five consecutive times. In the uplink, which side performs uplink large-packet service detection varies depending on whether a UE is involved in uplink fallback to LTE. If the UE is involved in uplink fallback to LTE, detection is performed by the LTE side, which then notifies the NR side of the detection result. If the UE is not involved in uplink fallback to LTE, detection is performed by the NR side. In the downlink, small-packet service detection is performed by the NR side. For details about uplink fallback to LTE, see *EPC-based NSA Basics*.

Table 7-3 lists the parameters related to the UE-traffic-model-based SCG release function.

Table 7-3 Parameters related to the UE-traffic-model-based SCG release function

Configuration Item	Parameter
Uplink large-packet service detection	**UL_LARGE_PKT_SERV_DECT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter
Notification of the uplink large-packet status of uplink fallback UEs to NR	**ULFB_UL_LRG_PKT_FLAG_TO_NR_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter
Traffic-volume-based NR SCG release	**TRF_VOL_BASED_NR_SCG_REL_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter
Interval	**NRDUCellSrsMeas**.***NsaUlToLteSinrTimeToTrig***
Traffic-volume-based SCG release UL SINR threshold	**NRCellNsaDcConfig**.***TfcBasedScgRelUlSinrThld***

7.1.6 Load Balancing

The intra-LTE load balancing mechanisms differ between NSA UEs and LTE-only UEs. The target frequency to which an NSA UE is handed over must be an NSA anchor frequency (that is, the anchoring priority must not be 0). If the target frequency is not an anchor frequency, no handover procedure will be initiated for the NSA UE.

To prevent NSA UEs from being handed over from anchor frequencies to non-anchor frequencies, inter-RAT load balancing does not take effect for NSA UEs.

For details about LTE load balancing, see *Intra-RAT Mobility Load Balancing* and *Inter-RAT Mobility Load Balancing* in eRAN feature documentation.

7.1.7 MBOCS

If the multi-band optimal carrier selection (MBOCS) feature is enabled on the LTE side but not enabled for NSA UEs, the **MBOCS_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter can be deselected to disable the MBOCS feature for NSA UEs, thereby preventing NSA UEs from being handed over from anchor frequencies to non-anchor frequencies. For details about MBOCS, see *Multi-band Optimal Carrier Selection* in eRAN feature documentation.

7.1.8 Spectrum Coordination

If spectrum coordination is enabled on the LTE side but not enabled for NSA UEs, the **SPCT_COORD_INTER_FREQ_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter can be deselected to disable spectrum coordination for NSA UEs, thereby preventing NSA UEs from being handed over from anchor frequencies to non-anchor frequencies. For details about spectrum coordination, see *LTE Spectrum Coordination* in eRAN feature documentation.

7.1.9 Intra-E-UTRAN Inter-eNodeB S1-based Handover Optimization

During an intra-E-UTRAN inter-eNodeB S1-based handover, if the size of a handover request message is so large that it exceeds the specification of the core network, the message may be truncated or rejected by the core network, leading to a handover failure. To avoid the handover failure, intra-E-UTRAN inter-eNodeB S1-based handover optimization is introduced. This function allows the eNodeB to control the size of the UE capability field carried in an S1-based handover request message by setting the **EnodebAlgoExtSwitch**.***LteS1HoReqUeCapbSize*** parameter based on the capability of the core network.

The processing mechanism of this function is as follows:

- If the **EnodebAlgoExtSwitch**.***LteS1HoReqUeCapbSize*** parameter is set to a value other than **0**, this function takes effect. Before initiating an S1-based handover request, the eNodeB checks whether the size of the UE capability field (indicating the EN-DC, NR, E-UTRAN, UTRAN, and GERAN capabilities) reported by the UE exceeds the parameter value.

- If the size of the UE capability field exceeds the parameter value, the S1-based handover request message contains only the E-UTRAN capability information.
- If the size of the UE capability field does not exceed the parameter value, the S1-based handover request message contains the complete UE capability field.

- If the **EnodebAlgoExtSwitch**.***LteS1HoReqUeCapbSize*** parameter is set to **0**, this function does not take effect. The S1-based handover request message contains the complete UE capability field.

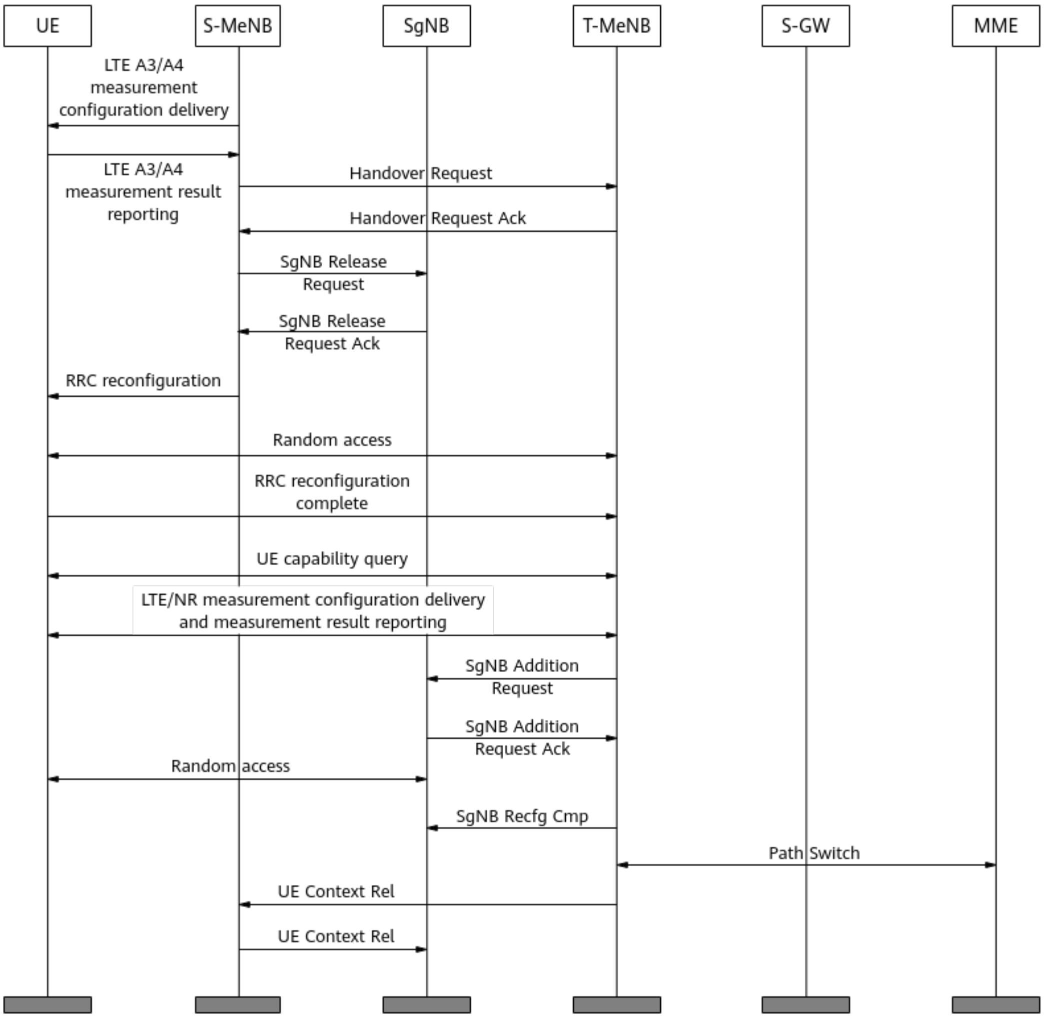
7.1.10 Optimized Inter-eNodeB Handover with SCG Addition

During an inter-eNodeB handover with SCG addition (**8.4 MeNB-Initiated Inter-MeNB Handover Without an SgNB Change**), the source eNodeB sends the target eNodeB a handover request message containing an *appliedFreqBandListFilter* IE in the UE capability information. If the LTE and NR frequency band combinations indicated by the *appliedFreqBandListFilter* IE are less than those to be queried by the target eNodeB, the target eNodeB cannot configure EN-DC frequency band combinations beyond the scope indicated by the IE for the UE. As a result, user experience is not optimal.

The optimized inter-eNodeB handover with SCG addition function is introduced to address this issue. This function is controlled by the **LTE_HO_WITH_SCG_ADD_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter. The function works as follows to configure an EN-DC band combination with optimal user experience for the UE:

When the **LTE_HO_WITH_SCG_ADD_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter is selected, this function takes effect. With this function, if the LTE and NR frequency band combinations indicated by the *appliedFreqBandListFilter* IE are less than those to be queried by the target eNodeB, the source eNodeB initiates a handover without SCG addition. After the target eNodeB receives an RRC reconfiguration complete message (indicating that the UE handover is complete), it initiates a UE capability query again based on its frequency band configurations. After receiving the EN-DC capabilities reported by the UE, the target eNodeB adds an SCG in a blind manner or based on measurement. For details about the signaling procedure, see **Figure 7-1**.

Figure 7-1 Optimized inter-eNodeB handover with SCG addition



If the **LTE_HO_WITH_SCG_ADD_OPT_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter is deselected, the target eNodeB adds an SCG during handover preparation. For details about the signaling procedure, see **Figure 8-4**.

7.1.11 Fast SCG Addition for NSA DC UEs in LTE and NR Co-Site Co-Coverage Scenarios

A large inter-site distance may result in discontinuous C-band coverage (for example, on high-speed railways). When LTE and NR cells are co-sited and provide the same coverage in such scenarios, SCGs may fail to be added in a timely manner for NSA-DC-capable UEs after the UEs are handed over to LTE anchor cells. To address this issue, Huawei eNodeBs now support fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios. This function enables NSA-DC-capable UEs to quickly detect NR coverage and have SCGs added after being handed over to LTE anchor cells, improving the user-perceived rate of CEUs.

This function is controlled by the **NsaDcMgmtConfig**.***NsaDcFastScgAddA1RsrpThld*** parameter. This function takes effect for a UE when this parameter is set to a value other than **255** and all the following conditions are met:

- The **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected for the LTE anchor cell.
- The UE is capable of NSA DC. That is, the capability information reported by the UE contains the **en-DC-r15:supported** field.
- The UE is not blacklisted for NSA DC.

After the UE is handed over to the LTE anchor cell, the eNodeB processes as follows depending on whether the anchor cell supports blind SCG addition (that is, whether the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected):

- If the anchor cell supports blind SCG addition, the eNodeB delivers event A1 measurement configurations of the serving cell to the UE. After receiving an event A1 measurement report from the UE, the eNodeB performs SCG addition based on the blind PSCell configuration. For details about blind PSCell configuration, see *EPC-based NSA Performance Enhancement*.
- If the anchor cell does not support blind SCG addition, the eNodeB delivers event B1 measurement configurations following a measurement-based PSCell configuration procedure. At the same time, the eNodeB delivers event A1 measurement configurations of the serving cell. For details about measurement-based PSCell configuration, see *EPC-based NSA Basics*. After the eNodeB delivers the event B1 measurement configurations, it starts a timer, with a length specified by the **NrScgFreqConfig**.***NrB1ReportWaitingTimer*** parameter.
  - If the eNodeB receives an event B1 measurement report before the timer expires, it adds an SCG based on the measurement report and deletes the events A1 and B1 measurement configurations.
  - If the eNodeB does not receive an event B1 measurement report but receives an event A1 measurement report before the timer expires, it does not process the measurement report and deletes the event A1 measurement configurations.
  - If the eNodeB does not receive any event B1 or A1 measurement report before the timer expires, it deletes the event B1 measurement configurations upon timer expiry. If, after the timer expires, it receives an event A1 measurement report, it deletes the event A1 measurement configurations and immediately delivers event B1 measurement configurations.

  This procedure does not affect the existing periodic SCG addition. For details about periodic SCG addition, see *EPC-based NSA Basics*.

NOTE

This function will not be triggered during intra- or inter-eNodeB handovers with SCG addition.

7.1.12 SCG Addition Only After NSA PCC Anchoring

In NSA networking, an SCG may be added for an NSA-capable UE during NSA PCC anchoring. If the anchoring fails, the SCG addition success rate decreases. The SCG addition only after NSA PCC anchoring function allows the eNodeB to add an SCG for an NSA-capable UE only after NSA PCC anchoring succeeds. This prevents SCG addition failures caused by anchoring failures as well as RRC connection reestablishments caused by UE compatibility issues. The specific scenarios are as follows:

- If an NSA-capable UE is in the LTE-only state before NSA PCC anchoring, the SCG addition policy is specified by the **SCG_ADD_AFTER_HANDOVER_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter.
  – If this option is selected, the eNodeB adds an SCG after PCC anchoring succeeds for the UE.
  – If this option is deselected, the eNodeB adds an SCG during PCC anchoring.
- For an NSA-capable UE in the NSA state before NSA PCC anchoring, the SCG addition policy is specified by the **NsaDcAlgoParam**.***NrScgAddAfterHoPolicy*** parameter.
  – If this parameter is set to **FAST_ADD**, the eNodeB does not add an SCG during PCC anchoring; when PCC anchoring is complete, the original SCG can be added without measurement once SCG addition is triggered.
  – If this parameter is set to **NORMAL_ADD**, the eNodeB does not add an SCG during PCC anchoring; when PCC anchoring is complete, blind or measurement-based SCG addition will be performed once SCG addition is triggered. For details about the preceding SCG addition procedures, see *EPC-based NSA Basics* and *EPC-based NSA Performance Enhancement*.
  – If this parameter is set to **OFF**, the eNodeB adds an SCG during PCC anchoring when an SCG has been added for the UE before the anchoring and the target cell supports SCG addition.

  The whitelist mechanism is supported in this scenario. If the **NR_SCG_ADD_AFTER_HO_OPT_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter is selected, this function takes effect only for whitelisted NSA-capable UEs that are in the NSA state before NSA PCC anchoring. If this option is deselected, this function takes effect for all NSA-capable UEs that are in the NSA state before NSA PCC anchoring.

7.1.13 SCell Configuration Policy During NSA Handover

During an LTE handover of an NSA UE that has the PSCell and an NR SCell added in NSA networking, the following may occur: the original PSCell is added, the original NR SCell is removed, and an attempt is made to blindly configure another NR SCell on the same frequency as the original NR SCell. In this case, if the original PSCell and the blind-configurable NR SCell do not cover the same area, this NR SCell may fail to be configured or this NR SCell may be removed after the LTE handover. For details about blind SCell configuration, see *Carrier Aggregation* in 5G RAN feature documentation.

The "SCell configuration policy during NSA handover" function supports different NR SCell configuration policies during an LTE handover. The policy is determined by the **NSA_HO_SCC_CFG_POL_SW** option of the **NRCellCaMgmtConfig.*CaStrategySwitch*** parameter.

- If this option is deselected, the eNodeB adds the original NR SCell or the blind-configurable NR SCell, whichever has a stronger signal strength.
- If this option is selected, the eNodeB adds the blind-configurable NR SCell.

7.1.14 Non-Gap-Assisted Coverage-based Inter-Frequency PSCell Change

In NSA networking, gap-assisted NR inter-frequency measurement is initiated during an inter-frequency PSCell change, which may affect the user-perceived rate. The "non-gap-assisted coverage-based inter-frequency PSCell change" function allows the SgNB to deliver A3/A5 measurement configurations only for the frequency of the NR SCell that has been configured for a UE without initiating gap-assisted NR inter-frequency measurements.

This function is controlled by the **COV_BASED_HO_NO_GAP_SW** option of the **NRCellFeatureSwitch**.***InterFreqHoExpSwitch*** parameter.

This function takes effect, and the non-gap-assisted A3/A5 measurement configurations delivered by the SgNB include the frequency of an NR SCell when the following conditions are met for that SCell:

- The RSRP of the NR SCell in the coverage-based A2 measurement report on the NR side, adjusted by subtracting half the value of the **NRCellServExp**.***InterHoNoGapRsrpOfs*** parameter, meets the A3 threshold or A5 threshold 2 for coverage-based inter-frequency handovers.

  For details about event A3 and event A5 for coverage-based inter-frequency handovers, see section "Measurement Configuration Delivery" under "Measurement-based Inter-Frequency Handover" in *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.
- The frequency of the NR SCell is not filtered out.
- The NR SCell is not filtered out.
- The **MOB_NO_GAP_MEAS_BAN_HO_SW** option of the **NRCellFreqRelation**.***FrequencyCtrlSwitch*** parameter is deselected for the frequency of the NR SCell.

Considering UE compatibility, the gNodeB can forbid non-gap-assisted coverage-based inter-frequency PSCell change for blacklisted UEs. This function is controlled by the **INTERF_HO_NO_GAP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter on the NR side.

7.1.15 High-Load-specific Handover

In certain scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (such as during concerts), channel variations may lead to frequent fluctuations in the measured signal quality of neighboring cells. This may trigger ping-pong handovers, subsequently resulting in degraded user experience for services. To mitigate this, the high-load-specific handover function is introduced.

This function is controlled by the **HIGH_LOAD_DIFF_HO_POL_SW** option of the **NRCellMobilityConfig**.***HoMeasAlgoSwitch*** parameter. When this option is selected and all the conditions for this function to take effect are met, high-load-specific handover takes effect. The **gNBScenarioDiffGrp** MO can be used for binding of different intra- or inter-frequency measurement parameter groups to UEs with different QCIs and operators, ensuring differentiated user experience. The conditions for the function to take effect are as follows:

- Cell requirements:
  – Number of RRC_CONNECTED UEs in the cell ≥ **NRCellMobilityConfig**.***HighLoadDiffHoUeNumThld***
  – Downlink PRB usage of the cell ≥ **NRCellMobilityConfig**.***HighLoadDiffHoDlPrbThld***
  – Uplink PRB usage of the cell ≥ **NRCellMobilityConfig**.***HighLoadDiffHoUlPrbThld***
- UE requirements: The **gNBScenarioDiffGrp** MO associated with the QCI of the UE's highest handover priority is present, and either the **NRCellIntraFHoMeaGrp** or **NRCellInterFHoMeaGrp** MO associated with the **gNBScenarioDiffGrp** MO is also present. **Table 7-4** describes the parameters in the **gNBScenarioDiffGrp** MO.

Table 7-4 Parameters in the **gNBScenarioDiffGrp** MO

Parameter	Description
gNBScenarioDiffGrp.***OperatorId***	Operator of the UE.
gNBScenarioDiffGrp.***ScenarioType***	Scenario type. Set this parameter to **HIGH_LOAD**.
gNBScenarioDiffGrp.***ScenarioDiffGrpId***	Set this parameter to the same value as the **NRCellQciBearer**.***ScenarioDiffGrpId*** parameter. Then, the QCI is associated with the **gNBScenarioDiffGrp** MO.
gNBScenarioDiffGrp.***IntraFreqHoMeasGroupId***	Set this parameter to the same value as the **NRCellIntraFHoMeaGrp**.***IntraFreqHoMeasGroupId*** parameter. Then, the **NRCellIntraFHoMeaGrp** MO is associated with the **gNBScenarioDiffGrp** MO.
gNBScenarioDiffGrp.***NsaInterFreqHoMeasGroupId***	Set this parameter to the same value as the **NRCellInterFHoMeaGrp**.***InterFreqHoMeasGroupId*** parameter. Then, the **NRCellInterFHoMeaGrp** MO is associated with the **gNBScenarioDiffGrp** MO.

7.2 Network Analysis

7.2.1 Benefits

- Non-gap-assisted B1 measurement

  This function allows the delivery of non-gap-assisted B1 measurement configurations to specific NSA UEs in LTE-only mode (without NR coverage). This reduces the number of gap-assisted measurements used for periodic measurement-based SCG addition, decreasing the impact of measurement gaps and increasing the user-perceived rates of these UEs.
- Intra-E-UTRAN inter-eNodeB S1-based handover optimization

This function increases the intra-E-UTRAN inter-eNodeB S1-based handover success rate if the capability indicated by the value of **EnodebAlgoExtSwitch**.***LteS1HoReqUeCapbSize*** does not exceed the processing capability of the core network.

- Optimized inter-eNodeB handover with SCG addition

  This function enables the eNodeB to configure EN-DC band combinations with larger total bandwidths for an NSA UE based on the EN-DC capability of the UE, improving user experience.

- gNodeB mobility control list

  This function helps maintain the inter-SgNB PSCell change success rate in LTE-NR NSA DC scenarios.

  Inter-SgNB PSCell change success rate in LTE-NR NSA DC scenarios = (**N.NsaDc.InterSgNB.PSCell.Change.Succ** / **N.NsaDc.InterSgNB.PSCell.Change.Att**) x 100%

- Fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios

  When LTE and NR cells are co-sited and provide the same coverage, this function enables timely SCG additions for CEUs, increasing the user-perceived rate of such UEs.

- SCG addition only after NSA PCC anchoring

  This function prevents SCG addition failures caused by anchoring failures as well as RRC connection reestablishments caused by UE compatibility issues.

- Non-gap-assisted coverage-based inter-frequency PSCell change

  This function increases the downlink user-perceived rate of CA UEs located in areas with good NR SCell coverage. Enabling this function is recommended when more than two NR frequencies are present, as it provides no benefits in other scenarios.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.2.3 Function Impacts**.

- User-experience-based SCG release
  - This function may reduce the 5G online duration of UEs, thereby impacting indicators such as the 5G online rate.
  - This function may reduce the number of B1 measurements, thereby lowering the proportion of 5G-capable UEs in the NSA state. The proportion is calculated as follows:
    - **L.Traffic.User.NsaDc.PCell.Avg** / **L.NsaDc.Capable.5gUser.RRC.Avg**. This formula applies to scenarios where UEs can use 5G services only after turning on their 5G switches and completing a service subscription.
    - **L.Traffic.User.NsaDc.PCell.Avg** / **L.NsaDc.Capable.User.RRC.Avg**. This formula applies to scenarios where UEs can use 5G services simply by turning on their 5G switches, without requiring a subscription.

- User-experience-based SCG release enhancement
  - This function may impact the 5G online rate and reduce the proportion of 5G-capable UEs in the NSA state.
  - This function may increase the number of SCG releases and the risk of service drops, affecting the values of certain counters:
    - The value of **N.NsaDc.SgNB.Rel** on the NR side may increase.
    - The values of **L.NsaDc.SgnbTrig.SgNB.NormRel** and **L.NsaDc.SgNB.Rmv.Att** on the LTE side may increase.
- UE-traffic-model-based SCG release
  - This function reduces the 5G online duration and the proportion of 5G-capable UEs in the NSA state.
  - This function may increase the uplink user-perceived rates of NSA UEs performing uplink large-packet services and downlink small-packet services.
- Optimized inter-eNodeB handover with SCG addition

  After this function takes effect, the target eNodeB delivers an additional UE capability query if the LTE and NR frequency band combinations indicated by the *appliedFreqBandListFilter* IE in the UE capability information of the handover request message are less than those to be queried by the target eNodeB. As a result, the SCG addition takes a longer time, and the downlink throughput of the NSA UE decreases within a short term. If the EN-DC combination selected for the NSA UE has a stronger capability, the downlink throughput of the UE will increase after a short-term decrease.
- gNodeB mobility control list

  This function enables a source base station to prohibit NSA UEs from being handed over to cells served by base stations indicated in a mobility control list. If all neighboring cells of this source base station are served by base stations indicated in the list, service drops will occur because no neighboring cells are available for handovers for a long time.
- Fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios
  - This function may shorten the SCG addition period for UEs, resulting in more frequent inter-frequency measurements. Additionally, uplink and downlink services may be negatively impacted in the absence of NR coverage.
  - This function may reduce the SCG addition success rate and increase the rate of MeNB-triggered abnormal service drops resulting from SCG addition failures.
- SCG addition only after NSA PCC anchoring
  - If the **SCG_ADD_AFTER_HANDOVER_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter is selected and an SCG addition is triggered before post-anchoring data forwarding is complete, the number of packets discarded due to forwarding failures increases.
  - If the **NsaDcAlgoParam**.***NrScgAddAfterHoPolicy*** parameter is set to **FAST_ADD**, SCG addition will be slightly delayed. Furthermore, if PDCP data forwarding triggered by anchoring is incomplete prior to SCG

addition, any unforwarded data will not be transferred to the new PDCP layer, resulting in data loss.

  - If the **NsaDcAlgoParam**.***NrScgAddAfterHoPolicy*** parameter is set to **NORMAL_ADD**, SCG addition will be significantly delayed, potentially leading to increased gap-assisted measurement overhead.

- SCell configuration policy during NSA handover
  - If the switch controlling this function is turned on, a blind-configurable cell is preferentially selected as an NR SCell. However, if its signal quality is poor or its bandwidth is small, user experience is affected.
  - If the switch controlling this function is turned off, the original NR SCell is added when the PSCell and the blind-configurable SCell do not cover the same area during an LTE handover of an NR CA UE. This increases the downlink throughput of the UE in the SCell. However, if the signal quality of the NR SCell measured by the UE is inaccurate, user experience is affected.
- Non-gap-assisted coverage-based inter-frequency PSCell change

  After this function takes effect, the downlink user-perceived rate of CA UEs in areas with poor NR SCell coverage may decrease. Additionally, this function may trigger camping policy changes that reduce handover and access success rates.
- High-load-specific handover

  After this function takes effect, the values of the following counters related to PSCell changes will change:
  - **N.NsaDc.InterSgNB.PSCell.Change.Att**
  - **N.NsaDc.InterSgNB.PSCell.Change.Succ**
  - **N.NsaDc.IntraSgNB.PSCell.Change.Att**
  - **N.NsaDc.IntraSgNB.PSCell.Change.Succ**
  - **N.NsaDc.InterSgNB.SgNB.Add.Att**
  - **N.NsaDc.InterSgNB.SgNB.Add.Ack**
  - **N.NsaDc.InterSgNB.SgNB.Add.Succ**
  - **N.NsaDc.InterSgNB.PSCell.Change.UeContextRel**
  - **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att**
  - **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Succ**
  - **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**
  - **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Succ**

7.3 Requirements

7.3.1 Licenses

For details about license requirements, see **5.3.1 Licenses** and **6.3.1 Licenses**.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA PCC anchoring based on NR coverage	• **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** set to **BASED_ON_NR_COVERAGE**	*EPC-based NSA Performance Enhancement*	If non-gap-assisted B1 measurement and band combination change triggered by NR inter-frequency handover are enabled, NSA PCC anchoring based on NR coverage must also be enabled on the LTE side.
LTE FDD LTE TDD	NSA DC capability	**NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*	Fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios requires that the NSA DC capability be enabled on the LTE side.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	NSA carrier combination selection based on user experience	• LTE: **NSA_LTE_FDD_SEL_OPT_SW** and **NSA_LTE_TDD_SEL_OPT_SW** options of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter • NR: **NSA_LTE_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter	*EPC-based NSA Experience Improvement*	If band combination change triggered by NR inter-frequency handover is enabled, NSA carrier combination selection based on user experience must also be enabled on the LTE side.
LTE FDD LTE TDD	Uplink large-packet service detection	**UL_LARGE_PKT_SERV_DECT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*NSA Mobility Management*	The "notification of the uplink large-packet status of uplink fallback UEs to NR" function requires that uplink large-packet service detection be enabled on the LTE side.
NR FDD Low-frequency NR TDD	Smart carrier selection	**SMART_CARR_SEL_SW** option of the **NRDUCellCollabServ**.***MultiCarrMgmtSw*** parameter	*Carrier Aggregation* in 5G RAN feature documentation	The "SCell configuration policy during NSA handover" function requires that smart carrier selection be enabled in the PSCell.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Carrier aggregation	**INTRA_BAND_CA_SW** or **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter selected	*Carrier Aggregation* in 5G RAN feature documentation	The "SCell configuration policy during NSA handover" function requires that carrier aggregation be enabled in the PSCell.
NR FDD Low-frequency NR TDD	Blind SCell configuration	**SCC_BLIND_CONFIG_SW** option of the **NRCellCaMgmtConfig.*****CaStrategySwitch*** parameter	*Carrier Aggregation* in 5G RAN feature documentation	The "SCell configuration policy during NSA handover" function requires that blind SCell configuration be enabled in the PSCell.
NR FDD Low-frequency NR TDD	Measurement-based inter-frequency handover	**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter	*SA Mobility Management in Connected Mode* in 5G RAN feature documentation	The "non-gap-assisted coverage-based inter-frequency PSCell change" function requires that measurement-based inter-frequency handover be enabled in the PSCell.

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	User-experience-based SCG release	**USER_EXP_SCG_REL_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter	*NSA Mobility Management*	User-experience-based SCG release enhancement is mutually exclusive with user-experience-based SCG release.

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA PCC anchoring based on NR coverage	**NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** set to **BASED_ON_NR_COVERAGE**	*EPC-based NSA Performance Enhancement*	During NSA PCC anchoring based on NR coverage, if the eNodeB receives an event A1 measurement report in response to the measurement configurations delivered for the fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios function, the eNodeB deletes the event A1 measurement configurations. In this case, the SCG addition procedure does not take effect.
NR FDD Low-frequency NR TDD	QCI-specific measurement parameter configuration	**NRCellQciBearer**.***IntraFreqHoMeasGroupId*** or **NRCellQciBearer**.***NsaInterFreqHoMeasGroupId***	*NSA Mobility Management*	When both QCI-specific measurement parameter configuration and high-load-specific handover are configured, the latter function preferentially takes effect.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

SSB time-domain position delivery: 5900 series base stations (macro base stations)

Other functions:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The compatible boards are listed below.

RAT	Board Type	Board Name	Option 3	Option 3x
LTE	Main control board	UMPTb	Supported	Supported
LTE	Main control board	UMPTe	Supported	Supported
LTE	Main control board	UMPTg/ UMPTga	Supported	Supported
LTE	Main control board	UMPTha	Supported	Supported
LTE	Baseband processing unit	All UBBP boards	Supported	Supported
LTE	Integrated board	UMDU	Supported	Supported
NR	Main control board	UMPTe	Supported	Supported
NR	Main control board	UMPTg/ UMPTga	Supported	Supported
NR	Main control board	UMPTha	Supported	Supported
NR	Baseband processing unit	UBBPg	Supported	Supported
NR	Baseband processing unit	UBBPfw1	Supported	Supported
NR	Baseband processing unit	UBBPh	Supported	Supported

RF Modules

This function does not depend on RF modules.

7.3.4 Others

- UE requirements
  - UEs must support NSA DC specified in 3GPP Release 15.
  - UEs must have subscribed to LTE and NR services.
  - UEs must match gNodeB and eNodeB versions.
- EPC requirements
  - The EPC must be CloudEPC to support Option 3 and Option 3x.
  - The EPC must support NSA DC. If the core network is provided by Huawei, see *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management* for details.
  - If NSA DC is enabled on an eNodeB, the connected MMEs need to support NSA DC. If a connected MME does not support NSA DC, the **MmeCapInfo**.***MmeNsaDcCapability*** parameter for this MME must be set to **NOT_SUPPORT**.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

For details about the data preparation for the basic functions related to NSA mobility management in connected mode, see **5.4.1.1 Data Preparation** and **6.4.1.1 Data Preparation**.

The following table describes the parameter that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure gap sharing.

Parameter Name	Parameter ID	Setting Notes
Measurement Gap Sharing Scheme	**NsaDcMgmtConfig**.***MeasGapSharingScheme***	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig**, **NrScgFreqConfig**, and **GlobalProcSwitch** MOs on the LTE side to configure measurement-based simultaneous LTE and NR cell changes.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	**NsaDcMgmtConfig**.*NsaDcAlgoExtSwitch*	It is recommended that the **SIMUL_LTE_NR_MEAS_BASED_HO_SW** option be selected to enable measurement-based simultaneous LTE and NR cell changes when LTE and NR cells are co-sited and provide the same coverage.
NSA DC LTE NR Simultaneous HO Threshold	**NrScgFreqConfig**.*NsaDcLteNrSimulHoThld*	This parameter specifies the threshold for an NR cell change. If the RSRP difference between a neighboring NR cell and the serving NR cell is greater than this threshold, the eNodeB determines that the neighboring NR cell can be selected as a target NR cell for simultaneous LTE and NR cell changes.
Protocol Procedure Support Switch	**GlobalProcSwitch**.*ProtocolSupportSwitch*	If measurement-based simultaneous LTE and NR cell changes are required, it is recommended that the **NsaDcLteMeasCtrlwithNbr-MeasSw** option be selected. Under this setting, the eNodeB includes a *reportAddNeighMeas* IE in delivered measurement configurations, and the UE reports the measurement results of the serving NR cell and neighboring NR cells along with intra-LTE measurement results.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** MO on the NR side to configure MeNB-initiated intra-SgNB cell change.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**NRCellNsaDcConfig**.*NsaDcAlgoSwitch*	In co-site deployment or blind configuration scenarios, it is recommended that the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option be selected if the SgNB Modification Request message sent to the SgNB during an intra-MeNB handover contains neighboring NR cell measurement results and an intra-SgNB cell change is expected to be triggered during the intra-MeNB handover.

The following table describes the parameters that must be set in the **gNodeBParam**, **gNBMobilityCommParam**, and **NRCellAlgoSwitch** MOs on the NR side and the **EnodebAlgoExtSwitch** MO on the LTE side to configure band combination change triggered by NR inter-frequency handover.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	**gNodeBParam**.*NsaDcOptSwitch*	Select the **INTERFREQ_HO_TRIG_BC_CHANGE_SW** option.
NSA Inter-Frequency Measurement Wait Timer	**gNBMobilityCommParam**.*NsaInterFreqMeasWaitTimer*	A non-zero value is recommended.
Multi-Frequency Algorithm Switch	**NRCellAlgoSwitch**.*MultiFreqAlgoSwitch*	Select the **UL_WEAK_COV_UE_PROTECT_SW** option.
NSA DC Algorithm Switch	**EnodebAlgoExtSwitch**.*NsaDcAlgoSwitch*	Select the **NO_NR_COV_EN_DC_RESEL_SW** option.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side and the **gNodeBParam** MO on the NR side to configure MeNB-initiated intra-SgNB inter-frequency handover.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch***	Select the **INTRA_SGNB_IF_MEAS_FILTER_SW** option.
NSA DC Optimization Switch	NR: **gNodeBParam**.***NsaDcOptSwitch***	Select the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option.

The following table describes the parameters that must be set in the **UeCompat** and **LnrFreqComb** MOs on the LTE side to configure non-gap-assisted B1 measurement for specific NSA UEs.

Parameter Name	Parameter ID	Setting Notes
White List Control Switch	**UeCompat**.***WhiteLstCtrlSwitch***	Select the **NO_GAP_B1_MEAS_SW_ON** option.
LTE-NR No Gap Freq Comb Blacklist	**UeCompat**.***LnrNoGapFreqCombBlacklist***	Select the **B1_NO_GAP_FREQ_COMB_ID_***N* option of this parameter, where *N* is equal to the value of the **LnrFreqComb**.***FreqCombId*** parameter.

The following table describes the parameters that must be set in the **LnrFreqComb** MO on the LTE side to blacklist a frequency combination for non-gap-assisted B1 measurement.

Parameter Name	Parameter ID	Setting Notes
Frequency Combination ID	**LnrFreqComb**.***FreqCombId***	Set this parameter to *N* if the **B1_NO_GAP_FREQ_COMB_ID_***N* option of the **UeCompat**.***LnrNoGapFreqCombBlacklist*** parameter is selected.
Downlink EARFCN	**LnrFreqComb**.***DlEarfcn***	Set this parameter to the LTE frequency in the combination for which the NSA UE does not support non-gap-assisted B1 measurement.

Parameter Name	Parameter ID	Setting Notes
Downlink NR-ARFCN	**LnrFreqComb**.***DlNrArfcn***	Set this parameter to the NR frequency in the combination for which the NSA UE does not support non-gap-assisted B1 measurement.

The following table describes the parameters that must be set in the **EnodeBAlgoExtSwitch** MO on the LTE side to configure protocol-based non-gap-assisted B1 measurement.

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	**EnodeBAlgoExtSwitch**.***MultiNetworkingOptionOptSw***	Select the **NR_B1_NO_GAP_SW** option.

The following table describes the parameters that must be set in the **NRDUCellAlgoSwitch** and **NRDUCellNsaDcConfig** MOs on the NR side to configure the user-experience-based SCG release function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch***	Select the **USER_EXP_SCG_REL_SW** option.
SCG Rel UL SRS SINR Thld	**NRDUCellNsaDcConfig**.***ScgRelUlSrsSinrThld***	• For NR FDD, SRS SINR has little impact on downlink performance, and therefore it is recommended that this parameter be set to **-200**. • For NR TDD, the initial recommended value is **-120**. • This threshold is closely related to the actual situation of the live network. It is recommended that this threshold be adjusted when the poor downlink performance on the live network is caused by the SRS SINR.

Parameter Name	Parameter ID	Setting Notes
SCG Rel DL CQI Thld	**NRDUCellNsaDcConfig**.***ScgRelDlCqiThld***	• The initial recommended value is **0**. • This threshold is closely related to the actual situation of the live network. You are advised to modify the configuration based on the requirements of the live network for the downlink rate.
User-Exp-based SCG Release Meas Num	**NRDUCellNsaDcConfig**.***UserExpScgRelMeasNum***	The value **5** is recommended.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure the SCG addition penalty duration for the user-experience-based SCG release function.

Parameter Name	Parameter ID	Setting Notes
SCG Addition Penalty Duration	**NsaDcMgmtConfig**.***ScgAddPenaltyDuration***	A value greater than **0** is recommended, to prevent an SCG from being added on the LTE side after being released.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** and **NRDUCellNsaDcConfig** MOs on the NR side to configure the user-experience-based SCG release enhancement function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**NRCellNsaDcConfig**.***NsaDcAlgoSwitch***	Select the **UL_DL_USER_EXP_SCG_REL_SW** option.

Parameter Name	Parameter ID	Setting Notes
User-Exp-based SCG Release A2 RSRP Thld	**NRCellNsaDcConfig**.***UserExpScgRelA2RsrpThld***	● The initial recommended value is **-115**. ● This threshold is closely related to the actual situation of the live network. It is recommended that this parameter be set to a value less than or equal to the event B1 threshold for SCG addition.
User-Exp-based SCG Release A1 RSRP Thld	**NRCellNsaDcConfig**.***UserExpScgRelA1RsrpThld***	● The initial recommended value is **-113**. ● This threshold must be at least 2 dB greater than the value of **NRCellNsaDcConfig**.***UserExpScgRelA2RsrpThld***.
SCG Rel UL SRS SINR Thld	**NRDUCellNsaDcConfig**.***ScgRelUlSrsSinrThld***	● For NR FDD, SRS SINR has little impact on downlink performance, and therefore it is recommended that this parameter be set to **-200**. ● For NR TDD, the initial recommended value is **-120**. ● This threshold is closely related to the actual situation of the live network. It is recommended that this threshold be adjusted when the poor downlink performance on the live network is caused by the SRS SINR.
SCG Rel DL CQI Thld	**NRDUCellNsaDcConfig**.***ScgRelDlCqiThld***	● The initial recommended value is **0**. ● This threshold is closely related to the actual situation of the live network. You are advised to modify the configuration based on the requirements of the live network for the downlink rate.

Parameter Name	Parameter ID	Setting Notes
User-Exp-based SCG Release Meas Num	**NRDUCellNsaDcConfig**.*UserExpScgRelMeasNum*	The value **5** is recommended.

The following table describes the parameters that must be set in the **UeCompat** and **EndcBandGapFreeCfg** MOs on the LTE side to configure non-gap-assisted measurement for LTE and NR band combinations in EN-DC.

Parameter Name	Parameter ID	Setting Notes
Whitelist Gap Free Combination Switch	**UeCompat**.*WhitelistGapFreeCombSwitch*	The **ENDC_GAP_FREE_COMB_ID_*X*** option of the **UeCompat**.*WhitelistGapFreeCombSwitch* parameter must be selected. *X* indicates the ID of the EN-DC band combination for non-gap-assisted measurements, that is, **EndcBandGapFreeCfg**.*EndcGapFreeCombId*.
EN-DC Gap Free Combination ID	**EndcBandGapFreeCfg**.*EndcGapFreeCombId*	Set this parameter based on the non-gap-assisted measurement capability of specific NSA UEs. For details about the configuration examples, see **7.1.1 Non-Gap-Assisted B1 Measurement**.
Band Type	**EndcBandGapFreeCfg**.*BandType*	Set this parameter based on the non-gap-assisted measurement capability of specific NSA UEs. For details about the configuration examples, see **7.1.1 Non-Gap-Assisted B1 Measurement**.
Band Index	**EndcBandGapFreeCfg**.*BandIndex*	Set this parameter based on the non-gap-assisted measurement capability of specific NSA UEs. For details about the configuration examples, see **7.1.1 Non-Gap-Assisted B1 Measurement**.

Parameter Name	Parameter ID	Setting Notes
Frequency Band ID	**EndcBandGapFreeCfg**.***FrequencyBandId***	Set this parameter based on the non-gap-assisted measurement capability of specific NSA UEs. For details about the LTE and NR frequency band IDs, see 3GPP TS 36.101 and TS 38.101. For details about the configuration examples, see **7.1.1 Non-Gap-Assisted B1 Measurement**.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side to configure the intra-E-UTRAN inter-eNodeB S1-based handover optimization function.

Parameter Name	Parameter ID	Setting Notes
LTE S1 Handover Request UE Capability Size	**EnodebAlgoExtSwitch**.***LteS1HoReqUeCapbSize***	Set this parameter based on the network plan. Before adjusting this parameter, check the processing capability of the EPC to process a handover request message. If the processing capability indicated by this parameter exceeds that of the EPC, the handover request may fail.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side to configure optimized inter-eNodeB handover with SCG addition.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch***	Set the **LTE_HO_WITH_SCG_ADD_OPT_SW** option based on the network plan.

The following table describes the parameter that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure the fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Fast SCG Add A1 RSRP Thld	**NsaDcMgmtConfig**.*NsaDcFastScgAddA1RsrpThld*	Set this parameter to a value other than **255** if the fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios function is required.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch**, **NsaDcAlgoParam**, and **UeCompat** MOs on the LTE side to configure the SCG addition only after NSA PCC anchoring function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**EnodebAlgoExtSwitch**.*NsaDcAlgoSwitch*	Select the **SCG_ADD_AFTER_HANDOVER_SW** option of this parameter if SCG addition failures caused by anchoring failures or RRC connection reestablishments caused by UE compatibility issues need to be prevented for NSA-capable UEs in the LTE-only state.
NR SCG Add After Handover Policy	**NsaDcAlgoParam**.*NrScgAddAfterHoPolicy*	You are advised to set this parameter to **FAST_ADD** if SCG addition failures caused by anchoring failures or RRC connection reestablishments caused by UE compatibility issues need to be prevented for NSA-capable UEs in the NSA state.
Whitelist Control Extension Switch 1	**UeCompat**.*WhitelistCtrlExtSwitch1*	Select the **NR_SCG_ADD_AFTER_HO_OPT_SW** option of this parameter if the SCG addition only after NSA PCC anchoring function needs to take effect only for whitelisted UEs.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side and the **gNodeBParam** and **NRDUCellSrsMeas** MOs on the NR side to configure the UE-traffic-model-based SCG release function.

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	**EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw***	Select the **UL_LARGE_PKT_SERV_DECT_SW** option.
NSA DC Algorithm Switch	**EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch***	Select the **ULFB_UL_LRG_PKT_FLAG_TO_NR_SW** option.
NSA DC Optimization Switch	**gNodeBParam**.***NsaDcOptSwitch***	Select the **TRF_VOL_BASED_NR_SCG_REL_SW** option.
Traffic-based SCG Release UL SINR Thld	**NRCellNsaDcConfig**.***TfcBasedScgRelUlSinrThld***	Set this parameter based on the network plan.

The following table describes the parameter that must be set in the **NRCellCaMgmtConfig** MO on the NR side to configure the "SCell configuration policy during NSA handover" function.

Parameter Name	Parameter ID	Setting Notes
CA Strategy Switch	**NRCellCaMgmtConfig**.***CaStrategySwitch***	If the PSCell and an NR SCell have been added for an NSA UE: ● To enable both the original PSCell and original NR SCell to be added during an LTE handover, deselect the **NSA_HO_SCC_CFG_POL_SW** option. ● To enable the original PSCell to be added and another NR SCell to be blindly configured during an LTE handover, select the **NSA_HO_SCC_CFG_POL_SW** option.

The following table describes the parameters that must be set in the **NRCellCaMgmtConfig** MO on the NR side to configure the non-gap-assisted coverage-based inter-frequency PSCell change function.

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Expand Switch	**NRCellFeatureSwitch**.***InterFreqHoExpSwitch***	To enable the non-gap-assisted coverage-based inter-frequency PSCell change function on the NR side, select the **COV_BASED_HO_NO_GAP_SW** option.
Inter-frequency Handover No Gap RSRP Offset	**NRCellServExp**.***InterHoNoGapRsrpOfs***	The value **4** is recommended. Set this parameter based on the network plan. If this parameter is set to an excessively small value, the number of coverage-based inter-frequency PSCell change attempts and the number of successful coverage-based inter-frequency PSCell changes may decrease. If this parameter is set to an excessively large value, the non-gap-assisted coverage-based inter-frequency PSCell change function may not take effect.
Frequency Control Switch	**NRCellFreqRelation**.***FrequencyCtrlSwitch***	If the non-gap-assisted coverage-based inter-frequency PSCell change function is enabled and measurement configurations are expected to be delivered only for the frequencies of one NR SCell, select the **MOB_NO_GAP_MEAS_BAN_HO_SW** option for other frequencies.
Blacklist Control Extension Switch	**gNBUeCompatSw**.***BlacklistCtrlExtSwitch***	To disable non-gap-assisted coverage-based inter-frequency handover for specific UEs, select the **INTERF_HO_NO_GAP_SW_OFF** option.

The following table describes the parameters that must be set in the **NRCellMobilityConfig**, **gNBScenarioDiffGrp**, and **NRCellQciBearer** MOs on the NR side to configure the high-load-specific handover function.

Parameter Name	Parameter ID	Setting Notes
Handover Measure Algorithm Switch	**NRCellMobilityConfig**.***HoMeasAlgoSwitch***	To reduce ping-pong handovers in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (such as during concerts), select the **HIGH_LOAD_DIFF_HO_POL_SW** option.
High Load Differentiated HO UE Num Threshold	**NRCellMobilityConfig**.***HighLoadDiffHoUeNumThld***	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	**NRCellMobilityConfig**.***HighLoadDiffHoDlPrbThld***	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	**NRCellMobilityConfig**.***HighLoadDiffHoUlPrbThld***	Set this parameter based on the network plan.
Scenario Differentiated Group ID	**gNBScenarioDiffGrp**.***ScenarioDiffGrpId***	Set this parameter based on the network plan.
Scenario Type	**gNBScenarioDiffGrp**.***ScenarioType***	To enable high-load-specific handover, set this parameter to **HIGH_LOAD**.
Operator ID	**gNBScenarioDiffGrp**.***OperatorId***	Set this parameter based on the network plan.
Intra-freq Handover Measurement Group ID	**gNBScenarioDiffGrp**.***IntraFreqHoMeasGroupId***	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	**gNBScenarioDiffGrp**.***NsaInterFreqHoMeasGroupId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Scenario Differentiated Group ID	**NRCellQciBearer**.***ScenarioDiffGrpId***	Set this parameter based on the network plan. If this parameter is set to a value other than **255**, the corresponding group identified by **gNBScenarioDiffGrp**.***ScenarioDiffGrpId*** must be already configured.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **5.4.1.2 Using MML Commands** and **6.4.1.2 Using MML Commands** and configure neighboring frequencies, external cells, and NSA mobility-related parameters on the LTE and NR sides.

Activation Command Examples

On the eNodeB side

```
//Enabling MeNB-initiated intra-SgNB inter-frequency handover
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=INTRA_SGNB_IF_MEAS_FILTER_SW-1;
//Enabling non-gap-assisted B1 measurement for specific NSA UEs
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-1;
//Configuring a combination of LTE and NR frequencies (for example, EARFCN 1500 and NR-ARFCN
636666) for which the NSA UE does not support non-gap-assisted B1 measurement
ADD LNRFREQCOMB: DlEarfcn=1500, DlNrArfcn=636666, FreqCombId=0;
//Blacklisting the combination of LTE and NR frequencies for which the NSA UE does not support non-gap-
assisted B1 measurement
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
LnrNoGapFreqCombBlacklist=B1_NO_GAP_FREQ_COMB_ID_0-1;
//Enabling protocol-based non-gap-assisted B1 measurement
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NR_B1_NO_GAP_SW-1;
//Configuring an NSA DC gap sharing policy
MOD NSADCMGMTCONFIG: LocalCellId=21, MeasGapSharingScheme=EQUAL_SPLITTING;
//Configuring measurement-based simultaneous LTE and NR cell changes
MOD NSADCMGMTCONFIG: LocalCellId=21,NsaDcAlgoExtSwitch=SIMUL_LTE_NR_MEAS_BASED_HO_SW-1;
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=636666, NsaDcLteNrSimulHoThld=0;
MOD GLOBALPROCSWITCH: ProtocolSupportSwitch=NsaDcLteMeasCtrlwithNbrMeasSw-1;
//(Optional) Enabling non-NR-coverage-triggered EN-DC combination reselection if EN-DC band
combination changes triggered due to weak NR coverage in the uplink or downlink is required
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch= NO_NR_COV_EN_DC_RESEL_SW-1;
//Setting the SCG addition penalty duration for the user-experience-based SCG release function
```



```
MOD NSADCMGMTCONFIG: LocalCellId=21, ScgAddPenaltyDuration=60;
//Configuring non-gap-assisted measurement for an EN-DC band combination that consists of LTE band 3, LTE band 10, NR band 14, and NR band 41 of which the UE can perform non-gap-measurement (For details about the UE whitelist configuration, see Terminal Awareness Differentiation.)
ADD ENDCBANDGAPFREECFG: EndcGapFreeCombId=0, BandIndex=1, BandType=LTE, FrequencyBandId=3;
ADD ENDCBANDGAPFREECFG: EndcGapFreeCombId=0, BandIndex=2, BandType=LTE, FrequencyBandId=10;
ADD ENDCBANDGAPFREECFG: EndcGapFreeCombId=0, BandIndex=9, BandType=NR, FrequencyBandId=14;
ADD ENDCBANDGAPFREECFG: EndcGapFreeCombId=0, BandIndex=14, BandType=GAP_FREE_NR_BAND, FrequencyBandId=41;
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhitelistGapFreeCombSwitch=ENDC_GAP_FREE_COMB_ID_0-1;
//Setting the LTE S1 Handover Request UE Capability Size parameter for intra-E-UTRAN inter-eNodeB S1-based handover optimization
MOD ENODEBALGOEXTSWITCH: LteS1HoReqUeCapbSize=50;
//Enabling optimized inter-eNodeB handover with SCG addition
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=LTE_HO_WITH_SCG_ADD_OPT_SW-1;
//Setting the RSRP threshold for event A1 related to fast SCG addition in NSA DC scenarios to enable fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcFastScgAddA1RsrpThld=-100;
//Enabling the SCG addition only after NSA PCC anchoring function
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=SCG_ADD_AFTER_HANDOVER_SW-1;
//Setting the policy on SCG addition after a handover
MOD NSADCALGOPARAM: NrScgAddAfterHoPolicy=FAST_ADD;
//Turning on the whitelist control switch for post-handover SCG addition
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhitelistCtrlExtSwitch1=NR_SCG_ADD_AFTER_HO_OPT_SW-1;
//(Optional, recommended) Enabling the UE-traffic-model-based SCG release function when an NSA UE with an SCG added has a lower uplink user-perceived rate than in the LTE-only state
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=UL_LARGE_PKT_SERV_DECT_SW-1, NsaDcAlgoSwitch=ULFB_UL_LRG_PKT_FLAG_TO_NR_SW-1;
```

On the gNodeB side

```
//Enabling MeNB-initiated intra-SgNB cell change
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=MENB_TRIG_INTRA_SGNB_CHANGE_SW-1;
//Enabling band combination change triggered by NR inter-frequency handover
MOD GNODEBPARAM: NsaDcOptSwitch=INTERFREQ_HO_TRIG_BC_CHANGE_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=7, MultiFreqAlgoSwitch=UL_WEAK_COV_UE_PROTECT_SW-1;
MOD GNBMOBILITYCOMMPARAM: NsaInterFreqMeasWaitTimer=3;
//Enabling MeNB-initiated intra-SgNB inter-frequency handover
MOD GNODEBPARAM: NsaDcOptSwitch=SIMU_INTRASGNB_INTERFREQ_HO_SW-1;
//Setting the switch for user-experience-based SCG release, corresponding thresholds, and number of measurement times for user-experience-based SCG releases
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, NsaDcAlgoSwitch=USER_EXP_SCG_REL_SW-1;
MOD NRDUCELLNSADCCONFIG: NrDuCellId=0, ScgRelDlCqiThld=0, ScgRelUlSrsSinrThld=-120, UserExpScgRelMeasNum=5;
//Setting the switch for user-experience-based SCG release enhancement, corresponding thresholds, and number of measurements for user-experience-based SCG release (which requires that user-experience-based SCG release be disabled first)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, NsaDcAlgoSwitch=USER_EXP_SCG_REL_SW-0;
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=UL_DL_USER_EXP_SCG_REL_SW-1, UserExpScgRelA2RsrpThld=-115, UserExpScgRelA1RsrpThld=-113;
MOD NRDUCELLNSADCCONFIG: NrDuCellId=0, ScgRelDlCqiThld=0, ScgRelUlSrsSinrThld=-120, UserExpScgRelMeasNum=5;
//(Optional, recommended) Enabling the UE-traffic-model-based SCG release function when an NSA UE with an SCG added has a lower uplink user-perceived rate than in the LTE-only state
MOD GNODEBPARAM: NsaDcOptSwitch=TRF_VOL_BASED_NR_SCG_REL_SW-1;
MOD NRCELLNSADCCONFIG: NrCellId=7, TfcBasedScgRelUlSinrThld=-50;
//(Optional, recommended) Turning off the switch controlling the "SCell configuration policy during NSA handover" function to allow both the original PSCell and original NR SCell to be added during an LTE handover
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=NSA_HO_SCC_CFG_POL_SW-0;
//(Optional, recommended) Enabling the non-gap-assisted coverage-based inter-frequency PSCell change function when gap-assisted measurement is not expected on the NR side during a coverage-based inter-frequency handover
MOD NRCELLFEATURESWITCH: NrCellId=7, InterFreqHoExpSwitch=COV_BASED_HO_NO_GAP_SW-1;
MOD NRCELLSERVEXP: NrCellId=7, InterHoNoGapRsrpOfs=4;
//(Optional) Selecting the MOB_NO_GAP_MEAS_BAN_HO_SW option for an NR SCell frequency (SSB frequency-domain position being 7800 used as an example) if only the frequencies of another NR SCell are expected to be included in the measurement configurations related to the non-gap-assisted coverage-based
```



```
inter-frequency PSCell change function
MOD NRCELLFREQRELATION: NrCellId=7, SsbFreqPos=7800,
FrequencyCtrlSwitch=MOB_NO_GAP_MEAS_BAN_HO_SW-1;
//(Optional) Disabling non-gap-assisted coverage-based inter-frequency PSCell change for UEs with
compatibility issues when non-gap-assisted coverage-based inter-frequency PSCell change is in effect
//(Optional) Disabling non-gap-assisted coverage-based inter-frequency PSCell change if UE information
with UeInfoIndex set to 1 does not exist (the UE information must be configured based on live network
conditions)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff1",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION2;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=INTERF_HO_NO_GAP_SW_OFF-1;
//(Optional) Disabling non-gap-assisted coverage-based inter-frequency PSCell change if UE information
with UeInfoIndex set to 1 exists
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=INTERF_HO_NO_GAP_SW_OFF-1;
//(Optional) Enabling the high-load-specific handover function to reduce ping-pong handovers in scenarios
where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (such as during
concerts). In the following MML command examples, OperatorId is set to 0, and both
NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId and NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId are
set to 1 for the desired gNodeB.
MOD NRCELLMOBILITYCONFIG: NrCellId=7, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
IntraFreqHoMeasGroupId=1, NsaInterFreqHoMeasGroupId=1;
MOD NRCELLQCIBEARER: NrCellId=7, Qci=9, ScenarioDiffGrpId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```
//Disabling protocol-based non-gap-assisted B1 measurement
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NR_B1_NO_GAP_SW-0;
//Disabling NSA UEs to report neighboring NR cell measurement results along with LTE measurement results
MOD GLOBALPROCSWITCH: ProtocolSupportSwitch=NsaDcLteMeasCtrlwithNbrMeasSw-0;
//Disabling measurement-based simultaneous LTE and NR cell changes
MOD NSADCMGMTCONFIG: LocalCellId=21,NsaDcAlgoExtSwitch=SIMUL_LTE_NR_MEAS_BASED_HO_SW-0;
//(Optional) Disabling non-NR-coverage-triggered EN-DC combination reselection
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch= NO_NR_COV_EN_DC_RESEL_SW-0;
//Disabling non-gap-assisted B1 measurement
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-0;
//Disabling the SCG addition penalty for the user-experience-based SCG release function
MOD NSADCMGMTCONFIG: LocalCellId=21, ScgAddPenaltyDuration=0;
//Disabling optimized inter-eNodeB handover with SCG addition
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=LTE_HO_WITH_SCG_ADD_OPT_SW-0;
//Disabling fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcFastScgAddA1RsrpThld=255;
//Disabling the SCG addition only after NSA PCC anchoring function
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=SCG_ADD_AFTER_HANDOVER_SW-0;
//Turning off the whitelist control switch for post-handover SCG addition
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch1=NR_SCG_ADD_AFTER_HO_OPT_SW-0;
//Disabling the UE-traffic-model-based SCG release function
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=UL_LARGE_PKT_SERV_DECT_SW-0,
NsaDcAlgoSwitch=ULFB_UL_LRG_PKT_FLAG_TO_NR_SW-0;
```

On the gNodeB side

```
//Disabling MeNB-initiated intra-SgNB cell change
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch= MENB_TRIG_INTRA_SGNB_CHANGE_SW-0;
//Disabling band combination change triggered by NR inter-frequency handover
MOD GNODEBPARAM: NsaDcOptSwitch=INTERFREQ_HO_TRIG_BC_CHANGE_SW-0;
```



```
MOD NRCELLALGOSWITCH: NrCellId=7, MultiFreqAlgoSwitch=UL_WEAK_COV_UE_PROTECT_SW-0;
MOD GNBMOBILITYCOMMPARAM: NsaInterFreqMeasWaitTimer=0;
//Disabling the user-experience-based SCG release function
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, NsaDcAlgoSwitch=USER_EXP_SCG_REL_SW-0;
//Disabling the user-experience-based SCG release enhancement function
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=UL_DL_USER_EXP_SCG_REL_SW-0;
//Disabling the UE-traffic-model-based SCG release function
MOD GNODEBPARAM: NsaDcOptSwitch=TRF_VOL_BASED_NR_SCG_REL_SW-0;
//Disabling blacklist control for the non-gap-assisted coverage-based inter-frequency handover function
MOD GNBUECOMPATSW: UeInfoIndex=1,ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=INTERF_HO_NO_GAP_SW_OFF-0;
//Disabling the non-gap-assisted coverage-based inter-frequency PSCell change function
MOD NRCELLFEATURESWITCH: NrCellId=7, InterFreqHoExpSwitch=COV_BASED_HO_NO_GAP_SW-0;
//Disabling the high-load-specific handover function
MOD NRCELLMOBILITYCONFIG: NrCellId=7, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-0;
```

7.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Monitoring Counters

- Check whether intra-frequency NR cell change has taken effect.

  If the values of the following counters are not 0, intra-frequency NR cell change has taken effect.

Counter ID	Counter Name	NE
1911820758	**N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att**	gNodeB
1911820760	**N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**	gNodeB

- Check whether inter-frequency NR cell change has taken effect.

  Observe the following counters to obtain the total number of intra- and inter-frequency NR cell change attempts.

Counter ID	Counter Name	NE
1911816748	**N.NsaDc.InterSgNB.PSCell.Change.Att**	gNodeB
1911820758	**N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att**	gNodeB
1911816750	**N.NsaDc.IntraSgNB.PSCell.Change.Att**	gNodeB
1911820760	**N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att**	gNodeB

If the result of **N.NsaDc.InterSgNB.PSCell.Change.Att** minus **N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att** or the result of **N.NsaDc.IntraSgNB.PSCell.Change.Att** minus **N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att** is not 0, NR inter-frequency cell change has taken effect.

- Check whether MeNB-initiated intra-SgNB cell change has taken effect.

  If the function of measurement-based simultaneous LTE and NR cell changes is enabled, check the values of the counters listed in the following table. If the values are not 0, MeNB-initiated intra-SgNB cell change has taken effect.

Counter ID	Counter Name	NE
1911827168	**N.NsaDc.MeNBTrigger.InterPSCell.Change.Att**	gNodeB
1911827167	**N.NsaDc.MeNBTrigger.InterPSCell.Change.Ack**	gNodeB
1911827166	**N.NsaDc.MeNBTrigger.InterPSCell.Change.Succ**	gNodeB

Message Tracing

1. Check whether the function of measurement-based simultaneous LTE and NR cell changes has taken effect.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**, and check whether the eNodeB sends a UE an RRC_CONN_RECFG message with a *reportAddNeighMeas* IE; choose **Trace Type** > **LTE** > **Application Layer** > **X2 Interface Trace**, and check whether the eNodeB sends the gNodeB an SgNB Addition Request or SgNB Modification Request message with IEs *CG-ConfigInfo* > *MeasResultList2NR* (indicating the measurement results of neighboring cells of the serving NR cell). If so, the function has taken effect.

7.4.3 Network Monitoring

Monitor the counters listed in the following table and compare them against the network plan to evaluate the feature performance.

- On the eNodeB side

  After NSA DC is enabled, observe the following counters to determine whether the handover success rate and service drop rate of NSA UEs are different from those of LTE-only UEs:

  – Outgoing handover success rate of NSA UEs = **L.NsaDc.HHO.ExecSuccOut** / **L.NsaDc.HHO.ExecAttOut** x 100%
  – Incoming handover success rate of NSA UEs = **L.NsaDc.PCell.Change.Succ** / **L.NsaDc.PCell.Change.Exec** x 100%
  – Abnormal service drop rate of NSA UEs = **L.NsaDc.E-RAB.AbnormRel** / **L.NsaDc.E-RAB.NormRel** x 100%

- On the gNodeB side

  – Observe the **N.NsaDc.InterSite.SgNB.Add.Att** counter to obtain the number of inter-site SgNB addition attempts in the LTE-NR NSA DC scenario.
  – Observe the **N.NsaDc.InterSite.SgNB.Add.Succ** counter to obtain the number of successful inter-site SgNB additions in the LTE-NR NSA DC scenario.

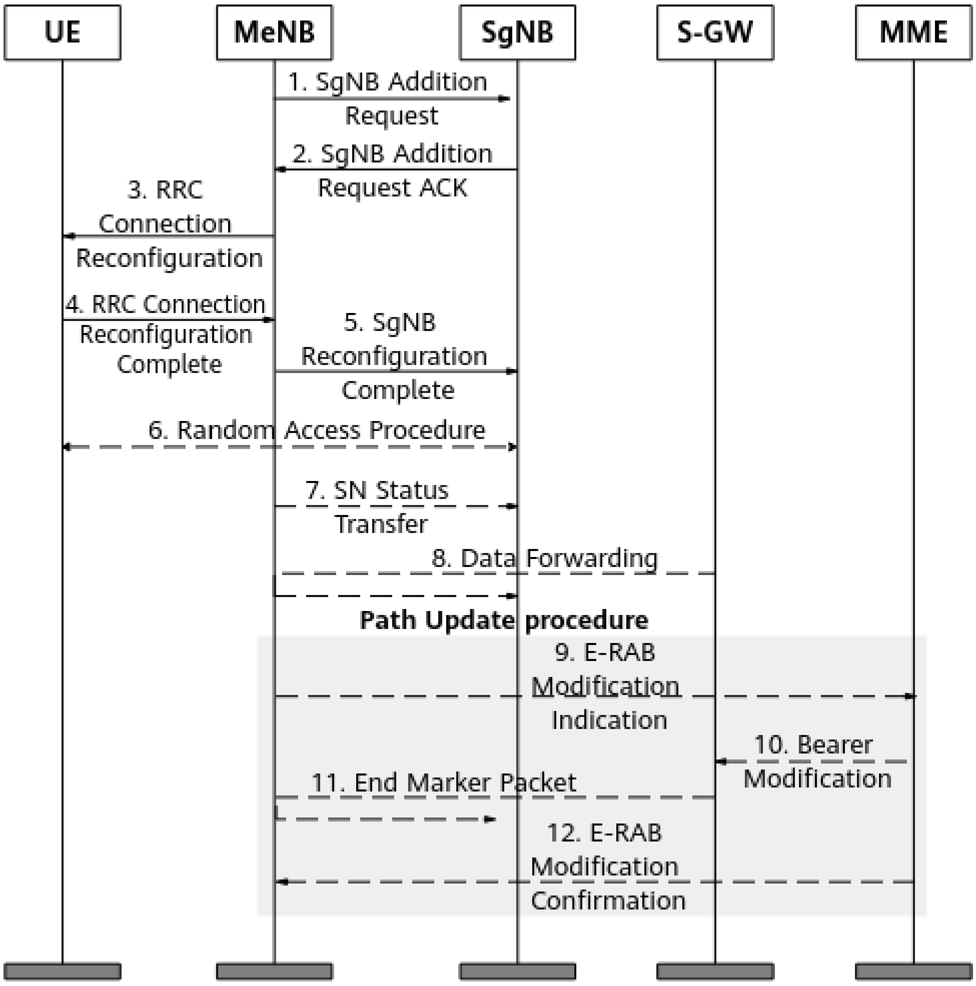
8 Appendix

This chapter describes the signaling procedures for mobility management in NSA networking.

8.1 MeNB-Initiated SgNB Addition

Figure 8-1 shows the SgNB addition procedure. For details about the triggering conditions and configuration, see *EPC-based NSA Basics*.

Figure 8-1 SgNB addition procedure



1. After receiving the B1 measurement report, the MeNB triggers an SgNB addition procedure. The MeNB adds the NR cells in the measurement report to the SgNB Addition Request message and sends this message to the SgNB. The SgNB selects the NR cell with the highest RSRP from the report. The request message carries the bearer type (MCG split bearer or SCG split bearer) and E-RAB information (E-RAB parameters and TNL transport address). The SCG-ConfigInfo IE in this message also includes the MCG configuration (DRB configuration, cell configuration, and SCG bearer encryption algorithm) and UE capabilities. The SgNB may reject the request. If the request is accepted, the corresponding radio bearer is established.

2. After the admission is complete and the SgNB allocates resources, the SgNB returns an SgNB Addition Request ACK message to the MeNB. The SCG-Config IE in this message carries the SCG radio resource configuration. For the MCG split bearer type, this IE contains the SgNB GTP Tunnel Endpoint address. For the SCG split bearer type, this IE contains the S1 DL TNL of the E-RAB.

3. The MeNB sends an RRC Connection Reconfiguration message to the UE. This message contains the NR RRC configuration message.

4. After receiving the RRC Connection Reconfiguration message, the UE completes reconfigurations. It then returns an RRC Connection Reconfiguration Complete message to the MeNB, including an NR RRC response message. If the UE fails to

complete the configurations specified in the RRC Connection Reconfiguration message, it performs the reconfiguration failure procedure.

5. The MeNB informs the SgNB that the UE has successfully completed the reconfiguration procedure by sending an SgNB Reconfiguration Complete message. This message includes the NR RRC response message.

6. If the bearers configured for the UE require SCG radio resources, the UE performs synchronization towards the SgNB PSCell and initiates random access to the SgNB PSCell.

7~8. In a bearer type change scenario, data forwarding is implemented between the MeNB and the SgNB to reduce the service interruption duration.

9~12. For SCG split bearers, user-plane paths between the SgNB and the EPC are updated. Specifically, the MeNB sends an E-RAB Modification Indication message to the core network, instructing the core network to set up the E-RAB S1-U interface with the SgNB.

For details, see chapter 10 "Multi-Connectivity operation related aspects" in 3GPP TS 37.340 V15.5.0.

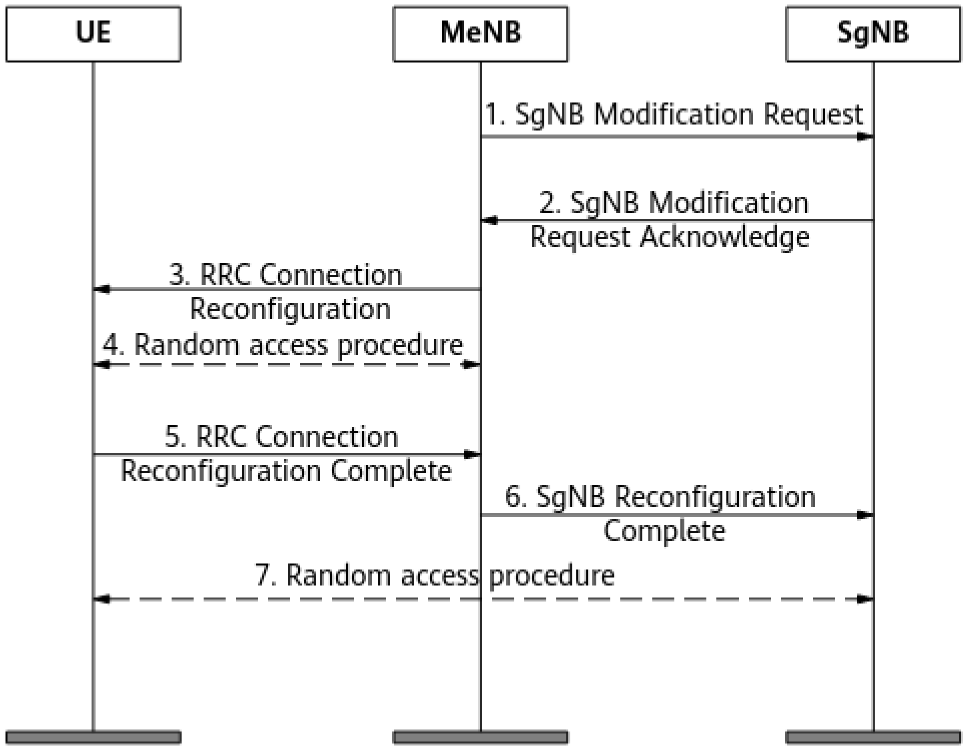
NOTE

- When the eNodeB sends an E-RAB Modification Indication message to the MME, if the MME does not respond or replies with a failure message, the UE network connection rolls back to LTE only after the timer for waiting for an S1 message expires. The eNodeB initiates an SgNB addition again only when the conditions for adding an SgNB are met.
- In NSA networking, LTE DRX and NR DRX of NSA UEs work independently.
  - After an SgNB is successfully added for an NSA UE, the LTE DRX parameters of this UE depend on the value of **CellQciPara**.***NsaDcDrxParaGroupId***. If the value is not **255**, the parameters in the **DRXPARAGROUP** MO associated with the **CellQciPara**.***NsaDcDrxParaGroupId*** parameter are used. If the value is **255**, the parameters in the **DRXPARAGROUP** MO associated with the **CellQciPara**.***DrxParaGroupId*** parameter are used.
  - After the SgNB of a UE is released, the LTE DRX parameters in the **DRXPARAGROUP** MO associated with the **CellQciPara**.***DrxParaGroupId*** parameter are used for this UE.

8.2 MeNB-Initiated Intra-MeNB Handover Without an SgNB Change

As shown in **Figure 8-2**, if NSA DC is enabled in the target MeNB cell and an X2 link has been established between the target MeNB and the SgNB, an intra-MeNB handover without an SgNB change can be performed.

Figure 8-2 MeNB-initiated intra-MeNB handover without an SgNB change



1~2. The MeNB delivers LTE intra-frequency or inter-frequency measurement configurations to the UE. The UE sends an intra-frequency or inter-frequency measurement report to the MeNB. After receiving the measurement report, the MeNB decides to trigger an intra-MeNB handover. The MeNB sends an SgNB Modification Request message to the SgNB, instructing the SgNB to update the encryption parameters. This message contains the change in UE context information such as encryption parameters used after an LTE cell handover. The SgNB completes reconfigurations and returns a response.

3~5. The UE is handed over to a new primary cell of LTE.

6~7. The UE initiates random access to a cell of the SgNB according to the reconfiguration information delivered by the MeNB.

NOTE

When the MeNB delivers LTE intra- or inter-frequency A3/A4/A5 measurement configurations to NSA UEs, whether it also delivers NR measurement configurations is determined by the **NsaDcLteMeasCtrlwithNbrMeasSw** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter, and whether it also delivers a *MaxReportRS-Index* IE is determined by the **MaxReportRsIndexSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter. For details about the *MaxReportRS-Index* IE, see section 5.5.5 "Measurement reporting" in 3GPP TS 36.331 V15.8.0.

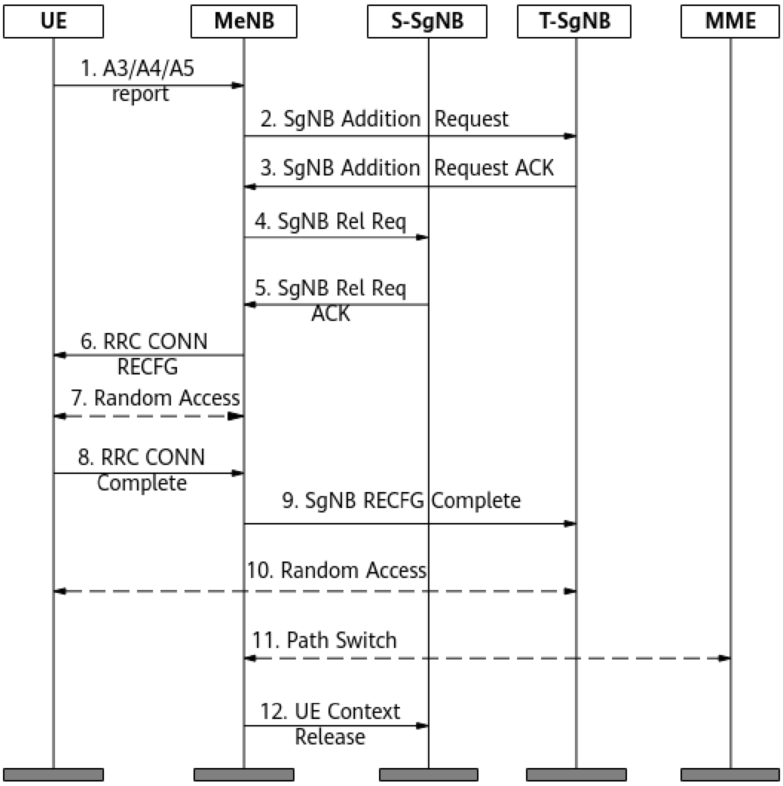
- If the **MaxReportRsIndexSwitch** option is selected, the base station includes the *MaxReportRS-Index* IE in measurement configurations delivered to the UE. After receiving the IE, the UE also reports the SSB index of the serving NR cell when reporting A3/A4/A5 measurement results. This ensures that the UE can perform non-contention-based random access on the NR side during an MeNB-initiated intra-MeNB handover without an SgNB change.
- If the **MaxReportRsIndexSwitch** option is deselected, the base station does not include the *MaxReportRS-Index* IE in measurement configurations delivered to the UE.

8.3 MeNB-Initiated Intra-MeNB Handover with an SgNB Change

If an X2 link has been set up between an MeNB and a target SgNB, an intra- or inter-SgNB change can be performed together with an intra-MeNB handover in the following scenarios, as shown in **Figure 8-3**.

- The intra-LTE measurement results reported by a UE contain the measurement results of the neighboring NR cells that have better signal quality than the serving NR cell.
- The blind PSCell configuration function is enabled in the target MeNB cell, and a blind-configurable neighboring NR cell on the highest-priority NR frequency is configured.

Figure 8-3 MeNB-initiated intra-MeNB handover with an SgNB change



1. The UE sends an LTE intra- or inter-frequency measurement report that contains the measurement results of the serving NR cell and neighboring NR cells. After receiving the measurement report, the MeNB decides to trigger an intra-MeNB handover.

2~3. The MeNB sends an SgNB Addition Request message to the target SgNB in either of the following conditions: (1) a neighboring NR cell with better signal quality than the serving NR cell is reported by the UE and served by the target SgNB; (2) a blind-configurable neighboring NR cell is configured for the target MeNB cell and served by the target SgNB.

4~5. The MeNB releases the source SgNB.

6~10. The UE performs random access to the new primary LTE cell and new NR cell based on the reconfiguration information delivered by the MeNB.

11~12. The MeNB sends a Path Switch message to the core network and sends a context release request to the source SgNB.

NOTE

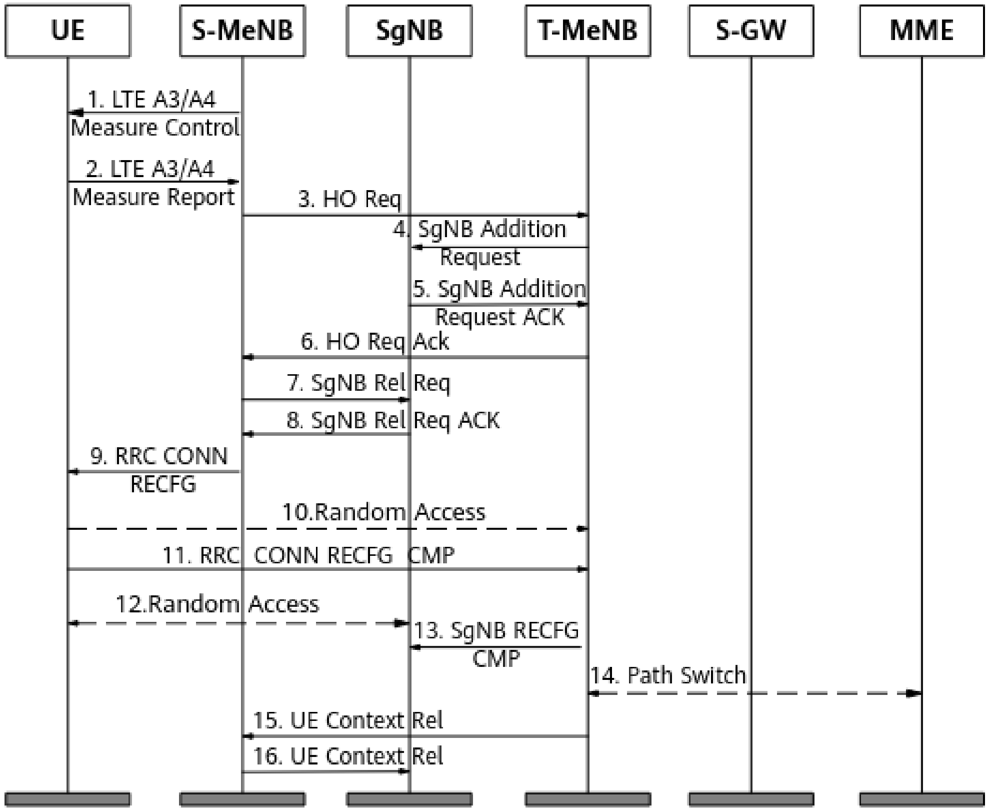
When the MeNB delivers LTE intra- or inter-frequency A3/A4/A5 measurement configurations to NSA UEs, whether it also delivers NR measurement configurations is determined by the **NsaDcLteMeasCtrlwithNbrMeasSw** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter, and whether it also delivers a *MaxReportRS-Index* IE is determined by the **MaxReportRsIndexSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter. For details about the *MaxReportRS-Index* IE, see section 5.5.5 "Measurement reporting" in 3GPP TS 36.331 V15.8.0.

- If the **MaxReportRsIndexSwitch** option is selected, the base station includes the *MaxReportRS-Index* IE in measurement configurations delivered to the UE. After receiving the IE, the UE also reports the SSB index of the serving NR cell when reporting A3/A4/A5 measurement results. This ensures that the UE can perform non-contention-based random access on the NR side during an MeNB-initiated intra-MeNB handover with an SgNB change.
- If the **MaxReportRsIndexSwitch** option is deselected, the base station does not include the *MaxReportRS-Index* IE in measurement configurations delivered to the UE.

8.4 MeNB-Initiated Inter-MeNB Handover Without an SgNB Change

As shown in **Figure 8-4**, if NSA DC is enabled in the target MeNB cell and an X2 link has been established between the target MeNB and the SgNB, an inter-MeNB handover without an SgNB change can be performed over the X2 interface.

Figure 8-4 MeNB-initiated inter-MeNB handover without an SgNB change



1~2. The source MeNB delivers LTE A3/A4 measurement configurations to the UE. The UE sends an A3/A4 measurement report. After receiving the report, the source MeNB decides to initiate an inter-MeNB handover.

3~8. The source MeNB sends a handover request to the target MeNB. The target MeNB performs SgNB addition. After the SgNB addition is complete, the source MeNB releases the SgNB. If blind PSCell configuration is enabled on the target MeNB and the blind PSCell configuration conditions are met, the target MeNB directly sends an SgNB Add Req message to the SgNB that serves the blind-configurable neighboring NR cell. If this SgNB is not the original SgNB, the procedure is called the inter-MeNB handover with an SgNB change.

9~13. The source MeNB sends an inter-eNodeB handover command to the UE. The UE accesses the target LTE cell and then initiates random access to the SgNB. The target MeNB returns a configuration complete message to the SgNB.

14~16. The target MeNB sends a Path Switch message to the core network and sends a context release request to the source MeNB. Then, the source MeNB sends the context release request to the SgNB.

NOTE

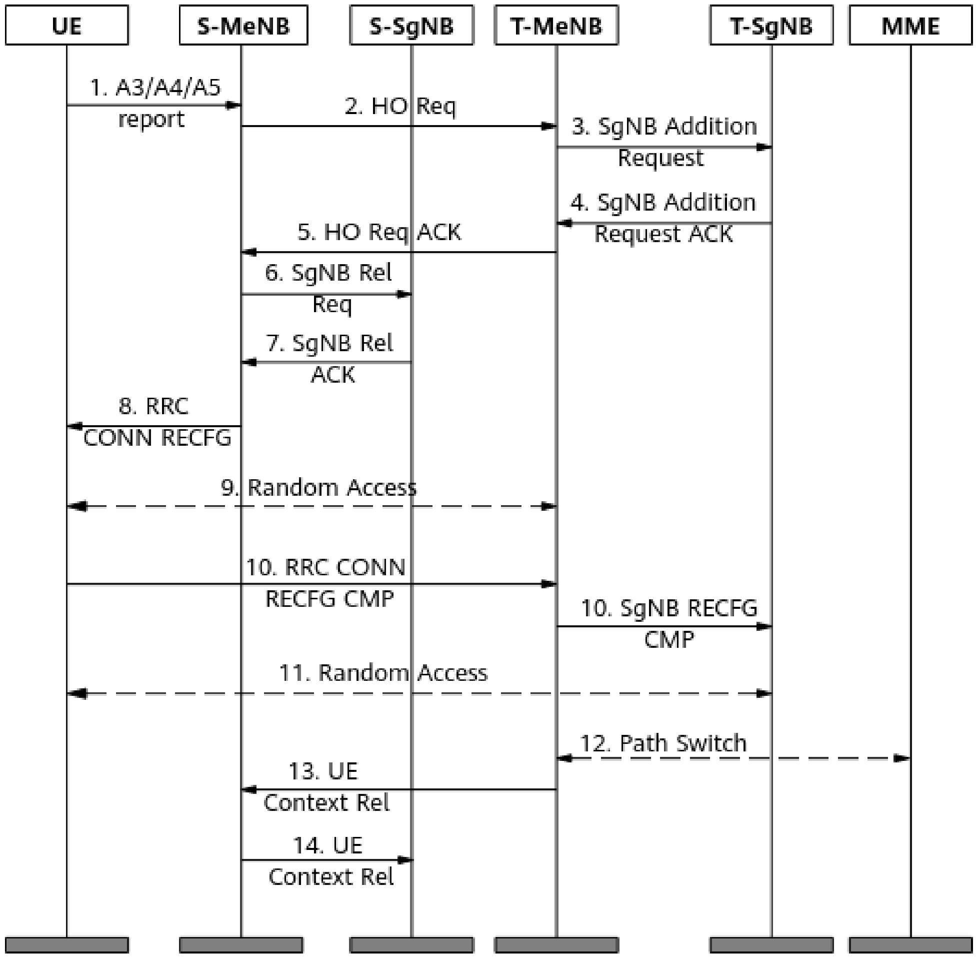
If no X2 link is set up between the target MeNB and SgNB in this procedure and the **X2_SON_SETUP_NO_NR_NRT_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter is selected, the target MeNB triggers X2 self-setup based on the gNodeB ID information contained in the handover request message, regardless of whether any cell under the SgNB is configured as a neighboring NR cell on the target MeNB. In this case, no SgNB is added during the handover. If this option is deselected, X2 self-setup can be triggered only when an SgNB cell is configured as a neighboring NR cell on the target MeNB.

8.5 MeNB-Initiated Inter-MeNB Handover with an SgNB Change

If an X2 link has been set up between a target MeNB and a target SgNB, an intra- or inter-SgNB change can be performed together with an inter-MeNB handover in the following scenarios, as shown in **Figure 8-5**.

- The intra-LTE measurement results reported by a UE contain the measurement results of the neighboring NR cells that have better signal quality than the serving NR cell. For details about this scenario, see **7.1.3 Measurement-based Simultaneous LTE and NR Cell Changes**.
- The blind PSCell configuration function is enabled in the target MeNB cell, and a blind-configurable neighboring NR cell on the highest-priority NR frequency is configured. For details about this scenario, see *EPC-based NSA Performance Enhancement*.

Figure 8-5 MeNB-initiated inter-MeNB handover with an SgNB change



1~2. The UE sends an LTE intra- or inter-frequency measurement report that contains the measurement results of the serving NR cell and neighboring NR cells. After receiving the measurement report, the source MeNB decides to trigger an inter-MeNB handover.

3~4. The target MeNB sends an SgNB Addition Request message to another SgNB (the target SgNB) in either of the following conditions: (1) a neighboring NR cell with better signal quality than the serving NR cell is reported by the UE and served by the target SgNB; (2) a blind-configurable neighboring NR cell is configured for the target MeNB cell and served by the target SgNB.

6~7. The target MeNB releases the source SgNB.

8~11. The UE performs random access to the target MeNB cell and target SgNB cell based on the reconfiguration information delivered by the source MeNB.

12~14. The target MeNB sends a Path Switch message to the core network and sends a context release request to the source MeNB. Then, the source MeNB sends the context release request to the source SgNB.

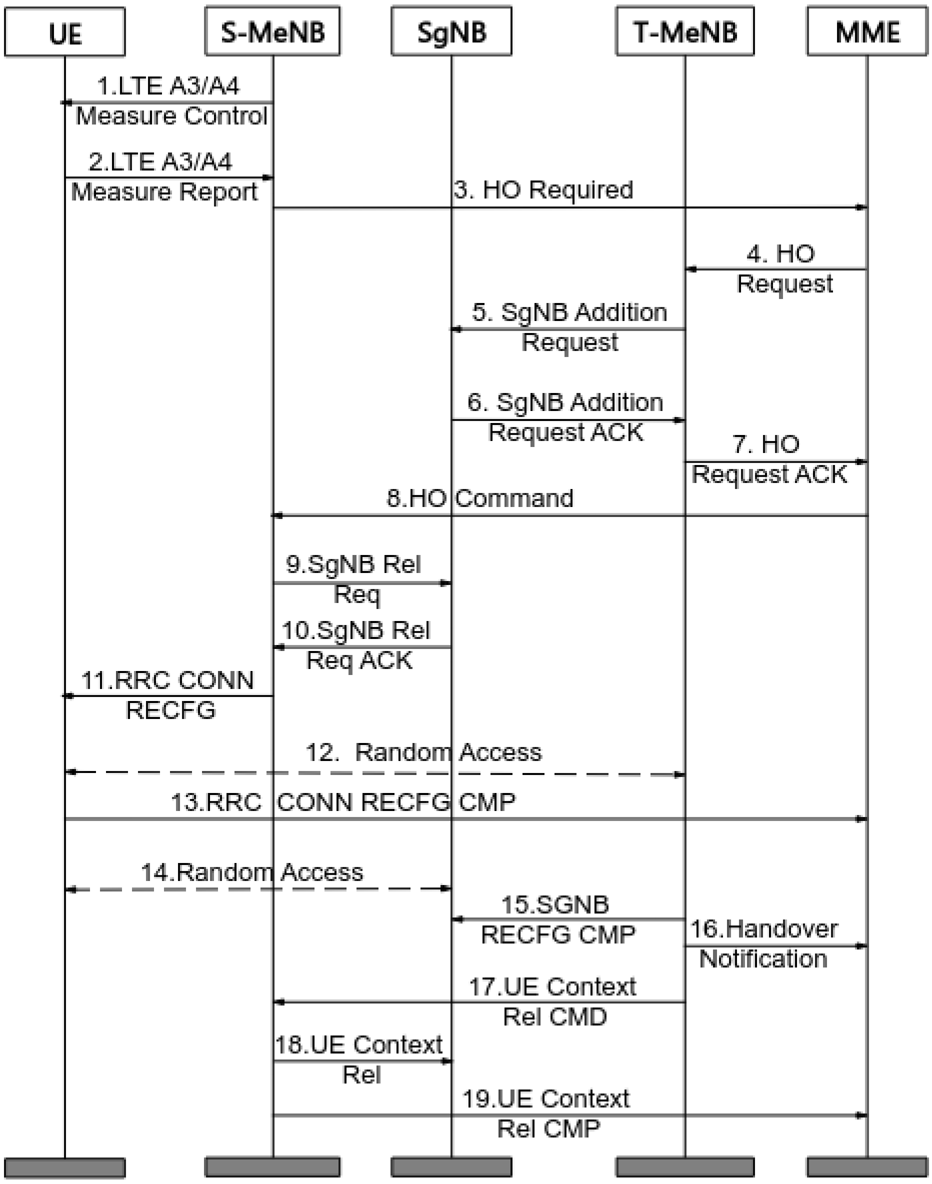
8.6 MeNB-Initiated S1-based Inter-MeNB Handover Without an SgNB Change

Figure 8-6 shows an S1-based inter-MeNB handover without an SgNB change. The handover can be performed if the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected on both the source MeNB and the target MeNB and the X2 link between the target MeNB and the SgNB has been set up.

NOTE

If the target eNodeB does not support NSA DC or does not support DC setup with the cells of the SgNB, the target eNodeB does not initiate an SgNB addition procedure.

Figure 8-6 MeNB-initiated S1-based inter-MeNB handover without an SgNB change



1~2. The source MeNB delivers LTE A3/A4 measurement configurations to the UE. The UE sends an A3/A4 measurement report. After receiving the report, the source MeNB decides to initiate an inter-MeNB handover.

3~10. The source MeNB sends an S1-based handover request to the core network. The target MeNB adds the SgNB. After the addition is complete, the source MeNB releases the SgNB.

- If the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter of the source MeNB is selected, then the source MeNB can carry the *UE Context Reference at the SgNB* IE (including the *Global en-gNB ID* IE and *SgNB UE X2AP ID* IE) to the target MeNB during an S1-based handover.
- If the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter of the target MeNB is selected and the SgNB selected by the target MeNB is the SgNB, then the target MeNB notifies the SgNB of the *SgNB UE X2AP ID* IE included in the *UE Context Reference at the SgNB* IE sent from the source MeNB. In this case, the SgNB retains the original UE context.
- If the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected on both the source MeNB and the target MeNB and if the target MeNB fails to add the SgNB in steps 5 and 6, then data forwarding from the SgNB to the target MeNB is performed for bearers in Option 3x.

NOTE

If the target MeNB selects another SgNB due to other features (for example, blind PSCell configuration), the target MeNB adds the SgNB through an SgNB change procedure. This is not controlled by the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter.

11~15. The source MeNB sends an inter-eNodeB handover command to the UE. The UE accesses the target LTE cell and then initiates random access to the SgNB. The target MeNB returns a configuration complete message to the SgNB.

16. The target MeNB sends a Handover Notification message to the core network.

17~19. The target MeNB sends a context release command to the source MeNB. The source MeNB sends a context release request to the SgNB. After the context release is complete, the source MeNB returns a context release complete message to the core network.

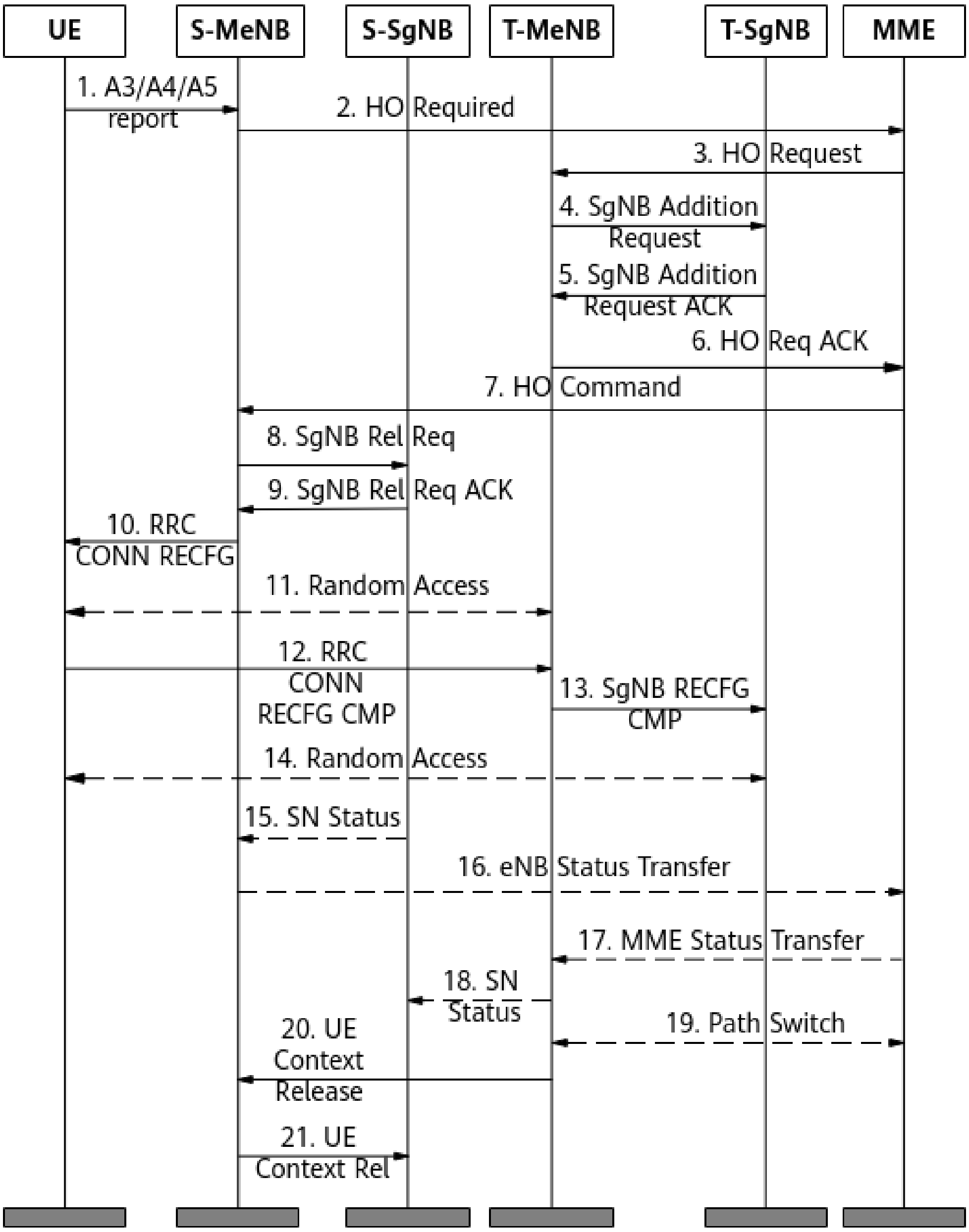
8.7 MeNB-Initiated S1-based Inter-MeNB Handover with an SgNB Change

Figure 8-7 shows an MeNB-initiated S1-based inter-MeNB handover with an SgNB change. The handover can be performed if the following conditions are met: (1) the **S1_HO_DATA_FORWARDING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected on both the source and target MeNBs; (2) the target MeNB is configured with blind-configurable neighboring NR cells, or the handover request in step 2 in the following figure indicates neighboring NR cells with stronger signals; (3) the preceding neighboring NR cells are served by an SgNB different from the source SgNB. Under these conditions, the target MeNB sends an SgNB addition request to the new SgNB.

NOTE

- If the target MeNB does not support NSA DC or does not support DC setup with the cells of the target SgNB, the target MeNB does not initiate an SgNB addition procedure.
- To prevent an inter-SgNB data forwarding failure caused by an SgNB change during S1-based inter-MeNB handover, the SgNB change is not triggered if the **S1_HO_PREP_SGNB_NO_CHANGE_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is selected.

Figure 8-7 MeNB-initiated S1-based inter-MeNB handover with an SgNB change



1. The source MeNB delivers LTE A3/A4/A5 measurement configurations to the UE. The UE sends an A3/A4/A5 measurement report. After receiving the report, the source MeNB decides to initiate an inter-MeNB handover.

2~9. The source MeNB sends an S1-based handover request to the core network. The target MeNB sends an SgNB addition request to the target SgNB. After the addition is complete, the source MeNB releases the source SgNB.

10~13. The source MeNB sends an inter-eNodeB handover command to the UE. The UE initiates random access to the target MeNB. The target MeNB returns a reconfiguration complete message to the target SgNB.

14. The UE initiates random access to the target SgNB.

15~18. The source SgNB sends an SN Status message to the source MeNB. The source MeNB sends an eNB Status Transfer message to the core network. The core network sends an MME Status Transfer message to the target MeNB. The target MeNB sends an SN Status message to the source SgNB.

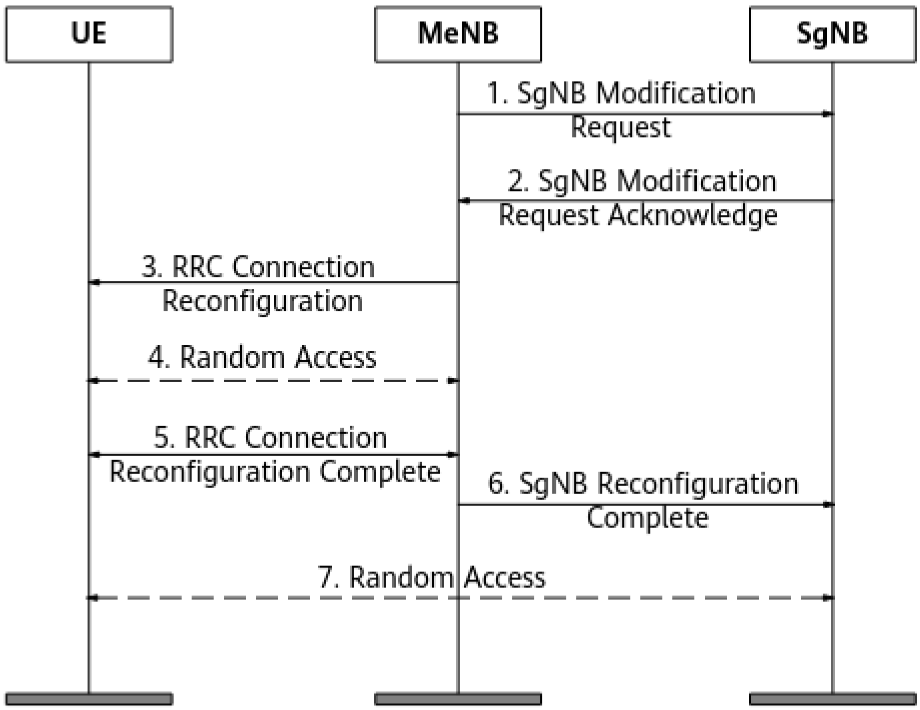
19~21. The target MeNB sends a Path Switch message to the core network and sends a context release request to the source MeNB. Then, the source MeNB sends the context release request to the source SgNB.

8.8 MeNB-Initiated SgNB Modification

An MeNB-initiated SgNB modification procedure, as shown in **Figure 8-8**, is used for bearer setup/modification/release or intra-SgNB cell change.

During an intra-MeNB handover, the MeNB triggers an SgNB modification procedure by sending an SgNB Modification Request message to instruct the SgNB PDCP layer to use the new key.

Figure 8-8 MeNB-initiated SgNB modification



The following uses SgNB modification triggered by an intra-MeNB handover as an example.

1. The MeNB sends an SgNB Modification Request message to the SgNB. This message contains information related to NR bearer setup/modification/release, neighboring NR cell measurement results, or a new key to be used by the SgNB PDCP layer.

NOTE

If the UE Context Modification Request message sent by the MME contains only the NR UE Security Capabilities IE, the MeNB records the IE and will not trigger the SgNB modification procedure immediately. In a future inter-MeNB handover, it will send the capabilities to the target MeNB.

2~3. After completing bearer setup/modification/release, SgNB cell change, or PDCP configuration by using the new key, the SgNB sends an SgNB Modification Request Acknowledge message to the MeNB. The MeNB sends a reconfiguration message to the UE. This message contains an LTE handover command and an NR PDCP reconfiguration message. If the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter is selected, the SgNB determines whether to trigger an intra-SgNB intra-frequency cell change based on the neighboring NR cell measurement results sent by the MeNB. If both the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter and the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter are selected on the NR side, the SgNB determines whether to trigger an intra-SgNB inter-frequency cell change based on the neighboring NR cell measurement results sent by the MeNB.

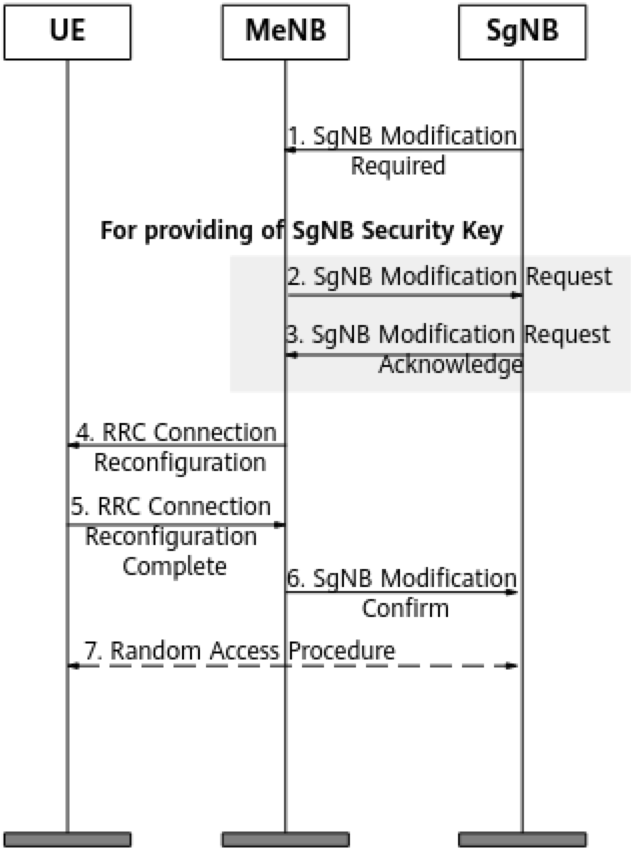
4~6. Upon reception of the reconfiguration message, the UE initiates random access to the target LTE cell and sends an RRC Connection Reconfiguration Complete message to the MeNB. The MeNB sends an SgNB Reconfiguration Complete message to the SgNB.

7. The UE initiates random access to the PSCell of the SgNB.

8.9 SgNB-Initiated SgNB Modification

SgNB-initiated SgNB modification is mainly used for NR inter-frequency measurement gap request, intra-NR resource configuration, or intra-SgNB PSCell change. The SgNB sends an RRC message to the MeNB and then to the UE by triggering a modification procedure, as shown in **Figure 8-9**. After receiving an intra-frequency or inter-frequency measurement report, the SgNB triggers an intra-base-station cell change procedure if the measured cell is served by the current SgNB.

Figure 8-9 SgNB-initiated SgNB modification



1. The SgNB sends an SgNB Modification Required message to the MeNB for NR inter-frequency measurement gap request, intra-NR resource configuration, or intra-SgNB PSCell change.

NOTE

Before an inter-frequency cell change is performed, a measurement gap configuration removal procedure must be performed. For details, see **6.1.2 PSCell Change Policies**.

2~3. If the SgNB sends an NR inter-frequency measurement gap request to the MeNB, the MeNB performs gap configuration after receiving the request and then sends an SgNB Modification Request message to the SgNB. After confirmation, the SgNB responds with an SgNB Modification Request Acknowledge message.

4~5. Operations identical with steps 3 and 4 of **8.1 MeNB-Initiated SgNB Addition** are performed.

6. After the UE completes reconfigurations, the MeNB sends an SgNB Modification Confirm message to the SgNB. This message contains the encoded NR RRC response message.

7. If the bearers configured for the UE require SCG radio resources, the UE synchronizes with the SgNB PSCell and initiates random access to the SgNB PSCell. Otherwise, the UE initiates uplink transmission after the new configuration takes effect.

For details, see chapter 10 "Multi-Connectivity operation related aspects" in 3GPP TS 37.340 V15.5.0.

NOTE

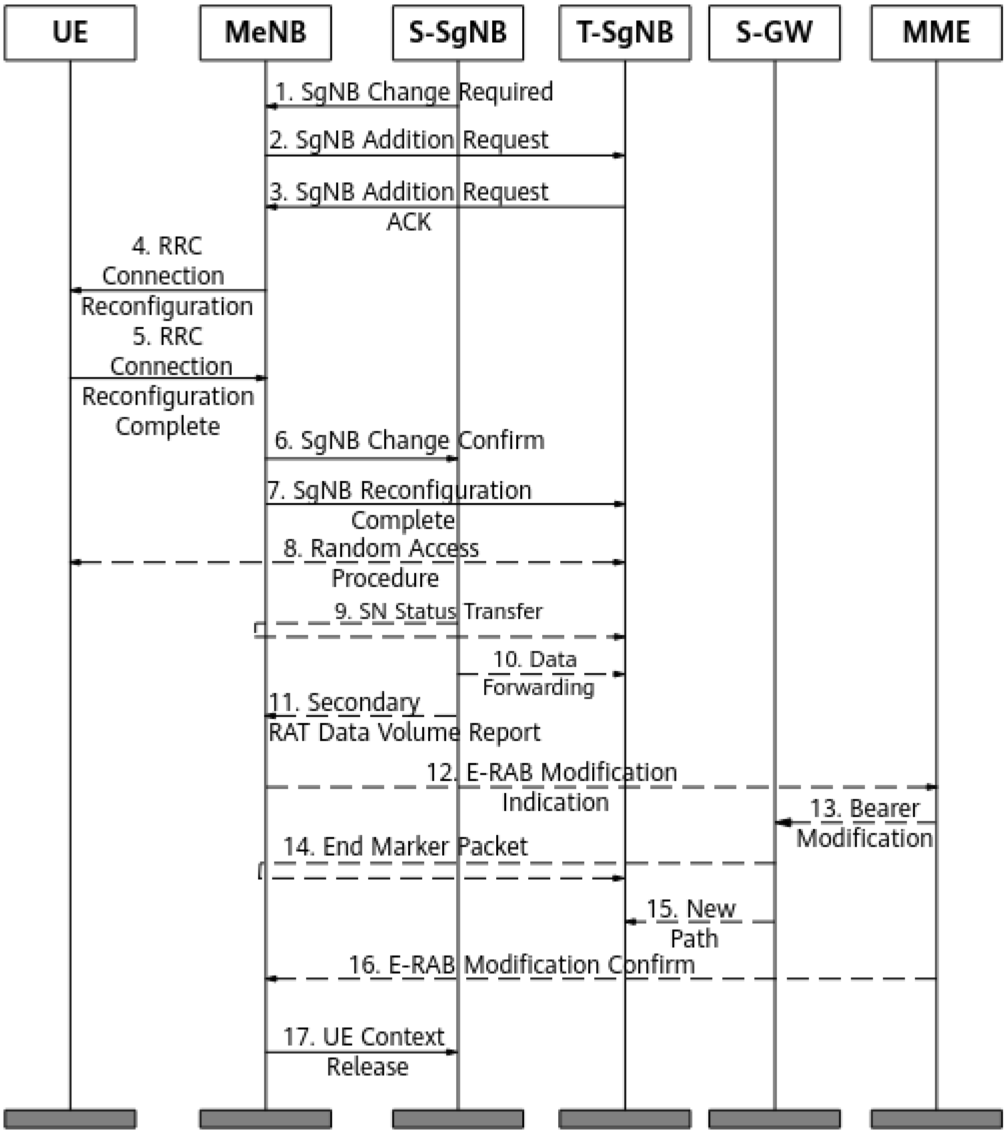
If an LTE handover is ongoing when the MeNB receives an SgNB Modification Required message from the SgNB, the LTE handover is preferentially processed, as stipulated in 3GPP TS 36.423. In this case, the MeNB returns an SgNB Modification Refuse message with the cause value "Message not Compatible with Receiver State" to the SgNB.

This may lead to a conflict between intra-SgNB handover and intra- or inter-MeNB handover, which in turn results in a decrease in the intra-SgNB handover success rate (indicated by **Intra-SgNB PSCell Change Success Rate (CU)**, which equals **N.NsaDc.IntraSgNB.PSCell.Change.Succ** / **N.NsaDc.IntraSgNB.PSCell.Change.Att** x 100%). To address this issue, optimization against conflicts between intra-SgNB handover and intra- or inter-MeNB handover can be enabled by selecting the **NSA_LNR_HO_CONFLICT_POLICY_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter. With this optimization, the handover request sent from the SgNB to the MeNB in the preceding scenario contains the handover status, thereby increasing the intra-SgNB handover success rate.

8.10 SgNB-Initiated SgNB Change

If an inter-SgNB PSCell change is required, the target PSCell can be an intra- or inter-frequency cell of a neighboring base station. **Figure 8-10** shows an SgNB change procedure.

Figure 8-10 SgNB change procedure



1. After receiving an intra-frequency or inter-frequency measurement report, the source SgNB selects an NR cell with the highest RSRP as the target NR cell for handover. The SgNB then triggers an SgNB change procedure by sending an SgNB Change Required message to the MeNB. This message contains the target SgNB ID and the measurement result.

NOTE

Before an inter-frequency cell change is performed, a measurement gap configuration removal procedure must be performed. For details, see **6.1.2 PSCell Change Policies**.

2~3. The MeNB sends an SgNB Addition Request message to the target SgNB, requesting the target SgNB to allocate resources to the UE. The message contains the UE-reported target SgNB cell measurement results obtained by the source SgNB.

4~5. Operations identical with steps 3 and 4 of **8.1 MeNB-Initiated SgNB Addition** are performed.

6. If the target SgNB successfully allocates resources to the UE, the MeNB sends an SgNB Change Confirm message to the source SgNB after the MeNB confirms that the resources of the source SgNB have been released.

7. If the RRC connection reconfiguration procedure is complete, the MeNB sends an SgNB Reconfiguration Complete message to the target SgNB to confirm reconfiguration completion. The message contains the encoded NR RRC message.

8. Operations identical with step 7 of **8.9 SgNB-Initiated SgNB Modification** are performed.

9~10. In a bearer type change scenario, data forwarding between SgNBs is required to reduce the service interruption duration.

11. The source SgNB reports the NR data volume to the MeNB.

12~16. Operations identical with steps 9 to 12 of **8.1 MeNB-Initiated SgNB Addition** are performed.

17. After receiving a UE Context Release message, the source SgNB releases the UE context.

For details, see chapter 10 "Multi-Connectivity operation related aspects" in 3GPP TS 37.340 V15.5.0.

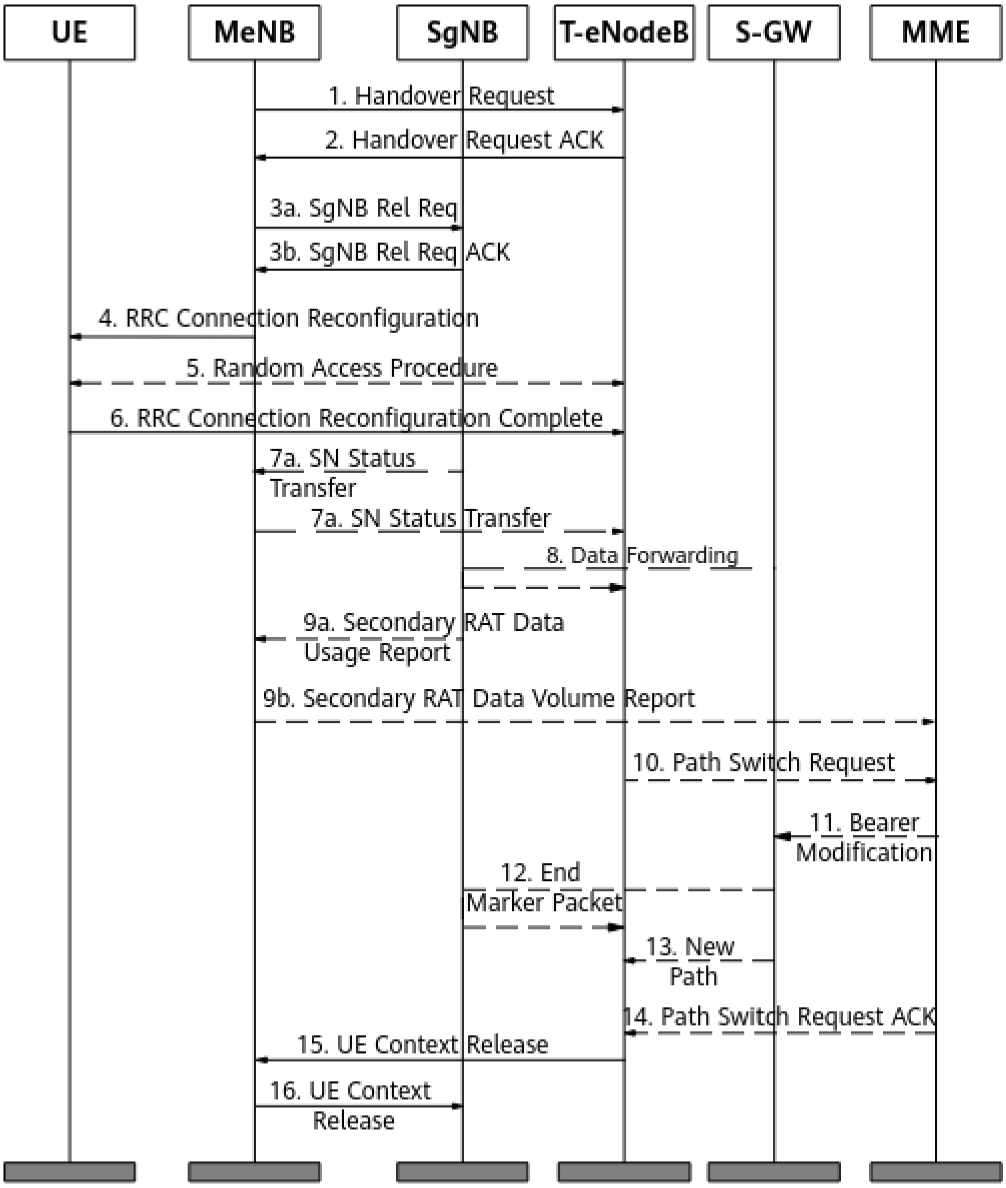
NOTE

If no X2 link is established between the MeNB and the target SgNB and the **X2_SON_SETUP_NO_NR_NRT_SW** option of the **GlobalProcSwitch**.***InterfaceSetupPolicySw*** parameter is selected, the MeNB triggers X2 self-setup based on the gNodeB ID carried in the SgNB Change Required message and returns an SgNB Change Refuse message to the source SgNB, no matter whether any cell under the target SgNB is configured as a neighboring cell on the MeNB. If this option is deselected, X2 self-setup can be triggered only when a target SgNB cell is configured as a neighboring cell on the MeNB.

8.11 MeNB-to-eNodeB Handover

When an NSA UE is handed over from the anchor MeNB to a non-anchor eNodeB, the MeNB releases the SCG, and the UE changes from the DC state to the LTE-only state. **Figure 8-11** shows the handover procedure.

Figure 8-11 MeNB-to-eNodeB handover



1. The MeNB sends a handover request to the T-eNodeB.

2. Upon reception of the handover request, the T-eNodeB makes handover preparations. Because the NSA DC function is not enabled on the T-eNodeB, the T-eNodeB does not trigger an SCG addition procedure. After handover preparations are complete, the T-eNodeB sends a Handover Request Acknowledge message to the MeNB.

3. The MeNB sends an SgNB Release Request to the SgNB and receives an acknowledgment message from the SgNB.

4~6. The UE performs an inter-eNodeB handover.

7~8. If a user-plane path exists between the SgNB and T-eNodeB, the SgNB directly forwards data to the T-eNodeB.

10~16. The T-eNodeB sends a path switch request to the core network and sends a UE context release request to the MeNB. The MeNB sends a UE context release request to the SgNB.

8.12 MeNB/SgNB-Initiated SgNB Release

Either the MeNB or the SgNB may trigger an SgNB release, depending on the scenario. For details, see **Figure 8-12** and **Figure 8-13**.

The MeNB initiates an SgNB release in the following scenarios:

- The UE is handed over from the anchor eNodeB (MeNB) to a non-anchor eNodeB.
- There is an SCG link fault.
- There is an X2-U link fault.
- The SgNB is in the inactive state.

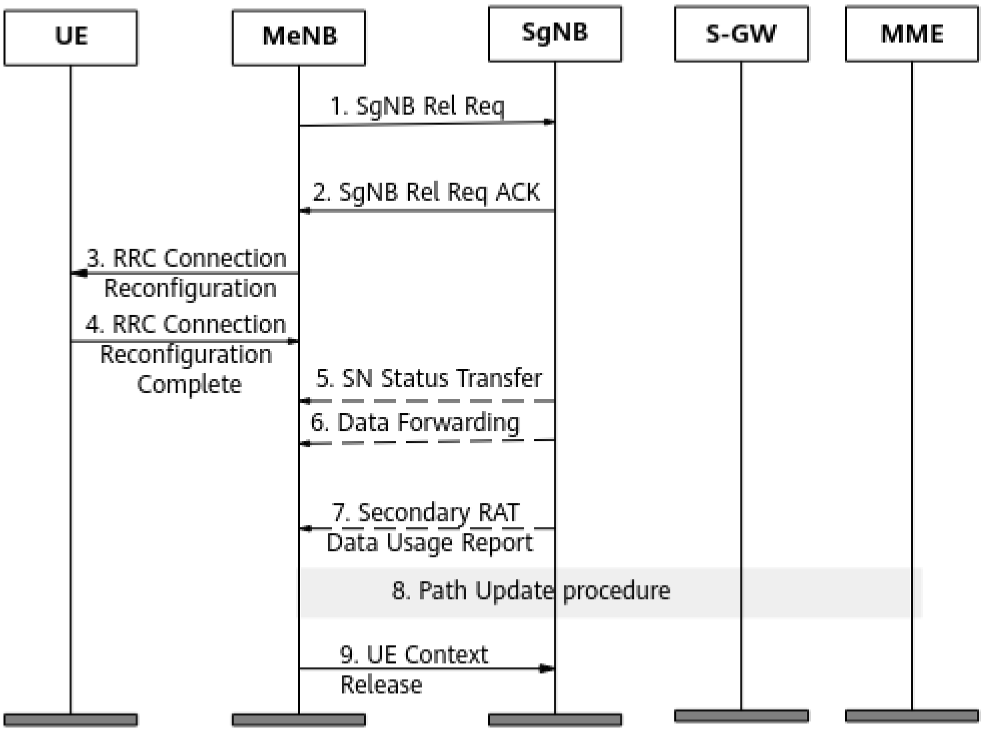
  When the UE inactivity timer on the NR side (specified by **NRDUCellQciBearer**.***UeInactivityTimer***) expires, the SgNB sends an SgNB Activity Notification message to the MeNB, indicating that the SgNB is in the inactive state. In this case, the value of the **EnodebAlgoExtSwitch**.***NrScgInactivityRelStrategy*** parameter determines how to trigger an SgNB release procedure.

  – If this parameter is set to **INSTANT_RELEASE**, the SgNB is released immediately.
  – If this parameter is set to **REL_UPON_LTE_INACTVTY_TMR_EXPN**, both the UE and the SgNB are released after the UE inactivity timer on the LTE side expires.

  NOTE

  If this parameter is set to **INSTANT_RELEASE**, the UE and the eNodeB send PDCP status reports to each other according to 3GPP TS 38.323 after the SgNB is released. As a result, the UE inactivity timer on the LTE side restarts, and the UE is released after the timer expires.

Figure 8-12 MeNB-initiated SgNB release



1. The MeNB sends an SgNB Release Request message to initiate an SgNB release procedure.

2. The SgNB sends an SgNB Release Request Acknowledge message to the MeNB to confirm that the SgNB can be released.

3~4. Operations identical with steps 3 and 4 of **8.1 MeNB-Initiated SgNB Addition** are performed.

5~6. In a bearer type change scenario, data forwarding is implemented between the MeNB and the SgNB to reduce the service interruption duration.

7. When the data volume reporting function is enabled, the SgNB reports NR data volume information to the MeNB.

8. A path update procedure is performed, which is required by the core network.

9. After receiving a UE Context Release message, the SgNB releases the UE context.

The SgNB initiates an SgNB release in the following scenarios:

- The RSRP or SSB SINR meets certain conditions.
  - When the signal quality of the PSCell keeps decreasing and no suitable neighboring cell is found for a PSCell handover, the PSCell can be deleted based on event A2. The threshold for event A2 is specified by the **NRCellNsaDcConfig**.***PscellA2RsrpThld*** parameter. If the **NRCellNsaDcConfigGrp**.***PscellA2RsrpThldOffset*** parameter is set for the highest-priority QCI of the UE, the RSRP measurement value needs to be less than **NRCellNsaDcConfig**.***PscellA2RsrpThld*** + **NRCellNsaDcConfigGrp**.***PscellA2RsrpThldOffset***. The time to trigger for event A2 is specified by the **NRCellNsaDcConfigGrp**.***PscellA2TimeToTrig***

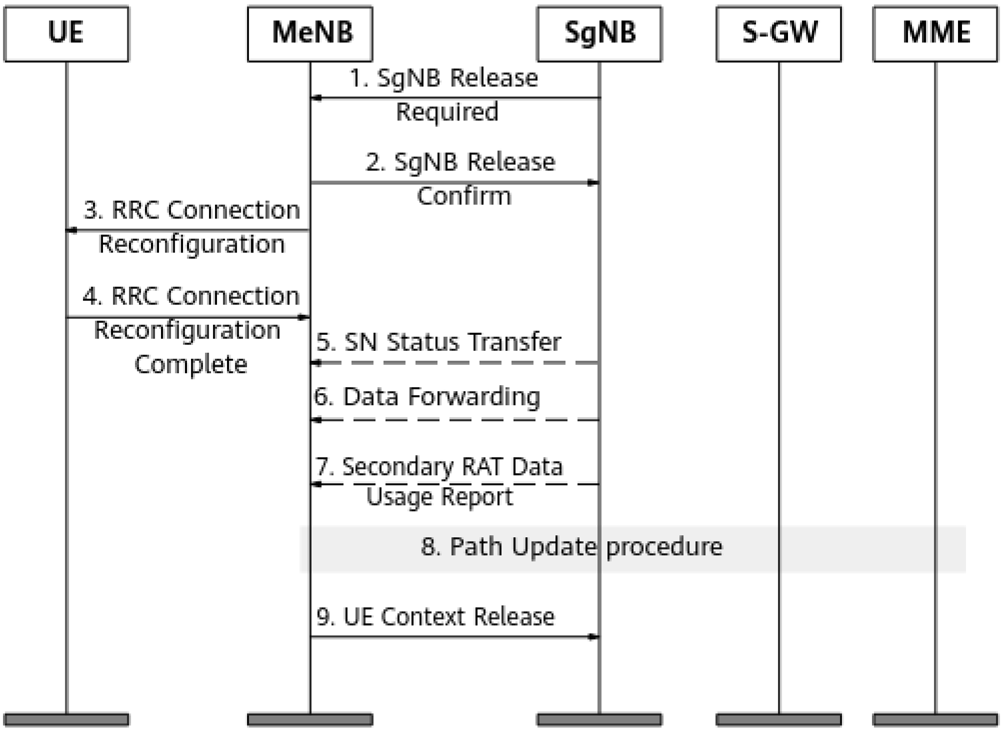
parameter. The hysteresis for event A2 is fixed at 1 dB and it is unconfigurable. If an NR cell is shared by multiple operators, the offset specified by the **NRCellOpPolicy**.***PscellA2RsrpThldOffset*** parameter can be added to the preceding A2 threshold so that different A2 thresholds can be set for different operators.

- If the **SINR_BASED_SCG_RELEASE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter is selected, the SgNB supports PSCell removal based on SINR A2. The A2 threshold is specified by the **NRCellNsaDcConfigGrp**.***PscellA2SinrThld*** parameter, and the other parameters are set to the same values as those for RSRP A2. This function enables the SgNB to initiate SgNB release based on SINR, but the MeNB still periodically initiates SgNB addition based on RSRP. It is recommended that the **NsaDcMgmtConfig**.***ScgAdditionInterval*** parameter be set to an appropriate value to prevent frequent SgNB addition and release.

  If NR-coverage-based NSA PCC anchoring is also activated on the eNodeB, the eNodeB adds an SgNB immediately after the previous SgNB is released. The **NR_COV_PCC_ANCHORING_DELAY_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter can be selected to prevent consecutive SgNB addition and release.

- The inactivity timer on the NR side expires. If the **EnodebAlgoExtSwitch**.***NrScgInactivityRelStrategy*** parameter is set to **NO_ACTIVITY_NOTIFICATION_REQ** on the LTE side and the inactivity timer specified by the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter expires on the NR side, the SgNB initiates an SgNB release.
- Packets sent from the GTP-U to the control plane are lost.

Figure 8-13 SgNB-initiated SgNB release



1. The SgNB sends an SgNB Release Required message to initiate an SgNB release procedure.

2. The MeNB sends an SgNB Release Confirm message to the SgNB to confirm the SgNB release. After receiving this message, the SgNB stops sending data to the UE.

3. The MeNB sends an RRC Connection Reconfiguration message to the UE to release the SgNB.

4. After receiving the RRC Connection Reconfiguration message, the UE completes the SgNB release reconfiguration and sends an RRC Connection Reconfiguration Complete message to the MeNB.

5~6. For SCG split bearers, data transmission is switched from the SgNB to the MeNB after the SgNB is released.

7. When the data volume reporting function is enabled, the SgNB reports NR data volume information to the MeNB.

8. For SCG split bearers, user-plane paths between the SgNB and the EPC are updated. Specifically, an E-RAB Modification Indication message is sent to the core network for switching the E-RAB S1-U interface from the SgNB to the MeNB.

9. The MeNB sends a UE Context Release message to the SgNB.

For details, see chapter 10 "Multi-Connectivity operation related aspects" in 3GPP TS 37.340 V15.5.0.

9 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 23.203: "Policy and charging control architecture"
- 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception"
- 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification"
- 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 Application Protocol (X2AP)"
- 3GPP TS 37.340: "NR; Multi-connectivity; Overall description; Stage-2"
- 3GPP TS 38.101: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.133: "NR; Requirements for support of radio resource management"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification"
- *License Control Item Lists (FDD)* in eRAN feature documentation
- *License Control Item Lists (CIoT)* in eRAN feature documentation
- *License Control Item Lists (TDD)* in eRAN feature documentation
- *LTE Spectrum Coordination* in eRAN feature documentation
- *MRO* in eRAN feature documentation
- *Multi-band Optimal Carrier Selection* in eRAN feature documentation
- *Idle Mode Management* in eRAN feature documentation
- *Mobility Management in Connected Mode* in eRAN feature documentation
- *TDM eICIC (FDD)* in eRAN feature documentation
- *Inter-RAT Mobility Load Balancing* in eRAN feature documentation
- *Intra-RAT Mobility Load Balancing* in eRAN feature documentation
- *Terminal Awareness Differentiation* in eRAN feature documentation

- *License Control Item Lists* in 5G RAN feature documentation
- *mmWave Beam Management (High-Frequency TDD)* in 5G RAN feature documentation
- *SA Mobility Management in Connected Mode* in 5G RAN feature documentation
- *SA Mobility Management in Idle Mode* in 5G RAN feature documentation
- *SA Mobility Management in Inactive Mode* in 5G RAN feature documentation
- *Multi-Frequency Smart Selection* in 5G RAN feature documentation
- *Multi-Operator Sharing* in 5G RAN feature documentation
- *Flexible User Steering* in 5G RAN feature documentation
- *ANR* in 5G RAN feature documentation
- *Standards Compliance* in 5G RAN feature documentation
- *Carrier Aggregation* in 5G RAN feature documentation
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *X2 and S1 Self-Management in NSA Networking*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Experience Improvement*
- *SRAN Networking and Evolution Overview*
- *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management*